

tRNA research 文献逐篇精读版

每篇文献 3 页：概览、方法 / 结果或综述论点、讨论与项目转化

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新版制作逻辑

Deck logic

覆盖 4 个子项目、60 篇入选文献；每篇固定 3 页，高度浓缩但保留方法、结果、讨论或综述论点。

实验文章：第 1 页讲研究问题，第 2 页讲材料 / 方法和核心结果，第 3 页讲讨论、局限与本项目可转化点。

综述文章：第 1 页讲背景与范围，第 2 页列主要论点，第 3 页列综合观点、争议和未来空白。

图像优先级：PMC 原始图 > PDF 首页缩略图 > 题录信息；所有图像页均保留来源标签。

Part 1. tRNA 的生物功能与结构特征

Biological functions and structural characteristics of tRNA

本部分包含 15 篇文献，每篇 3 页。

1. RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.
2. A virally encoded t RNA neutralizes the PARIS antiviral defence system.
3. t RNA modifications tune m6A-dependent m RNA decay.
4. An engineered UGA suppressor t RNA gene for disease-agnostic AAV delivery.
5. AAV-delivered UGA suppressor t RNA for disease-agnostic in vivo gene therapy.
6. t RNA as an assembly chaperone for a macromolecular transcription-processing complex.
7. Structural basis of t RNA recognition by the m3C RNA methyltransferase METTL6 in complex with...
8. A methyltransferase-independent role for METTL1 in t RNA aminoacylation and oncogenic transfor...
9. Translational regulation by oxidative desulfuration of t RNA modifications.
10. Structure of fungal t RNA ligase Trl1 with RNA reveals conserved substrate-binding principles.
11. In vivo structure profiling reveals human cytosolic and mitochondrial t RNA structurome and in...
12. Structural basis of t RNA recognition by the widespread OB fold.
13. Deciphering t RNA repertoires and translation coordination during mouse early embryogenesis by...
14. Mechanistic basis for PYROXD1-mediated protection of the human t RNA ligase complex against ox...
15. AAV-delivered engineered suppressor t RNA rescues visual function in mice with an inherited re...

1.1 文献概览： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.

Nature (2026) | Experimental | DOI 10.1038/s41586-025-09852-9

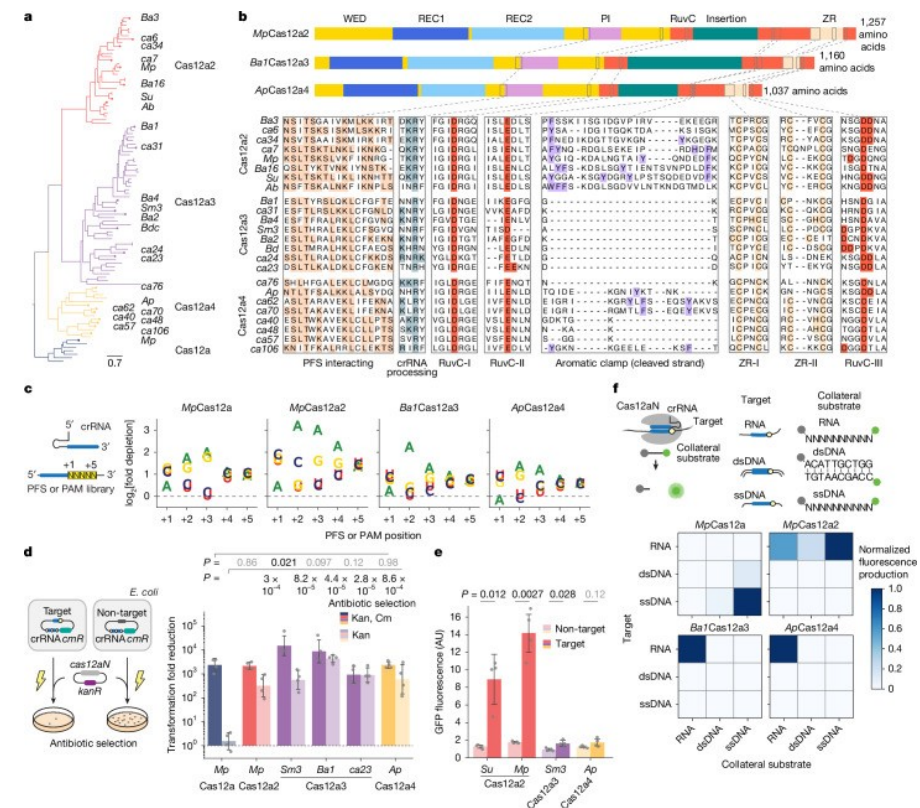
研究问题 / 背景

In all domains of life, t RNAs mediate the transfer of genetic information from m RNAs to proteins.

As their depletion suppresses translation and, consequently, viral replication, t RNAs represent long-standing and increasingly recognized target...

Here we report Cas12a3 effector nucleases from type V CRISPR-Cas adaptive immune systems in bacteria that preferentially cleave t RNAs after rec...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.1 材料方法与结果： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immuni...

Nature (2026) experimental | DOI 10.1038/s41586-025-09852-9

材料 / 方法 / 分析路线

This file contains Supplementary Figs.

This file contains the appearance of CRISPR arrays and spacer acquisition genes near cas12a3 and cas12a4 genes.

In some cases, cas genes occur on contigs too short to assess the presence of adjacent CRISPR arrays or acquisition genes.

This file contains a list of strains, plasmids and RNA and DNA sequences used in this study.

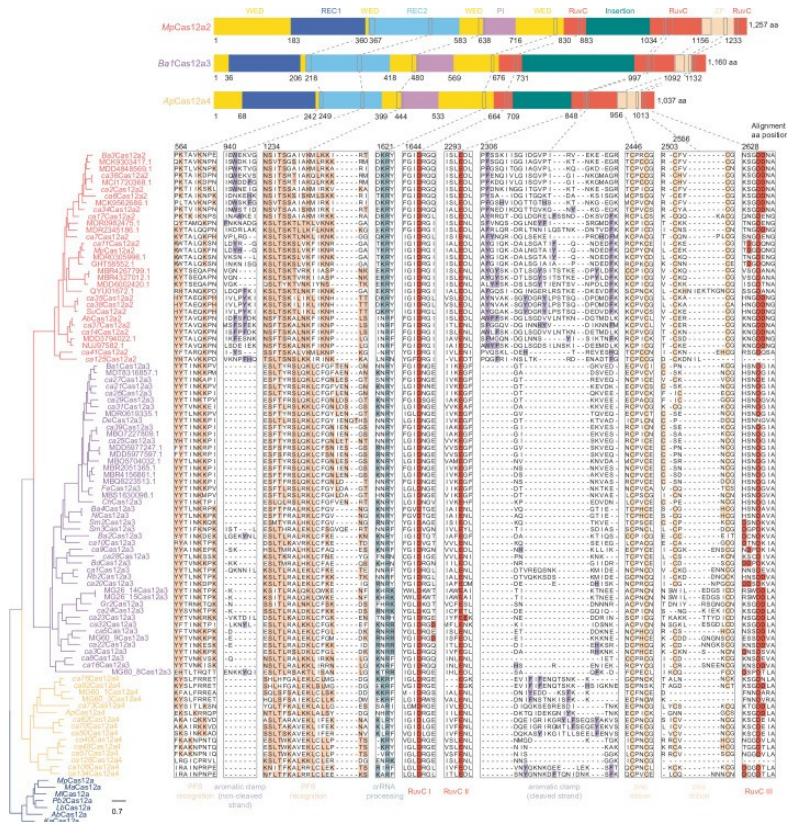
核心结果

are the averages of independent experiments (n = 2).

Note that the PAM for Mp Cas12a is the reverse complement of the sequence commonly reported for Cas12a nucleases (YYV).

d, Schematic (left) and quantification (right) of the assessment of plasmid clearance versus growth arrest in E. coli based on variations of a plasmid interference assay.

Plasmid clearance and growth arrest were differentiated on the basis of antibiotic selection for the target plasmid or the target plasmid and th...



immunity. Nature (2026) | experimental | DOI 10.1038/s41586-025-09852-9

1.2 文献概览: A virally encoded t RNA neutralizes the PARIS antiviral defence system.

1.2 | Nature (2024) | experimental | DOI 10.1038/s41586-024-07874-3

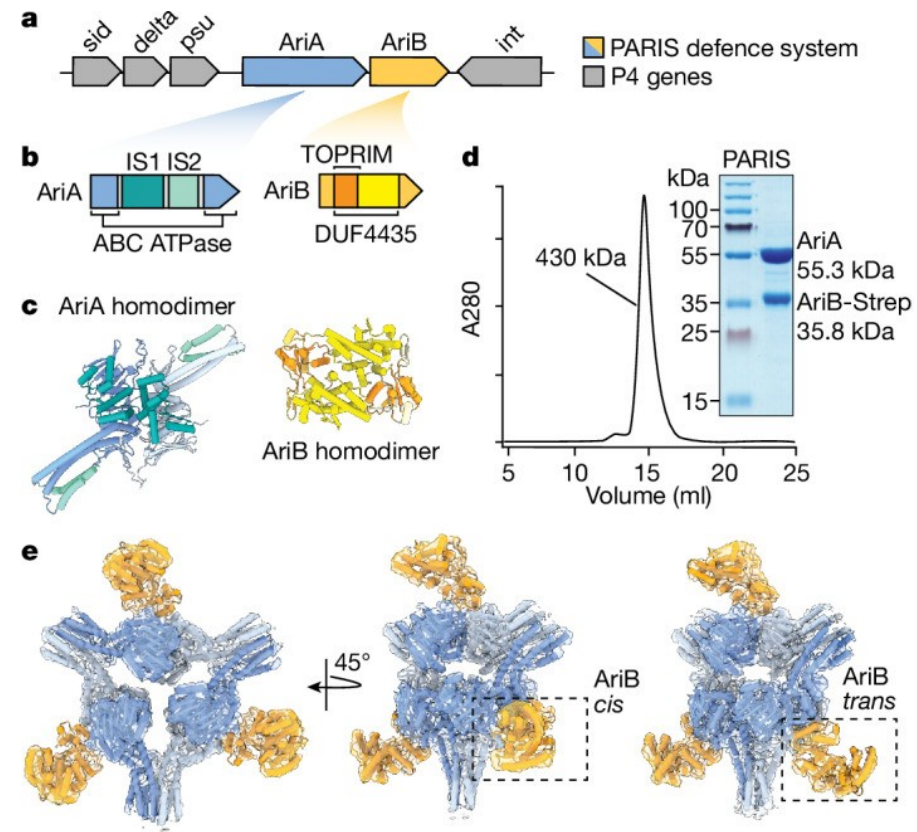
研究问题 / 背景

Viruses compete with each other for limited cellular resources, and some deliver defence mechanisms that protect the host from competing genetic...

The phage antirestriction induced system (PARIS) is a defence system, often encoded in viral genomes, that is composed of a 55 k Da ABC ATPase (...)

However, the mechanism by which Ari A and Ari B function in phage defence is unknown.

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.2 材料方法与结果: A virally encoded t RNA neutralizes the PARIS antiviral defence system...

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材料 / 方法 / 分析路线

This file contains Supplementary Tables 1–3 and legends for Supplementary Videos 1 and 2.

Masked local refinement was used to generate a high-resolution map of one asymmetric unit (Ari A2–Ari B1). Density for ATPyS is evident in both NBDs of the Ari A homodimer.

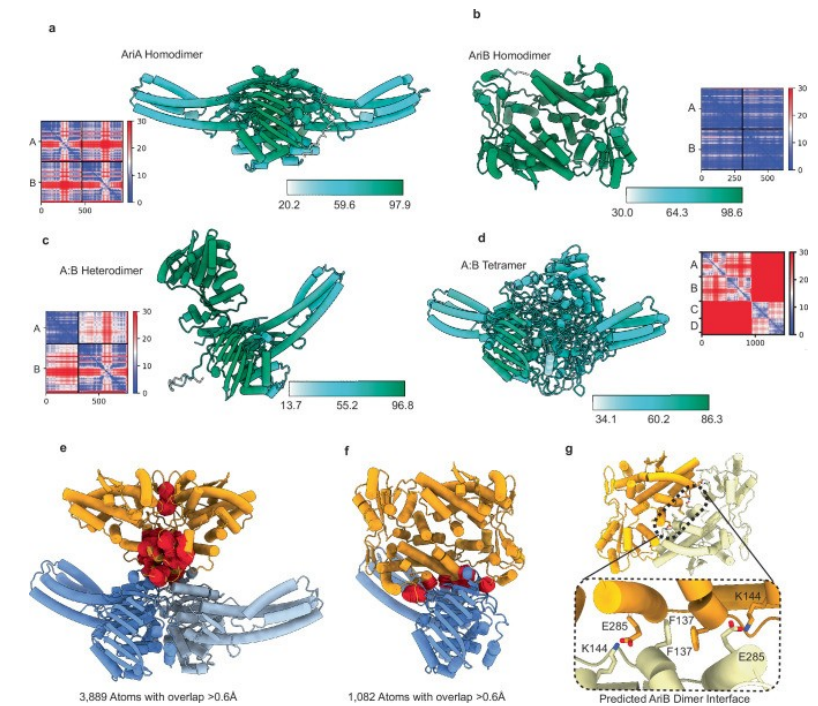
A series of electrostatic interactions link Ari B to Ari A.

核心结果

a , Total DNA was extracted from E. coli MG1655 carrying plasmids p FR85 (PARIS) and p FD250 (P Phl F -Ocr) at different time points after Ocr induction with or without DAPG (n = 3...

Samples were run on a 1% agarose TAE gel.

b , Total RNA was extracted from E. coli MG1655 carrying plasmid p FR85 (PARIS) or p FR66 (sf GFP), and p FD250 (P Phl F -Ocr) at different time points after induction of Ocr expe...



1.2 讨论与局限: A virally encoded t RNA neutralizes the PARIS antiviral defence system.

讨论与局限

Our results demonstrate how Ari A forms a propeller-shaped hexamer capable of binding three Ari B monomers.

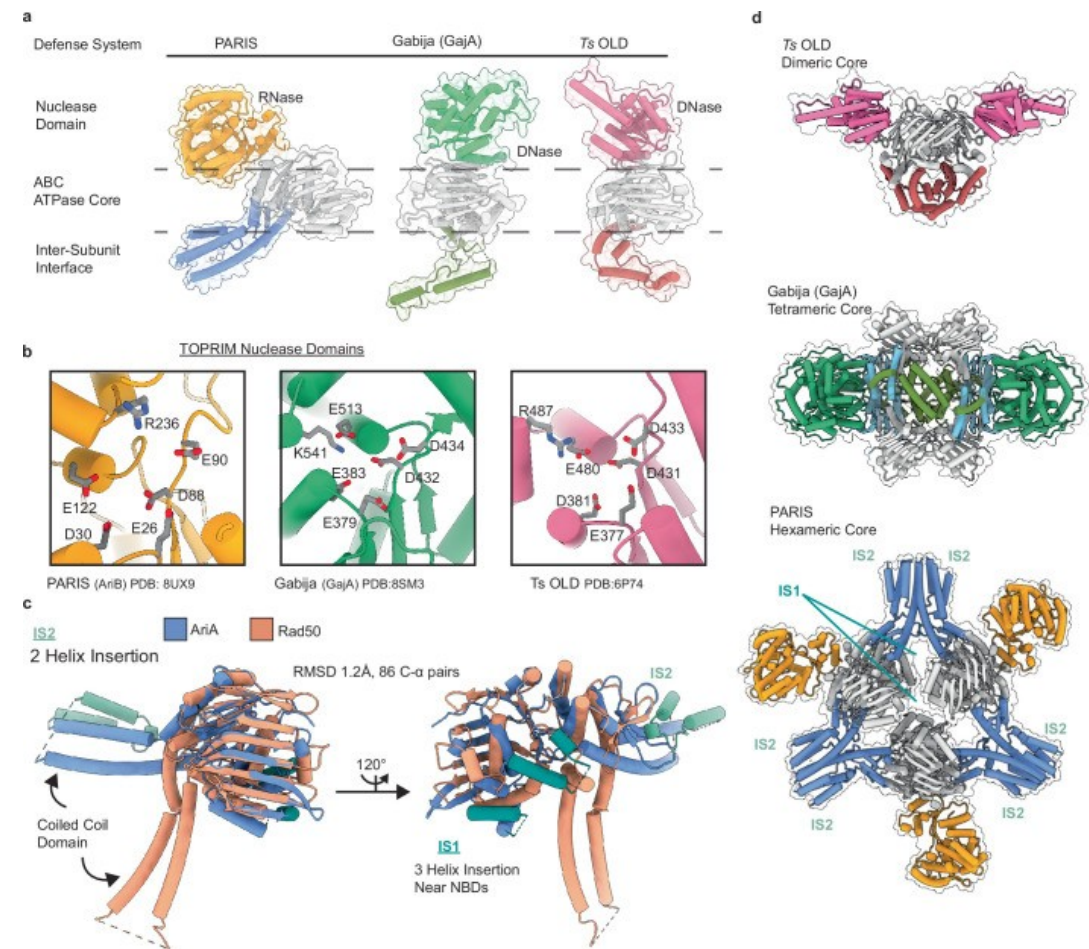
Ari A is an ATP-dependent sensor that detects viral immune suppressors (for example, Ocr), which trigger the release of Ari B.

Released Ari B assembles into a homodimer that cleaves specific host t RNAs, resulting in rapid interruption of translation followed by loss of...

As opposed to prototypical toxin–antitoxin systems, Ari B has modest toxicity when expressed alone.

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.3 文献概览： t RNA modifications tune m6A-dependent m RNA decay.

1.3 | Cell (2025) | experimental | DOI 10.1016/j.cell.2025.04.013

研究问题 / 背景

Chemically modified nucleotides in m RNA are critical regulators of gene expression, primarily through interactions with reader proteins that bi...

Here, we present a mechanism by which the epitranscriptomic mark N6-methyladenosine (m6A) is read by t RNAs during translation.

Codons that are modified with m6A are decoded inefficiently by the ribosome, rendering them "non-optimal" and inducing ribosome collisions on ce...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.3 材料方法与结果： t RNA modifications tune m6A-dependent m RNA decay.

1.3 | Cell (2025) | experimental | DOI 10.1016/j.cell.2025.04.013

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

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This couples m RNA translation to decay. 5-Methoxycarbonylmethyl-2-thiouridine (mcm5s2U) in the t RNA anticodon loop counteracts this effect.

1.3 讨论与局限： t RNA modifications tune m6A-dependent m RNA decay.

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可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.4 文献概览： An engineered UGA suppressor t RNA gene for disease-agnostic AAV delivery.

Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-025-02982-5

研究问题 / 背景

Suppressor transfer RNAs (sup-t RNAs) have the potential to rescue nonsense mutations in a disease-agnostic manner and are an alternative therap...

Among all human pathogenic nonsense variants, approximately 20% arise from C-to-T transitions that convert the CGA arginine codon into a UGA sto...

While recombinant adeno-associated virus (r AAV) has been successfully used to deliver a UAG-targeting sup-t RNA gene in vivo, extending this ap...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

nature biotechnology

8

Article

<https://doi.org/10.1038/s41587-025-02982-5>

An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery

Received: 8 October 2025

Accepted: 10 December 2025

Published online: 19 January 2026

Check for updates

Mengyao Xu^{1,6}, Hao Liu^{1,6}, Jiaming Wang^{1,7}, Andre F. C. Vieira^{1,8,9}, Xuntao Zhou¹, Nan Liu^{1,7}, Jialing Liang¹, Ailing Du¹, Xiupeng Chen^{1,7}, Ruxiao Xing¹, Yang Yang^{1,7}, Maria P. Gonzalez-Perez^{1,7}, Vikas Kumar^{1,7} & Dan Wang^{1,8,9}✉

Suppressor transfer RNAs (sup-tRNAs) have the potential to rescue nonsense mutations in a disease-agnostic manner and are an alternative therapeutic approach for many rare and ultrarare disorders. Among all human pathogenic nonsense variants, approximately 20% arise from C-to-T transitions that convert the CGA arginine codon into a UGA stop codon. While recombinant adeno-associated virus (rAAV) has been successfully used to deliver a UAG-targeting sup-tRNA gene in vivo, extending this approach to UGA-targeting sup-tRNA genes has posed unique challenges related to rAAV vector production. Here, we demonstrate that an engineered UGA-sup-tRNA gene, designed with transcriptional regulatory elements, can be efficiently packaged into rAAV for in vivo delivery. A single administration in mouse models of two distinct lysosomal storage disorders restores enzymatic activity to approximately 10% of normal levels. Comparative analysis reveals differential sup-tRNA expression and aminoacylation patterns across tissue types, which correlate with enhanced therapeutic effects. The applied rAAV-based agents and engineering strategies expand the potential therapeutic scope of sup-tRNA therapies.

Nonsense mutations are point mutations that convert a sense codon into a premature termination codon (PTC) in the mRNA, causing two deleterious events during translation: nonsense-mediated mRNA decay (NMD) and cessation of full-length protein synthesis. Nonsense suppression^{1–4}, also known as stop codon readthrough therapy, allows translation machineries to decode the PTC during protein synthesis as they normally decode sense codons, thereby overcoming its detrimental effects without changing the DNA or mRNA sequences. As nonsense mutations account for approximately 11% of human pathogenic mutations, nonsense suppression has the potential to treat a large population in a gene-agnostic and disease-agnostic manner. This is an attractive feature considering the challenges associated with developing bespoke gene therapies or mutation-specific gene editing therapies for individual rare and ultrarare diseases with a small population^{5,6}.

PTCs, like normal stop codons in mammalian cells, can be broadly categorized into three classes: UAA, UAG and UGA. By mutating the anticodon of a human lysine transfer RNA (tRNA) for decoding UAG, Kan and colleagues generated a suppressor tRNA (sup-tRNA) and demonstrated that it could readthrough a UAG PTC underlying β⁰ thalassemia, partially restoring the full-length β-globin protein in *Xenopus oocyte*⁷. Since then, a body of literature has documented the effectiveness of sup-tRNAs in overcoming various pathogenic nonsense mutations in cell culture models⁸. Recent breakthroughs in developing in vivo sup-tRNA therapy benefited from the nucleic acid delivery technologies that have advanced in the gene therapy field^{9–11}. Albers and colleagues showed that lipid nanoparticles (LNPs) could deliver an in vitro transcribed UGA-targeting sup-tRNA (UGA-sup-tRNA) to the mouse liver and lung and mediated efficient readthrough of the PTC

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Nature Biotechnology

Source: Nature biotechnology 2026; PMID 41555020; PMCID NA; DOI 10.1038/s41587-025-02982-5

1.4 材料方法与结果: An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery

Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-025-02982-5

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting sum- maries, source data, extended data, supplementary information, acknowledgements, peer r...
Therapeutics based on stop codon readthrough.
Suppression of nonsense mutations by new emerging technologies.
Targeting shared molecular etiologies to accelerate drug development for rare diseases.

核心结果

, h NFS-R8 and m NFS-R8 restored higher GLuc activity in serum than U6-R8, which outperformed U6-R7 and U6-R17 (Fig.
Quantifi- cation of the EGFP–FLAG protein in liver lysates by western blot more consistently showed the superior performance of h/m NFS-R8 over...
1d), which was corroborated by immunohistochemistry analysis with liver tissue sections (Fig.
Taken together, these results identified R8 in NFS as the leading can- didate to readthrough UGA PTC and we chose to focus on 2×h NFS-R8 in the...
Silencing UGA-sup-t RNA expression to improve r AAV production We previously showed that standard triple transfection in HEK293 cells resulted i...

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Article

<https://doi.org/10.1038/s41587-025-02982-5>

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Nature Biotechnology

Source: Nature biotechnology 2026; PMID 41555020; PMCID NA; DOI 10.1038/s41587-025-02982-5

1.4 讨论与局限： An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery...

讨论与局限

Here, we report the development of AAV-No STOP(UGA) that installs arginine at a pathogenic UGA PTC during translation and, therefore, can potent...

It builds on multiple optimization steps to improve PTC readthrough efficiency, r AAV pro- duction yield and vector genome homogeneity.

These efforts led us to demonstrate efficacy in two disease mouse models, underscoring the gene-agnostic and disease-agnostic feature of sup-t R...

Using a mouse model with a nonsense mutation in a native, nuclear gene is a rigorous experimental approach to evaluate in vivo sup-t RNA therapy.

可用于本项目的提炼

与本子项目的关系： 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

nature biotechnology

Article

Article

Received: 8 October 2025

Accepted: 10 December 2025

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Check for updates

https://doi.org/10.1038/s41587-025-02982-5

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✉Department of Genetic and Cellular Medicine, University of Massachusetts Chan Medical School, Worcester, MA, USA. ²Department of Biochemistry and Molecular Biotechnology, University of Massachusetts Chan Medical School, Worcester, MA, USA. ³Mass Spectrometry Facility, University of Massachusetts Chan Medical School, Shrewsbury, MA, USA. ⁴Borch Department of Medicinal Chemistry and Molecular Pharmacology, College of Pharmacy, Purdue University, West Lafayette, IN, USA. ⁵RNA Therapeutics Institute, University of Massachusetts Chan Medical School, Worcester, MA, USA. ⁶These authors contributed equally: Mengyao Xu, Hao Liu. ✉e-mail: Dan.Wang@umassmed.edu

Nature Biotechnology

1.5 文献概览： AAV-delivered UGA suppressor t RNA for disease-agnostic in vivo gene thera...

Source: Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-026-02999-4

研究问题 / 背景

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.5 材料方法与结果： AAV-delivered UGA suppressor t RNA for disease-agnostic in vivo gene...

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材料 / 方法 / 分析路线

核心结果

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1.5 讨论与局限： AAV-delivered UGA suppressor t RNA for disease-agnostic in vivo gene th...

Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-026-02999-4

讨论与局限

可用于本项目的提炼

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1.6 文献概览： t RNA as an assembly chaperone for a macromolecular transcription-processing...

Source: Nature structural & molecular biology (2025) | review | DOI 10.1038/s41594-025-01653-y

研究问题 / 背景

Transfer RNAs (t RNAs) are widely recognized for their role in translation. Here, we describe a previously unidentified function of t RNA as an assembly chaperone. During poxviral infection, t RNAGln/Arg lacking the anticodon mcm5s2U34 modification is specifically sequestered from the cellular t RNA pool to...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

nature structural & molecular biology

Article

<https://doi.org/10.1038/s41594-025-01653-y>

tRNA as an assembly chaperone for a macromolecular transcription-processing complex

Received: 27 November 2024

Accepted: 14 July 2025

Published online: 4 September 2025

Check for updates

Julia Bartuli^{1,2}, Stefan Jungwirth¹, Manisha Dixit^{1,3}, Takumi Okuda^{1,3}, Johannes Patrick Zimmermann^{1,3}, Matthias Erlacher^{1,3}, Tao Pan^{1,3}, Asisa Volz⁴, Alexander Hüttenhofer¹, Bettina Warscheid^{1,3}, Claudia Höbartner^{1,3}, Clemens Grimm^{1,3} & Utz Fischer^{1,3}✉

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Transfer RNAs (tRNAs) are best known for their role in decoding mRNA codons and translating them into proteins¹. However, certain tRNAs are also involved in a variety of noncanonical functions including nutrient sensing², splicing³, transcription⁴, apoptosis⁵ and scaffolding⁶, as reviewed by Su et al.⁷. In these contexts, tRNAs or their fragments can act as antisense decoys, protein modulators, primers or sensors. Recently, human tRNA^{Arg}⁸ and, to a lesser extent, tRNA^{Gln}⁹ have been identified in a cellular context that is not consistent with any of the established functions of tRNAs. Specifically, these tRNAs (termed tRNA^{Gln/Arg}) throughout this manuscript) were found to be a stoichiometric component of a macromolecular RNA polymerase complex, known as complete vRNAP, which forms in cells upon infection with the prototypic poxvirus vaccinia¹⁰. This virus belongs to the diverse group of nucleocytoplasmic large DNA viruses, comprising double-stranded DNA viruses that express their genome within the cytoplasm of their host using their own gene expression machinery^{11,12}. The megadeltaon complete vRNAP unit integrates the poxviral core RNA polymerase (core vRNAP), composed of eight Rpo subunits, with early transcription factors Rsp94, VETF-α, VETF-β, NPH-1 and E11, the capping enzyme dimer DJ/D2 and host tRNA^{Gln/Arg}.

We showed previously that recruited tRNA^{Gln/Arg} is uncharged and tethers associated factors to vRNAP through interactions with Rsp94, NPH-1 and VETF-β (ref. 8). Consistent with its biochemical composition, complete vRNAP acts as an autonomous early transcription unit

¹Department of Biochemistry 1, Theodor Boveri-Institute, University of Würzburg, Würzburg, Germany. ²Institute of Organic Chemistry, University of Würzburg, Würzburg, Germany. ³Department of Biochemistry 2, Theodor Boveri-Institute, University of Würzburg, Würzburg, Germany. ⁴Institute of Genomics and Rhinomics, Biocenter, Medical University of Innsbruck, Innsbruck, Austria. ⁵Department of Biochemistry and Molecular Biology, University of Chicago, Chicago, IL, USA. ⁶University of Veterinary Medicine Hannover (TiHo), Institute of Virology, Hannover, Germany. ⁷Helmholtz Institute for RNA-based Infection Research (HIR), Helmholtz Centre for Infection Research (HZI), Würzburg, Germany. ⁸Present address: National Heart Lung and Blood Institute, National Institutes of Health, Bethesda, MD, USA. ⁹These authors contributed equally: Julia Bartuli, Clemens Grimm. ✉e-mail: clemens.grimm@uni-wuerzburg.de; utz.fischer@uni-wuerzburg.de

Nature Structural & Molecular Biology | Volume 32 | November 2025 | 2349–2358

2349

1.6 综述主要论点: t RNA as an assembly chaperone for a macromolecular transcription-pro...

671 | Nature structural & molecular biology (2025) | review | DOI 10.1038/s41594-025-01653-y

主要论点

Coordinated assembly of Rap94, NPH-I and E11 with v RNAP by t RNAGln/Arg To test whether t RNAGln/Arg acts as an assembly chaperone rather than...

We isolated a minimal version of v RNAP, consisting of the eight Rpo subunits, only as a presumed starting point of assembly.

This unit, along with recombinant transcription factors (Extended Data Fig. 1a) and t RNAGln/Arg isolated from complete v RNAP, was then used to study the formation of complete v RNAP in vitro (Fig.

Cosedimentation and gel mobility shift experiments showed that [32P]t RNAGln/Arg alone failed to bind minimal v RNAP (Fig.

tRNA as an assembly chaperone for a macromolecular transcription-processing complex

Received: 27 November 2024
Accepted: 14 July 2025
Published online: 4 September 2025
[Check for updates](#)

Julia Bartuli^{1,2}, Stefan Jungwirth¹, Manisha Dixit^{1,2}, Takumi Okuda^{1,2}, Johannes Patrick Zimmermann^{1,2}, Matthias Erlacher^{1,2}, Tao Pan^{1,2}, Asisa Volz¹, Alexander Hüttenhofer¹, Bettina Warscheid¹, Claudia Höbartner^{1,2}, Clemens Grimm^{1,2} & Utz Fischer^{1,2}✉

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1.6 综合观点与空白: t RNA as an assembly chaperone for a macromolecular transcription-proce...

综合观点与空白

In this study, we dissected the entire assembly pathway of complete v RNAP through a combination of biochemical and structural (cryo-EM) approach... Our findings identify t RNAGln/Arg as assembly chaperones for a macromolecular transcription and processing complex and, thus, add a novel aspect... Assembly chaperones are a heterogeneous group of factors that join components with otherwise little or no affinity to each other and typically U... They have, however, no further role with respect to the function of the assembled unit and often dissociate from the assembled particle.

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

nature structural & molecular biology

Article

https://doi.org/10.1038/s41594-025-01653-y

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Nature Structural & Molecular Biology | Volume 32 | November 2025 | 2349–2358

2349

Source: Nature structural & molecular biology 2025; PMID 40908366; PMCID PMC12618233; DOI 10.1038/s41594-025-01653-y

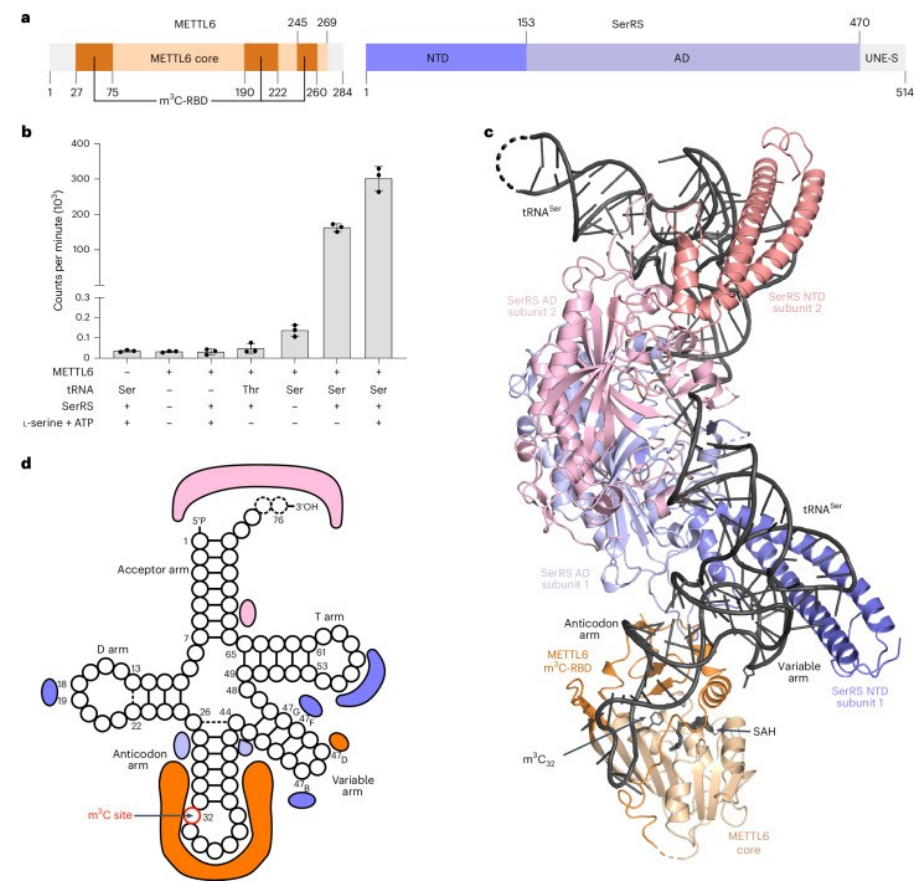
1.7 文献概览: Structural basis of t RNA recognition by the m3C RNA methyltransferase MET...

Nature structural & molecular biology (2024) | experimental | DOI 10.1038/s41594-024-01341-3

研究问题 / 背景

Methylation of cytosine 32 in the anticodon loop of t RNAs to 3-methylcytosine (m3C) is crucial for cellular translation fidelity. Misregulation of the RNA methyltransferases setting this modification can cause aggressive cancers and metabolic disturbances. Here, we report the cryo-electron microscopy structure of the human m3C t RNA methyltransferase METTL6 in complex with seryl-t RNA synthetase (S...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.7 材料方法与结果： Structural basis of t RNA recognition by the m3C RNA methyltransferas...

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer r...

RNA nucleotide methylation: 2021 update.

Rapid t RNA decay can result from lack of nonessential modifications.

Retrograde transfer RNA nuclear import provides a new level of t RNA quality control in *Saccharomyces cerevisiae*.

核心结果

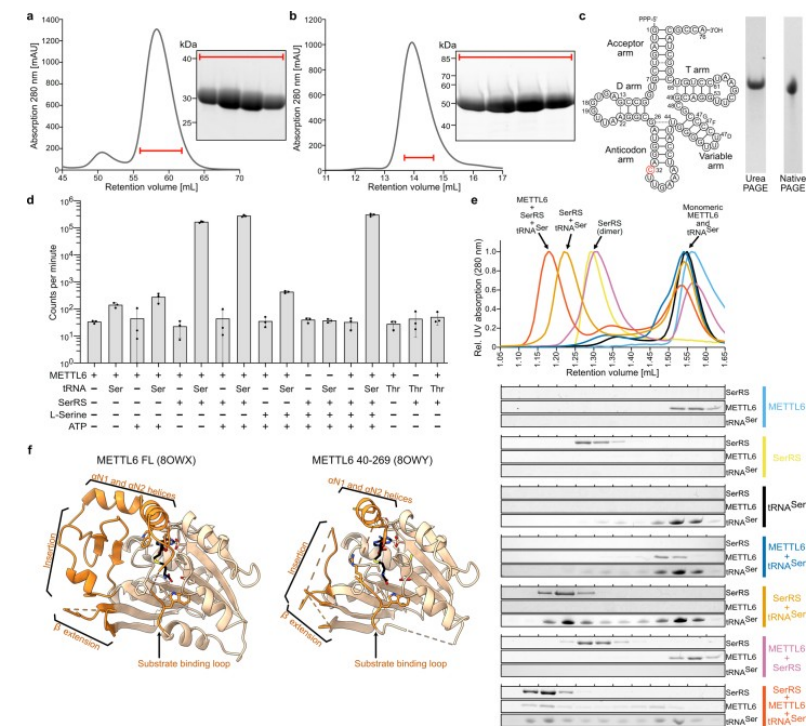
, we identified the methyltransferases METTL11A and METTL11B, which methylate the N-terminal α -amino group of proteins, as relatives with a si...

Overall, our METTL6–Ser RS–t RNASer complex structure identifies the m3C methyltransferase-specific RNA-binding domain, which differs from so-fa...

Acceptor arm D arm T arm Anticodon arm m3C site Variable arm 32 26 22 13 7 65 49 48 53 61 1 76 18 19 44 5'P 3'OH 47B 47D 47F 47G d t RNASer t RN...

1 | The structure of the Ser RS–t RNA–METTL6 complex.

a, Domain organization of human METTL6 and Ser RS.



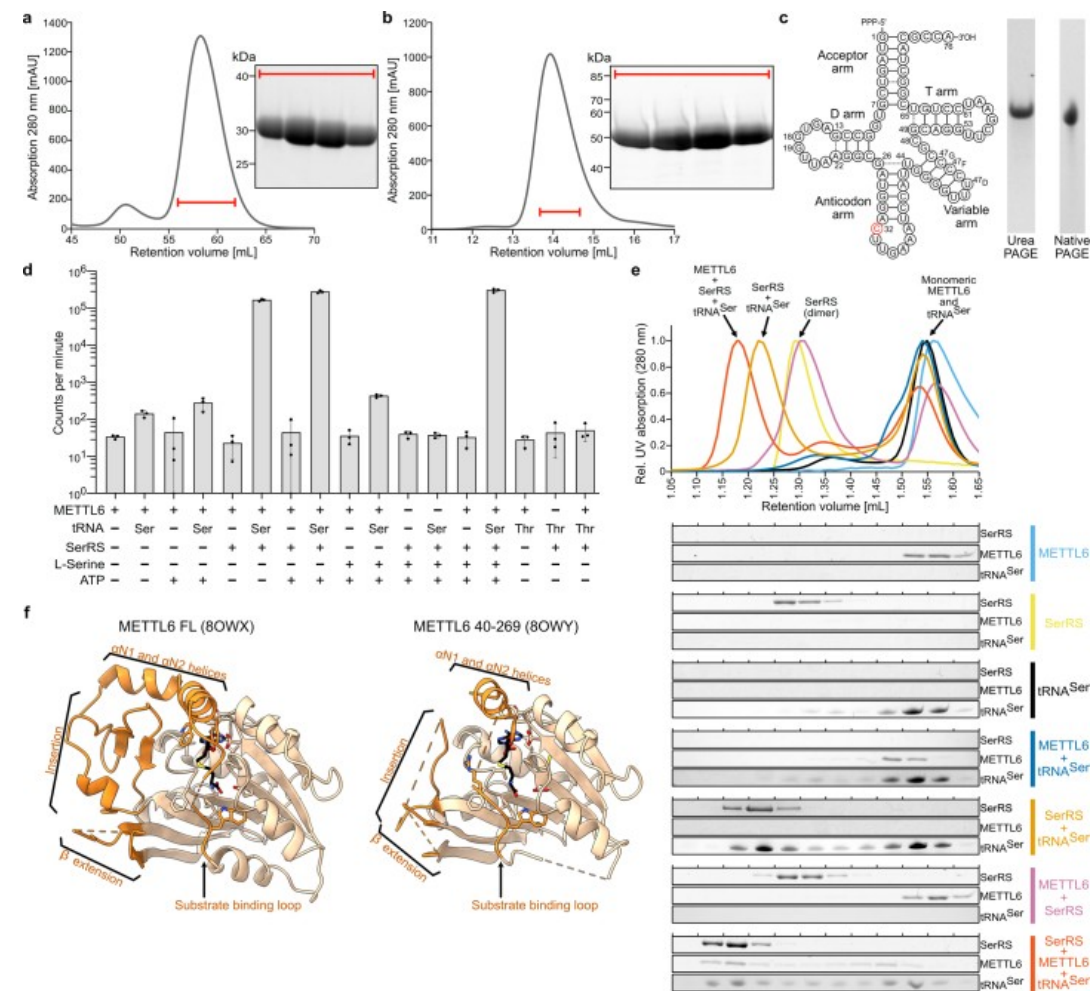
1.7 讨论与局限: Structural basis of t RNA recognition by the m3C RNA methyltransferase...

讨论与局限

In this work, we present the cryo-EM structure of METTL6 in complex with Ser RS bound to the co-products of the methyl transfer, the m3C32-mod... We identify the tripartite t RNA-binding domain of METTL6 and show that Ser RS is necessary as a t RNA selection factor that augments METTL6 act... Aminoacylation-independent functions of aminoacylases have been identified, most commonly as gene-regulatory auto-feedback mechanisms53–56. Ser RS plays a noncanonical role in vascular development that is independent of its aminoacylation activity57–59.

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.8 文献概览: A methyltransferase-independent role for METTL1 in t RNA aminoacylation an...

研究问题 / 背景

Cancer genomes are shaped by mutational processes that change the sequence or structure of DNA, coupled to selection for mutations that increase...

1 The identification of recurrent mutations affecting individual genes led to the discovery of many of the best-known oncogenes and tumor suppre...

2 – 5 However, genetic lesions affecting multiple genes, such as aneuploidy or focal copy-number alterations of chromosomal segments, are highly...

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.8 材料方法与结果： A methyltransferase-independent role for METTL1 in t RNA aminoacylati...

8 | Molecular cell | 2025 | Experimental | DOI 10.1016/j.molcel.2025.01.003

材料 / 方法 / 分析路线

Original plasmids used in this manuscript have been submitted to addgene.org (see key resources table for unique identifiers).

We assessed DNA copy number in human 12q-amplified liposarcomas by leveraging single nucleotide polymorphism (SNP) arrays performed on a cohort...

Six of these 50 cases lacked evidence of 12q amplification, and so were excluded from our analysis (cases excluded: PT1DD, PT13DD, PT20DD, PT21D...

Minimal amplicons were defined as a region of genomic amplification, defined as log2 DNA copy number ratio ≥ 1 , in probes spanning at least 100...

核心结果

Amplification of chromosomal material derived from 12q13-15 is common in human cancer and believed to result in overexpression of multiple colla...

To define the oncogenes involved, we overexpressed genes recurrently amplified in human liposarcoma using a zebrafish model of the disease.

We found several genes whose overexpression collaborated with AKT in sarcomagenesis, including the t RNA methyltransferase METTL1.

This was surprising, because AKT phosphorylates METTL1 to inactivate its enzymatic activity.

Indeed, phosphomimetic S27D or catalytically dead alleles phenocopied the oncogenic activity of wild-type METTL1.

1.8 讨论与局限: A methyltransferase-independent role for METTL1 in t RNA aminoacylation...

讨论与局限

The oncogenes driving selection for some of the most common copy-number aberrations in human cancer remain incompletely defined. Here, we used a zebrafish model to define oncogenes within 12q13–15 amplicons, a common aberration in multiple tumor types, including liposarcom...

6 , 12 , 32 We confirmed the long-suspected roles of CDK4 , MDM2 , and HMGA2 as bona fide liposarcoma oncogenes and also found genes whose contr...

We were most surprised to find that overexpression of METTL1 collaborates with AKT activation in sarcomagenesis because AKT phosphorylates and i...

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.9 文献概览: Translational regulation by oxidative desulfuration of t RNA modifications.

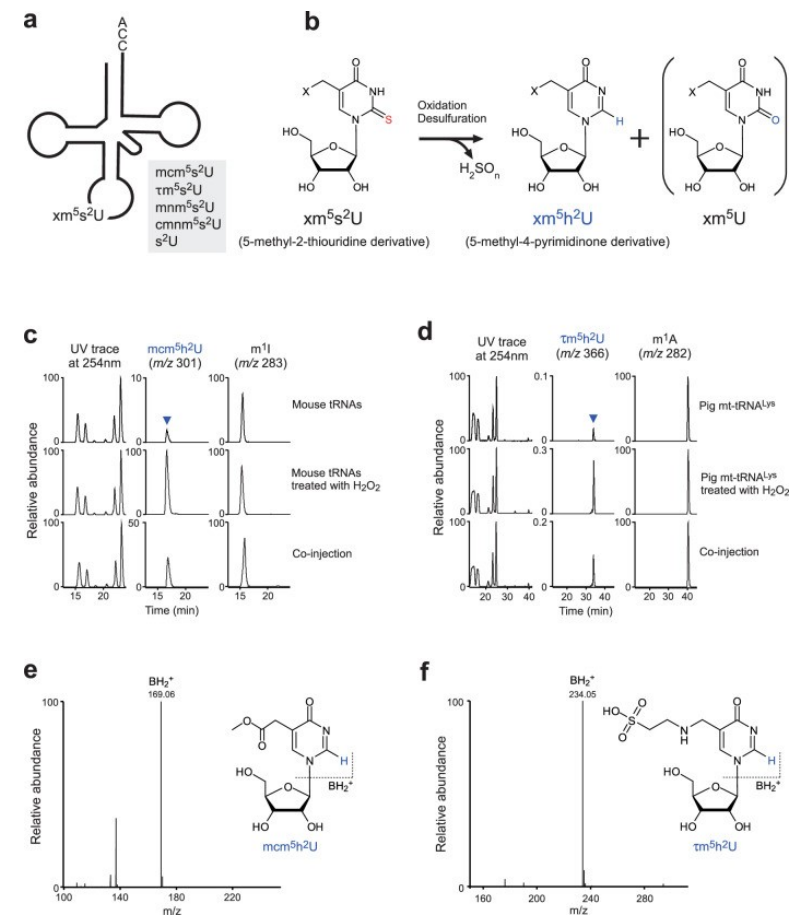
研究问题 / 背景

Higher-order biological processes emerge from sophisticated regulation of gene expression at multiple stages at both transcriptional and post-tr...

In protein synthesis, transfer RNA (t RNA) anticodons recognize and decode messenger RNA (m RNA) codons on the ribosome.

t RNAs undergo the most extensive and diverse range of post-transcriptional modifications, particularly within the anticodon region.

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.9 材料方法与结果: Translational regulation by oxidative desulfuration of t RNA modifica...

Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-70126-7

材料 / 方法 / 分析路线

Description of Additional Supplementary Files

核心结果

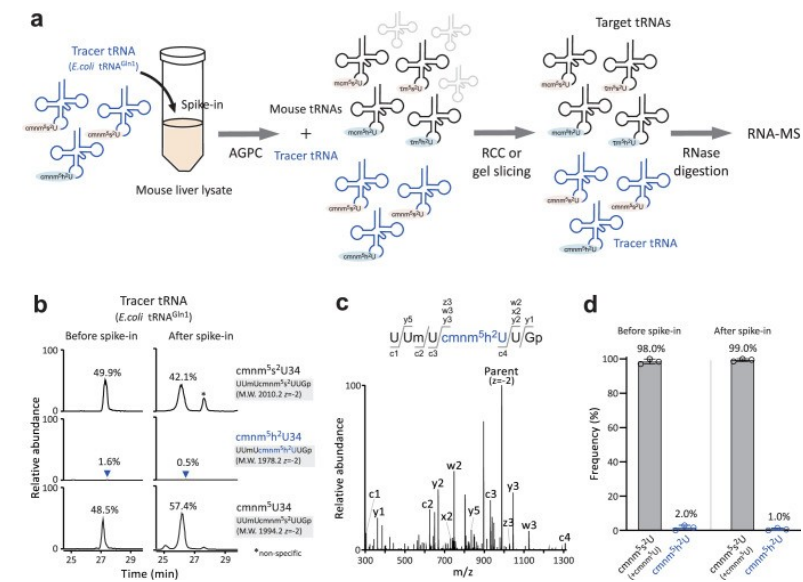
Detection of h2U derivatives in mammalian and bacterial t RNAs We reviewed our previous liquid chromatography-mass spectrometry (LC-MS) data on...

In LC-MS nucleoside analyses of total RNA from mouse livers, we clearly detected unassigned nucleosides with m/z 301 (Fig.

In addition, we also detected nucleosides with m/z 366 in mitochondrial t RNAs isolated from pig liver (Fig.

Collision- induced dissociation (CID) of these nucleosides generated base- related ions (BH2 +) with loss of unmodified ribose (Fig.

1e, f), sug- gesting that the 32 Da loss originates from their bases.



1.9 讨论与局限: Translational regulation by oxidative desulfuration of t RNA modifications

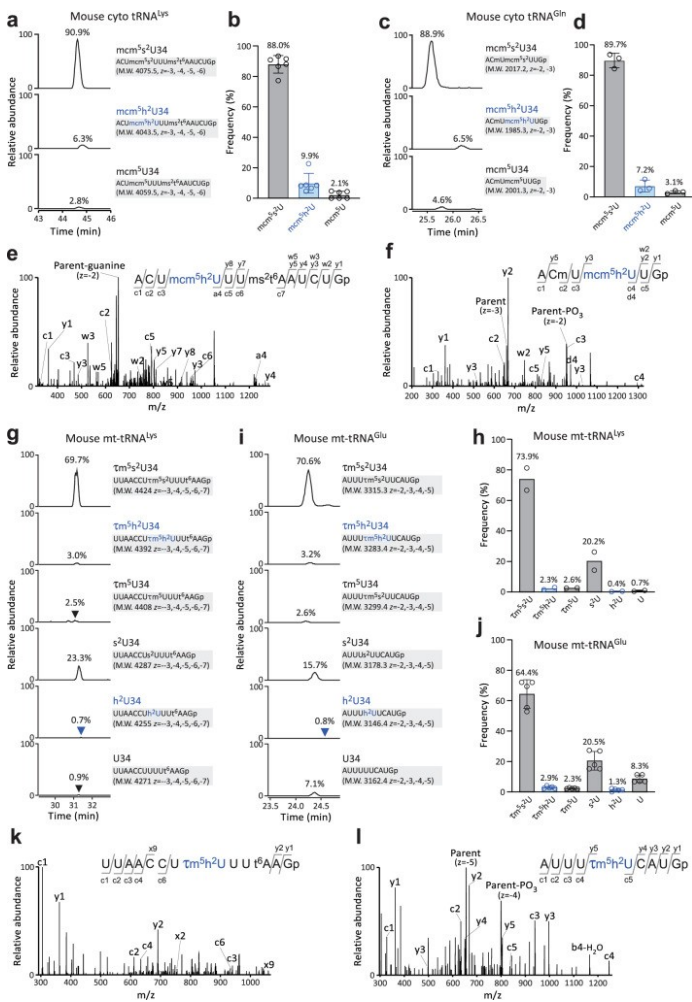
Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-70126-7

讨论与局限

Through a spike-in experiment, we demonstrated that 2-thiouridine derivatives in t RNA are desulfurized in cells and converted to h 2 U derivati...

3b, d), and τm 5 h 2 U was detected in only a few percent of mitochondrial t RNAs with τm 5 s 2 U (Fig.

However, the abundance of h 2 U derivatives varies in different cell types. Approximately 15% of t RNA Lys in He La cells contained mcm 5 h 2 U, whereas little was detected in A549 and T24 cells (Supplementary Fig.



可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

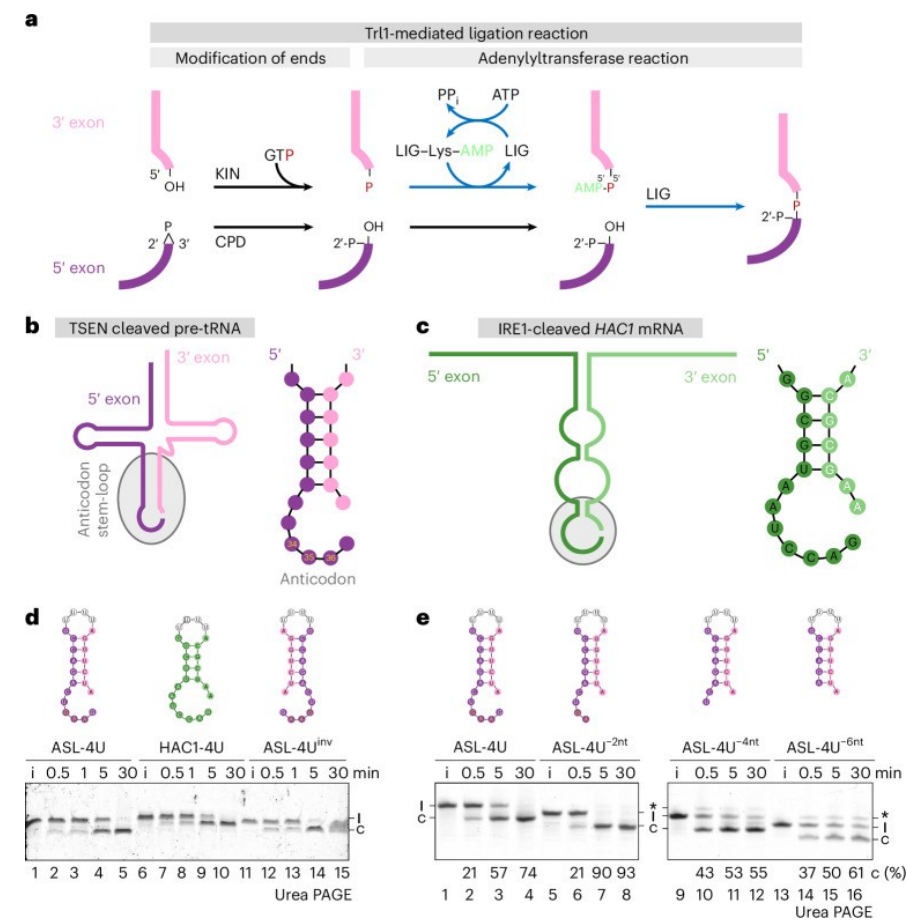
1.10 文献概览: Structure of fungal t RNA ligase Trl1 with RNA reveals conserved substrate...

Nature structural & molecular biology (2025) | experimental | DOI 10.1038/s41594-025-01589-3

研究问题 / 背景

RNA ligases play a vital role in RNA processing and maturation, including t RNA splicing, RNA repair and the unfolded protein response (UPR). In fungi and plants, the tripartite t RNA ligase Trl1 catalyzes the joining of TSEN-cleaved pre-t RNA exon halves. Trl1 also functions as ligase in the non-conventional HAC1 mRNA splicing during the UPR.

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.10 材料方法与结果: Structure of fungal t RNA ligase Trl1 with RNA reveals conserved subs...

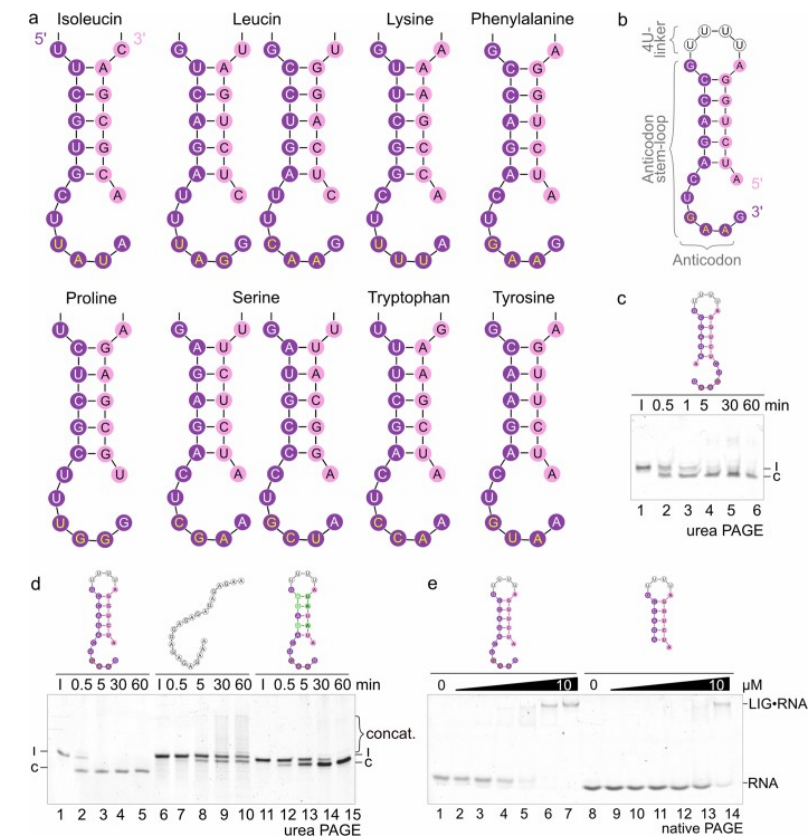
Nature structural & molecular biology (2025) | experimental | DOI 10.1038/s41594-025-01589-3

材料 / 方法 / 分析路线

After linking the backbone of nucleotides 31 and 32, we energy-minimized the conformation of the protein-unbound nucleotides 32–34 to yield a bo...
The ends of the starting model matched an activated (that is, adenylylated) physiological substrate (2'-P, 3'-OH and 5'-App RNA).
The resulting model showed that Trl1-LIG can accommodate a cleaved t RNA-ASL in its binding cleft as a hairpin structure (Fig).
All base-pairing interactions in the hairpin remained intact, except for the first A-U base pair toward the loop, which was also unpaired in the...

核心结果

in a 5'-5' RNA-adenylylate (App RNA).
Finally, the LIG domain catalyzes Received: 12 January 2024 Accepted: 12 May 2025 Published online: 25 June 2025 Check for updates
1Heidelberg U...
2Laboratory of Bioinformatics and Protein Engineering, International Institute of Molecular and Cell Biology in Warsaw, Warsaw, Poland.
e-mail: jirka.peschek@bzh.uni-heidelberg.de
Nature Structural & Molecular Biology | Volume 32 | September 2025 | 1657–1668 1658 Article
<https://...>
The specific role of the CTD within Trl1-type RNA ligases has remained elusive.



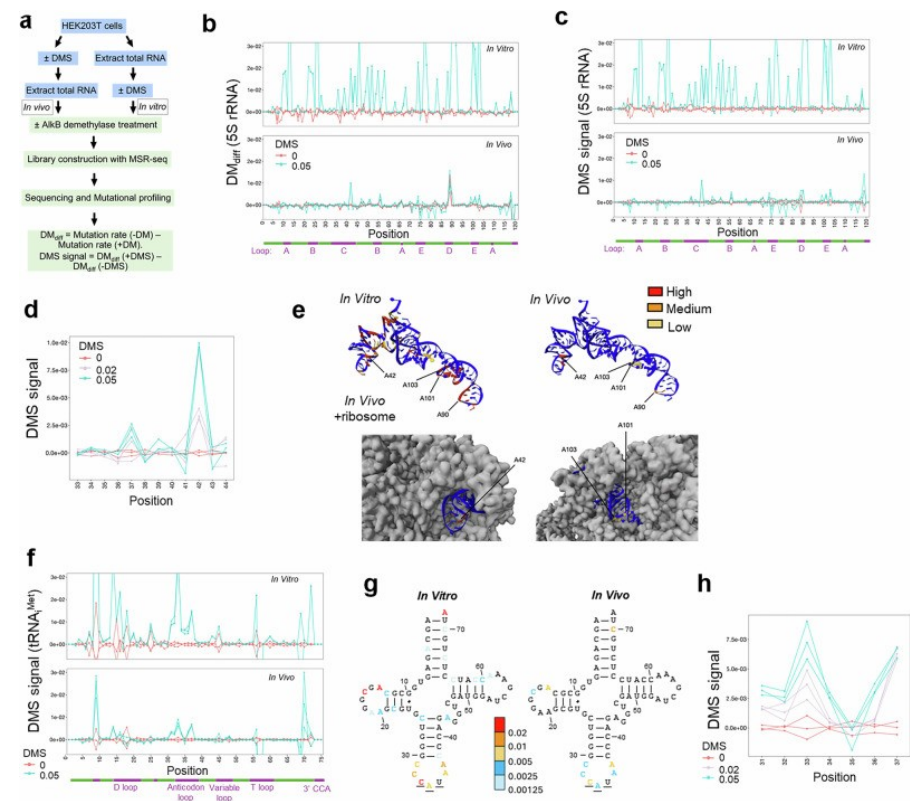
1.11 文献概览: In vivo structure profiling reveals human cytosolic and mitochondrial tRNA...

Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

研究问题 / 背景

Transfer RNA (t RNA) is the most abundant cellular RNA family in terms of copy numbers.
It not only folds into defined structures but also has complex cellular interaction networks involving aminoacyl-t RNA synthetases, translation...
The human t RNAome is comprised of chromosomal-encoded t RNAs with a large sequence diversity and mitochondrial-encoded t RNAs with A/U-rich seq...

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



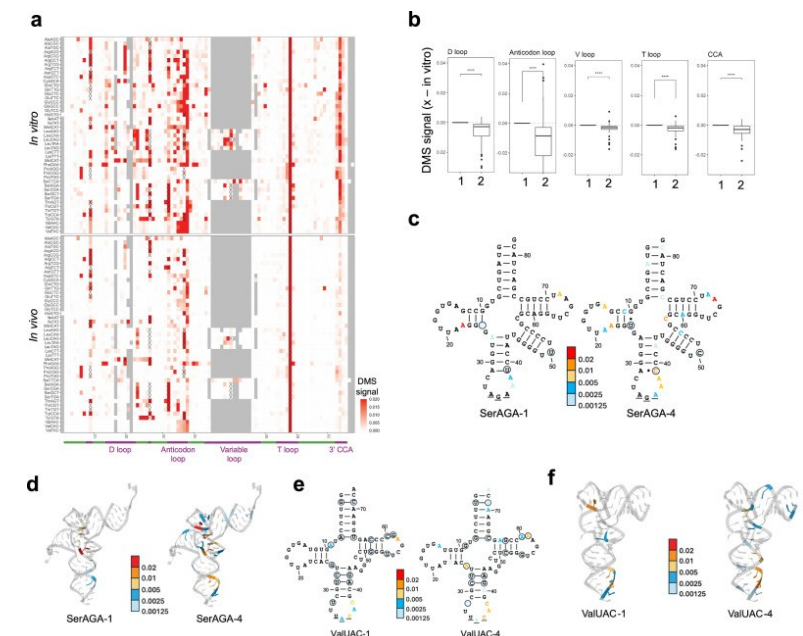
1.11 材料方法与结果: In vivo structure profiling reveals human cytosolic and mitochondrial...

材料 / 方法 / 分析路线

, the in vivo structures of t RNAs have been largely taken for granted to resemble their in vitro forms. However, t RNAs have many well-characterized protein interactions and modifications in vivo, which led us to hypothesize that their in vivo str... applied t RNA structure-seq to examine in vivo structures of E. coli t RNAs and revealed dynamics of t RNA folding under heat stress²⁰.

核心结果

reveal variations of t RNA structurome and dynamic interactome that have functional consequences in translational regulation. RNAs are both carriers of genetic information and adapters that mediate biomolecular interactions. In cells, RNAs fold into secondary and tertiary structures to detect metabolites^{1–3}, recruit proteins^{4,5}, and regulate phase separation⁶. Stable RNA structures also impact RNA–RNA interactions and affect RNA stabilities. In biological processes, including transcription, mRNA splicing, and translation, RNA structures play universal and profound regulatory roles.



1.11 讨论与局限: In vivo structure profiling reveals human cytosolic and mitochondrial

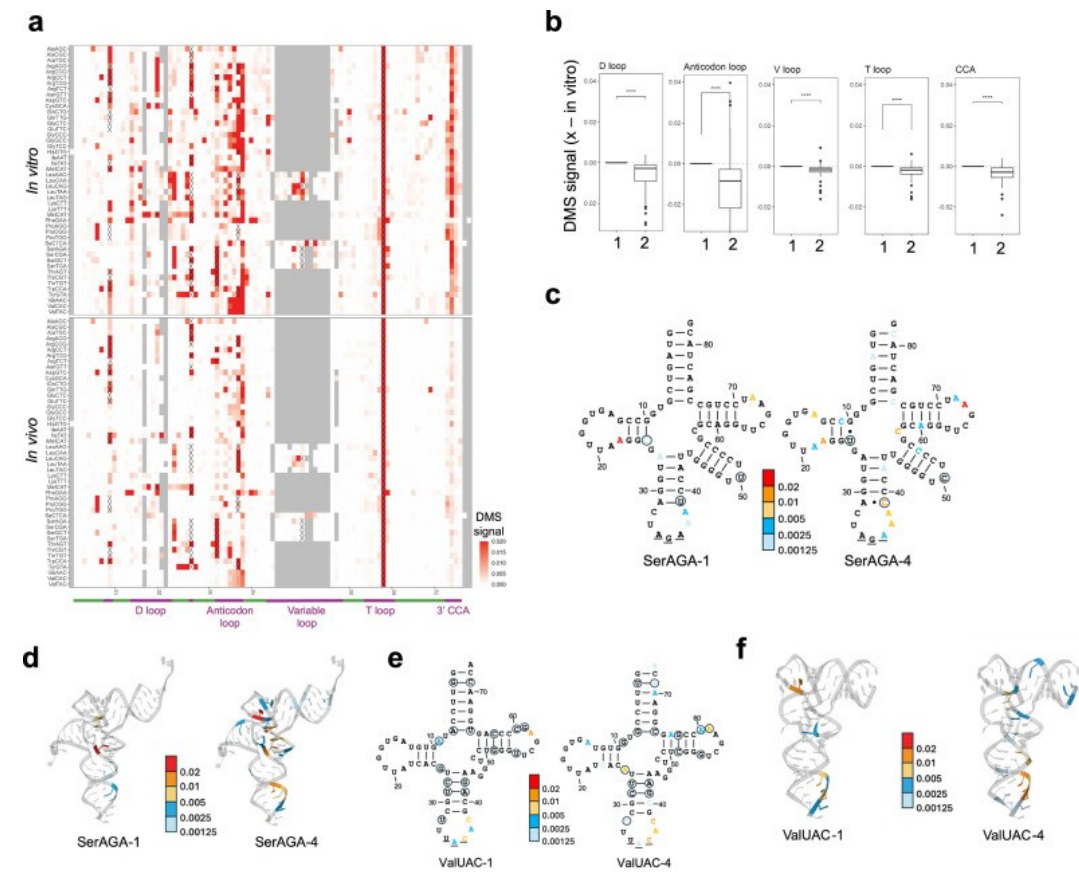
t11. Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

讨论与局限

The rapid development of high-throughput sequencing methods for highly structured RNAs made it possible to examine transcriptome- Fig. 6 | Mitochondrial t RNA structure and interaction in response to arsenite stress.

The parameters used here are: Δ DMS signal = DMS signal_{in vivo}_As O2 – DMS signal_{in vivo}_control.

a Heatmap showing the positional Δ DMS signals on mt-t RNAMet, red: higher, blue: lower DMS signal with arsenite.



可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

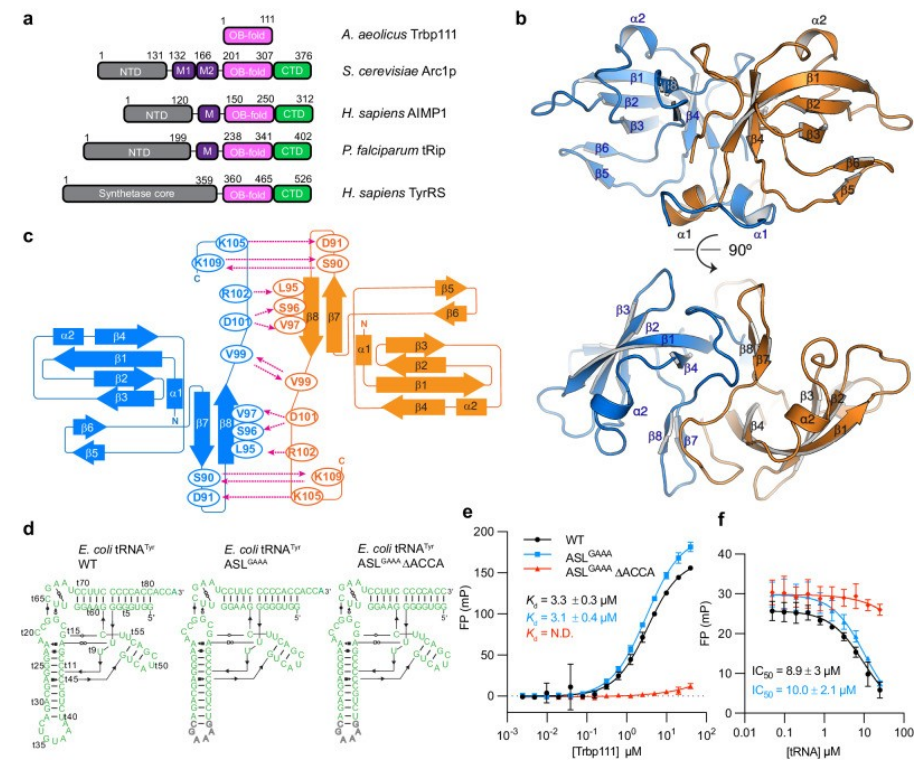
1.12 文献概览: Structural basis of t RNA recognition by the widespread OB fold.

1.12 | Nature communications (2024) | experimental | DOI 10.1038/s41467-024-50730-1

研究问题 / 背景

t RNAs are among the most abundant and essential biomolecules in the cell. Besides decoding the genetic code on the ribosome, t RNAs and their fragments also perform key regulatory functions in transcriptional and trans... They are further exploited by retroviruses such as HIV-1 to serve as obligatory viral replication primers and regulators of virion assembly 2 –...

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.12 材料方法与结果： Structural basis of t RNA recognition by the widespread OB fold.

1.12 | Nature communications (2024) | experimental | DOI 10.1038/s41467-024-50730-1

材料 / 方法 / 分析路线

from n biologically independent replicates) from ITC measurements.

n = 4 for WT, $\Delta 3'A$; n = 2 for ASLGAAA, 3'UGGU, ASLGAAA/5'GC, ASLGAAA/5'GGCC, G18C/G19C/ Δ ACCA; n = 3 for Δ ACCA, ASLGAAA/ Δ ACCA; n = 9 for G18C/G19...

Fold-change values relative to WT t RNA are indicated.

Article <https://doi.org/10.1038/s41467-024-50730-1> Nature Communications| (2024) 15:6385 8 as RNase P and RNase Z, we changed the stem length of...

核心结果

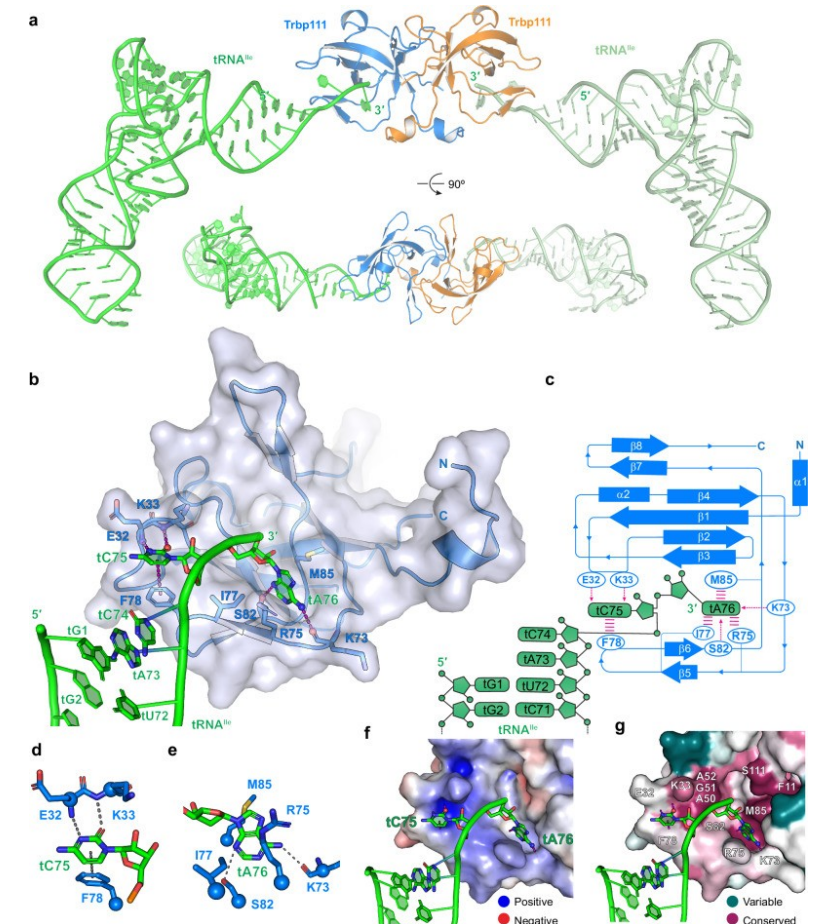
Trbp111 employs an extraordinarily robust homodimer interface The crystal structure of *Aquifex aeolicus* Trbp111 was previously reported at 2.5 Å...

Here we report another crystal structure of Trbp111 at 2.3 Å in space group C2221 which exhibits substantially improved backbone and sidechain g...

As described, Trbp111 forms a homodimer where each protomer adopts a canonical OB-fold architecture resembling the Greek key motif, in which five...

The core OB-fold comprises residues 16–89, with the flanking N-terminal (1–15) and C-terminal (90–111) residues of each protomer interacting in a...

The interface involves an extensive network of hydrogen bonds, salt bridges, hydrophobic interactions, aromatic stacking, and cation- π interacti...



1.12 讨论与局限: Structural basis of t RNA recognition by the widespread OB fold.

1.12 | Nature communications (2024) | experimental | DOI 10.1038/s41467-024-50730-1

讨论与局限

The main findings of this work are: (1) Contrasting previous models, Trbp111 recognizes exclusively the universal 3' end of mature t RNAs.

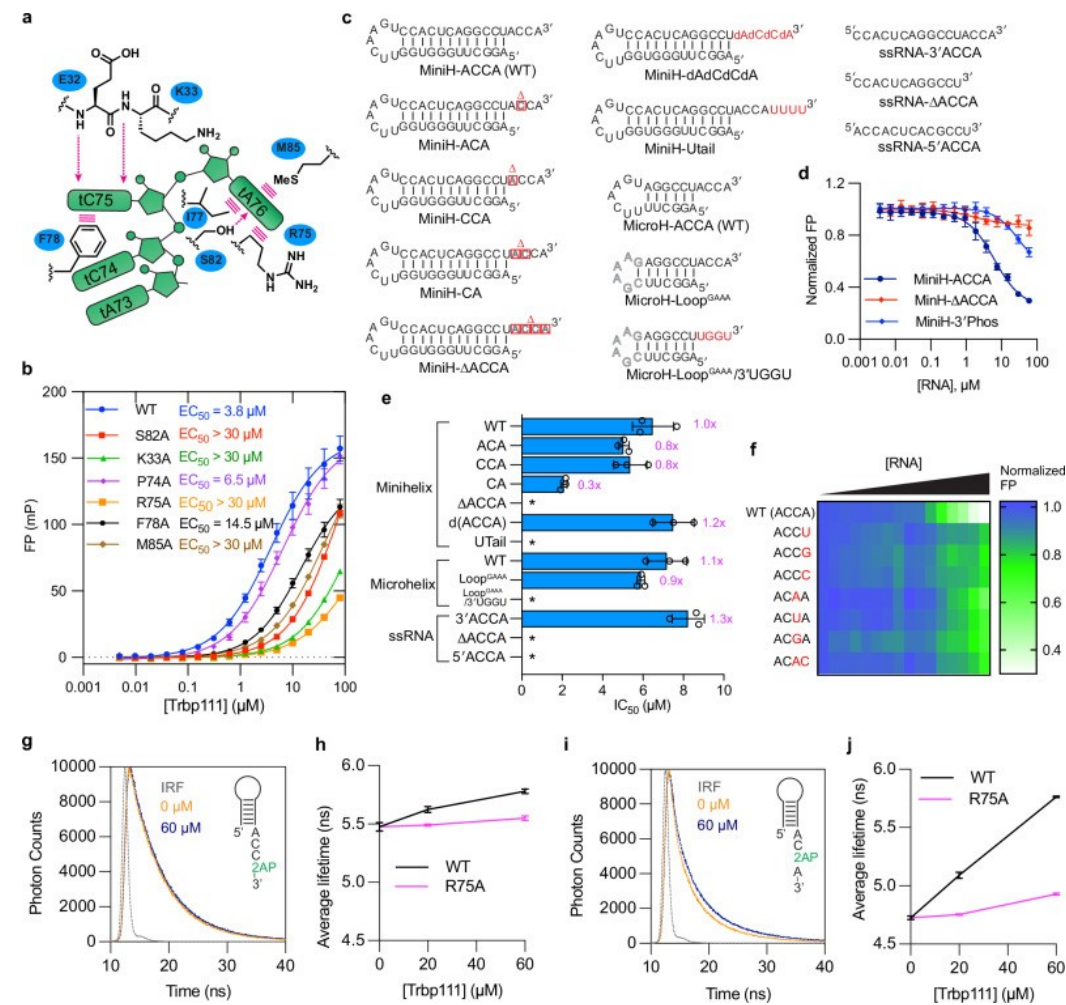
(2) Using its OB-fold, Trbp111 precisely recognizes the single-stranded structure, 3' terminal location, and specific sequence of the 3' CA dinu...

(3) Arc1p binds t RNA 3' end similarly as Trbp but also engages the t RNA body, likely via its basic middle domain 29 – 31 .

This secondary contact may have shifted or otherwise constrained its OB-t RNA 3' end interaction giving rise to the modest differences in the ef...

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.13 文献概览: Deciphering t RNA repertoires and translation coordination during mouse ea...

2026 | Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

研究问题 / 背景

Translational control is vital during the maternal-to-zygotic transition (MZT), yet the landscape of embryonic transfer RNA (t RNA) pools has be...

Here, we develop Optimized Reaction for Accurate Capture of Low-input Entities t RNA Sequencing (ORACLE-t RNAseq), enabling robust t RNA profili...

We map t RNA landscapes from oocyte to blastocyst, identifying a distinct transition to embryonic t RNA repertoires and upregulation of t RNA ps...

与本子项目的关系：该文可用于解释 **tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。**

1.13 材料方法与结果: Deciphering t RNA repertoires and translation coordination during mou...

13 | Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取; 本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

Translational control is vital during the maternal-to-zygotic transition (MZT), yet the landscape of embryonic transfer RNA (t RNA) pools has be...

Here, we develop Optimized Reaction for Accurate Capture of Low-input Entities t RNA Sequencing (ORACLE-t RNAseq), enabling robust t RNA profil...

We map t RNA landscapes from oocyte to blastocyst, identifying a distinct transition to embryonic t RNA repertoires and upregulation of t RNA ps...

Integrated multi-omics analyses reveal that zygotic t RNA gene activation coincides with zygotic genome activation (ZGA) and correlates with H3K...

By coupling ORACLE-t RNAseq with Ribo-seq, we demonstrate that embryonic t RNA anticodon pools coordinate with high translation-efficiency gene...

1.13 讨论与局限: Deciphering t RNA repertoires and translation coordination during mouse...

Source: Nature communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

讨论与局限

Translational control is vital during the maternal-to-zygotic transition (MZT), yet the landscape of embryonic transfer RNA (t RNA) pools has be...

Here, we develop Optimized Reaction for Accurate Capture of Low-input Entities t RNA Sequencing (ORACLE-t RNAseq), enabling robust t RNA profil...

We map t RNA landscapes from oocyte to blastocyst, identifying a distinct transition to embryonic t RNA repertoires and upregulation of t RNA ps...

Integrated multi-omics analyses reveal that zygotic t RNA gene activation coincides with zygotic genome activation (ZGA) and correlates with H3K...

可用于本项目的提炼

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。

1.14 材料方法与结果: Mechanistic basis for PYROXD1-mediated protection of the human t RNA...

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting sum- maries, source data, extended data, supplementary information, acknowledgements, peer r...

Characterizing expression and processing of precursor and mature human t RNAs by hydro-t RNAseq and PAR-CLIP.

t RNA introns: presence, processing, and purpose.

Precise excision of intervening sequences from precursor t RNAs by a membrane-associated yeast endonuclease.

Source: Nature structural & molecular biology 2025; PMID 40069351; PMCID PMC12263435; DOI 10.1038/s41594-025-01516-6

核心结果

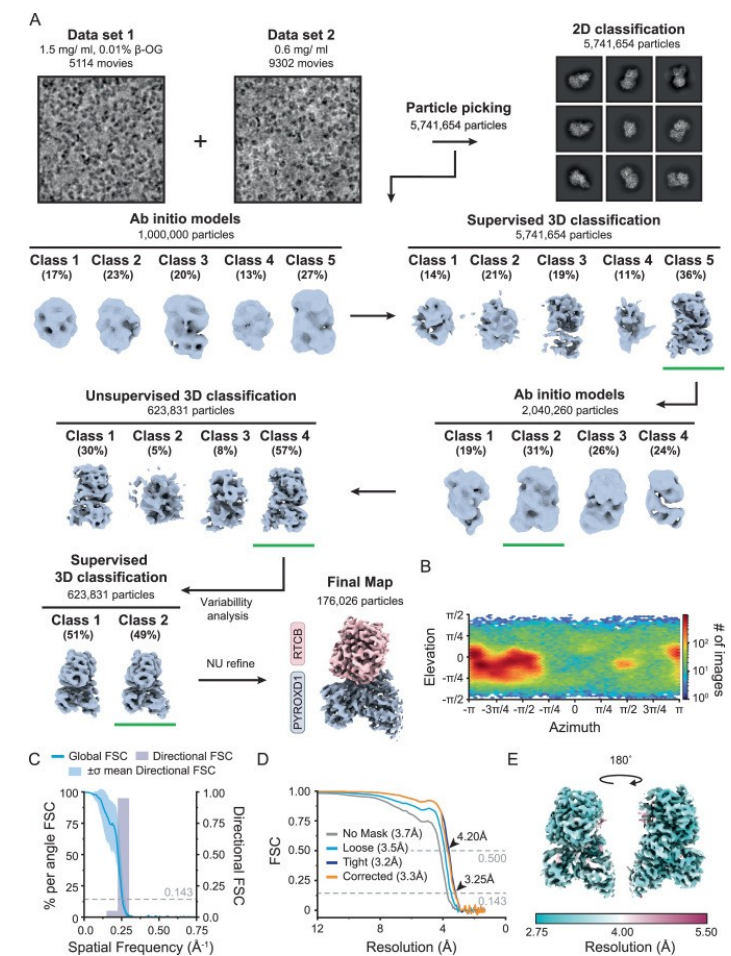
Molecular architecture of the RTCB–PYROXD1 complex To elucidate the mechanism underlying the protective function of the flavoprotein PYROXD1 in...

Co-precipitation of recombinant PYROXD1 by immobilized RTCB required the presence of 0.5 m M NADH (Extended Data Fig.

1a), in agreement with our prior work demonstrating that the interaction of RTCB and PYROXD1 is NAD(P)H-dependent and that NAD(P)+ does not supp...

Furthermore, PYROXD1 co-precipitation was observed in the presence of a variety of divalent metal cations (Ca²⁺, Co²⁺, Cu²⁺, Mg²⁺, Mn²⁺, Ni²⁺ an...

1b), confirming that the interaction is divalent cation-dependent¹⁹.



1.14 讨论与局限: Mechanistic basis for PYROXD1-mediated protection of the human tRNA li...

讨论与局限

The RNA ligase activity of the tRNA-LC is essential for cellular RNA metabolism.

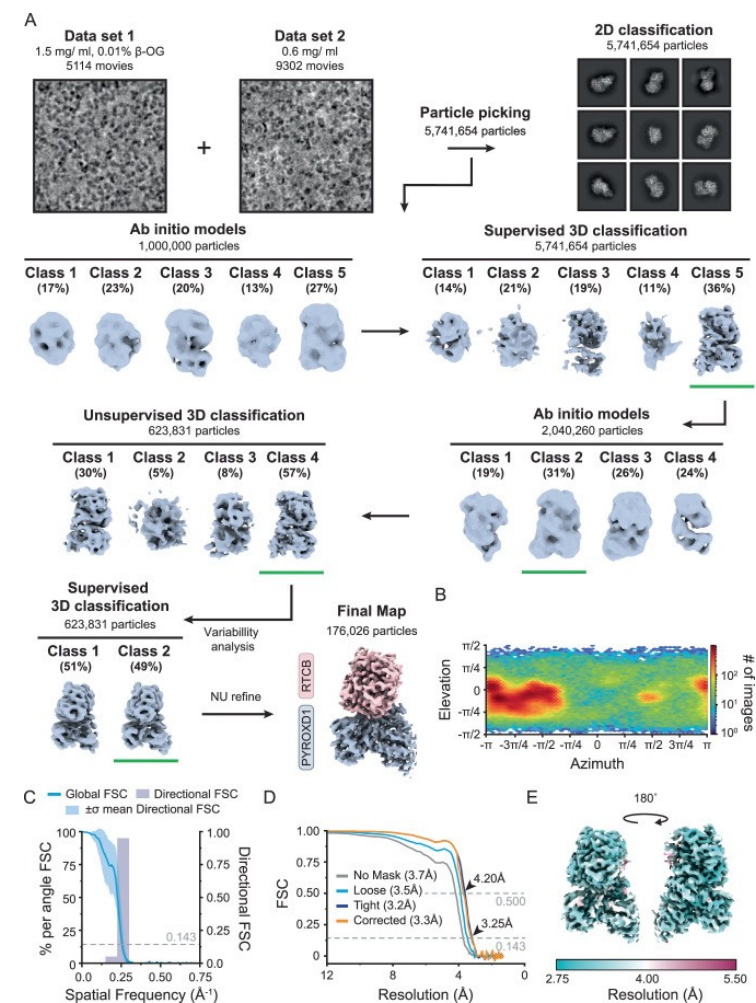
The enzymatic mechanism of RTCB, the catalytic subunit of tRNA-LC, has anaerobic origins^{16,17} and is dependent on a highly reactive cysteine...

In metazoa, RTCB has co-evolved with the oxidoreductase PYROXD1, which protects RTCB from oxidative inactivation, whereas in land plants and f...

In this work, we combined single-particle cryo-EM with biochemical experiments to reveal the mechanistic basis for the protective function of PY...

可用于本项目的提炼

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.15 文献概览: AAV-delivered engineered suppressor tRNA rescues visual function in mice...

15 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-66176-y

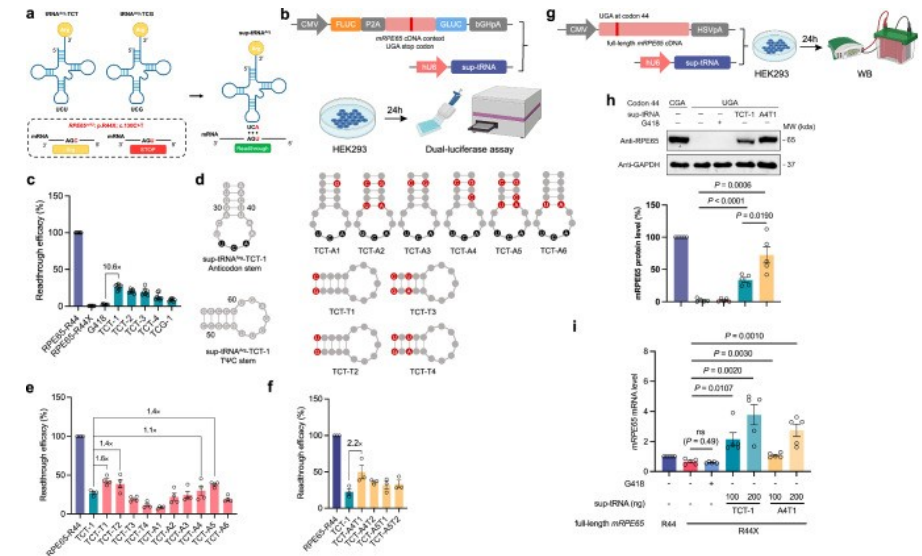
研究问题 / 背景

Inherited retinal diseases (IRDs) are a heterogenous group of retinal degenerative diseases with at least 300 causative genes identified to date...

In-frame nonsense mutations account for approximately 18.5% of human IRDs-related mutations 2 .

m RNAs bearing premature stop codon (PTC) are incapable of producing full-length proteins and frequently activate nonsense mediated decay (NMD)...

与本子项目的关系: 该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



1.15 材料方法与结果： AAV-delivered engineered suppressor t RNA rescues visual function in...

1.15 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-66176-y

材料 / 方法 / 分析路线

Description of Additional Supplementary Files

Additional Supplementary Files

Additional Supplementary Files

Additional Supplementary Files

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核心结果

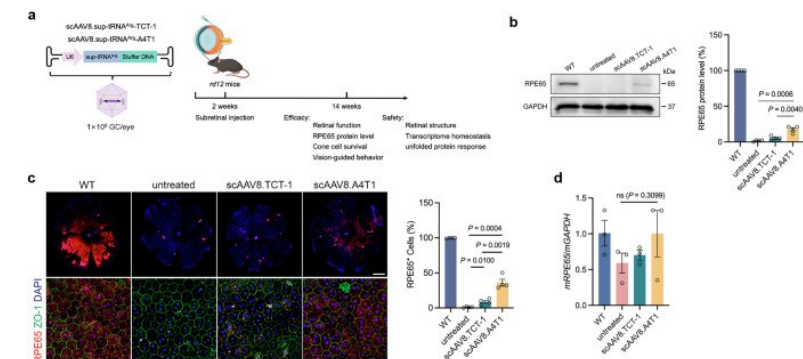
in cell culture models, challenges remain for in vivo delivery, particularly regarding efficacy, stability, and safety.

In this study, we engineer the body sequence of sup- t RNAArg to enhance readthrough efficiency at clinically relevant PTCs in the RPE65 and ABCA...

By using the self-complementary adeno-associated virus (sc AAV), we achieve restoration of RPE65 protein expression in up to 50.2% retinal pigme...

Notably, the engineered sup-t RNAArg significantly restores retinal function in mice, with effects lasting for at least 36 weeks.

Furthermore, sc AAV8.sup-t RNAArg exhibits minimal retinal toxicity and had negligible effects on global readthrough at native termination codon...



1.15 讨论与局限: AAV-delivered engineered suppressor t RNA rescues visual function in mi...

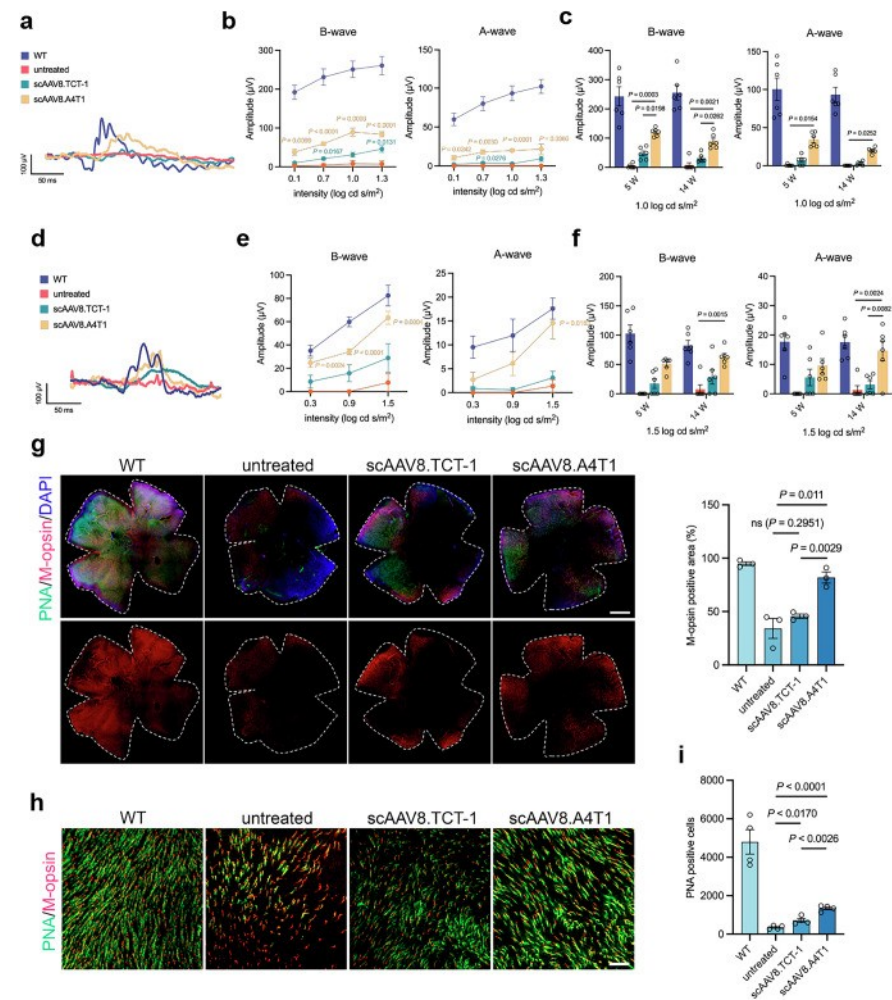
in mi... | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-66176-y

讨论与局限

In this proof-of-concept study, we demonstrated the therapeutic potential of sup-t RNA Arg in suppressing nonsense mutation that causes IRDs in... We screened eleven near-cognate sup-t RNA Arg and identified the most efficient one, sup-t RNA Arg -TCT-1. By applying a body sequence engineering strategy based on sup-t RNA Arg -TCT-1, we generated an optimized variant, sup-t RNA Arg -A4T1, which re... Using an sc AAV-based gene delivery platform, we successfully restored retinal function and vision-guided behavior in rd12 mice without signific...

可用于本项目的提炼

与本子项目的关系：该文可用于解释 tRNA 结构、修饰或互作如何转化为翻译 / 免疫 / 疾病功能。



Part 2. tRNA 分析工具与算法

tRNA analysis tools and algorithms

本部分包含 15 篇文献，每篇 3 页。

1. RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.
2. AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.
3. Design, construction, and functional characterization of a t RNA neochromosome in yeast.
4. Quantitative analysis of t RNA abundance and modifications by nanopore RNA sequencing.
5. Quantifying t RNA abundance by sequencing.
6. High-resolution quantitative profiling of t RNA abundance and modification status in eukaryote...
7. A ligation-independent sequencing method reveals t RNA-derived RNAs with blocked 3' termini.
8. A pro-metastatic t RNA fragment drives Nucleolin oligomerization and stabilization of its boun...
9. Deciphering t RNA repertoires and translation coordination during mouse early embryogenesis by...
10. In vivo structure profiling reveals human cytosolic and mitochondrial t RNA structurome and in...
11. Genome-wide profiling of t RNA modifications by Induro-t RNAseq reveals coordinated changes.
12. t RNA renovatio: Rebirth through fragmentation.
13. t RNA modification dynamics from individual organisms to metaepitranscriptomics of microbiomes.
14. Nanopore sequencing of intact aminoacylated t RNAs.
15. Quantitative t RNA-sequencing uncovers metazoan tissue-specific t RNA regulation.

2.1 文献概览： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.

Nature (2026) | Experimental | DOI 10.1038/s41586-025-09852-9

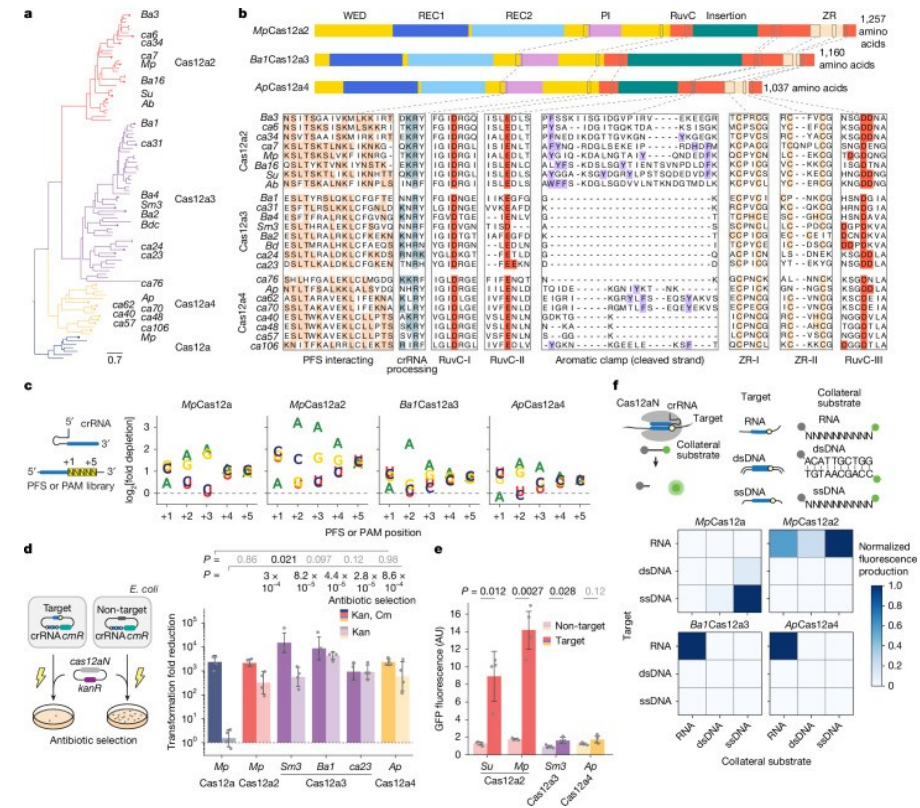
研究问题 / 背景

In all domains of life, t RNAs mediate the transfer of genetic information from m RNAs to proteins.

As their depletion suppresses translation and, consequently, viral replication, t RNAs represent long-standing and increasingly recognized target...

Here we report Cas12a3 effector nucleases from type V CRISPR-Cas adaptive immune systems in bacteria that preferentially cleave t RNAs after rec...

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



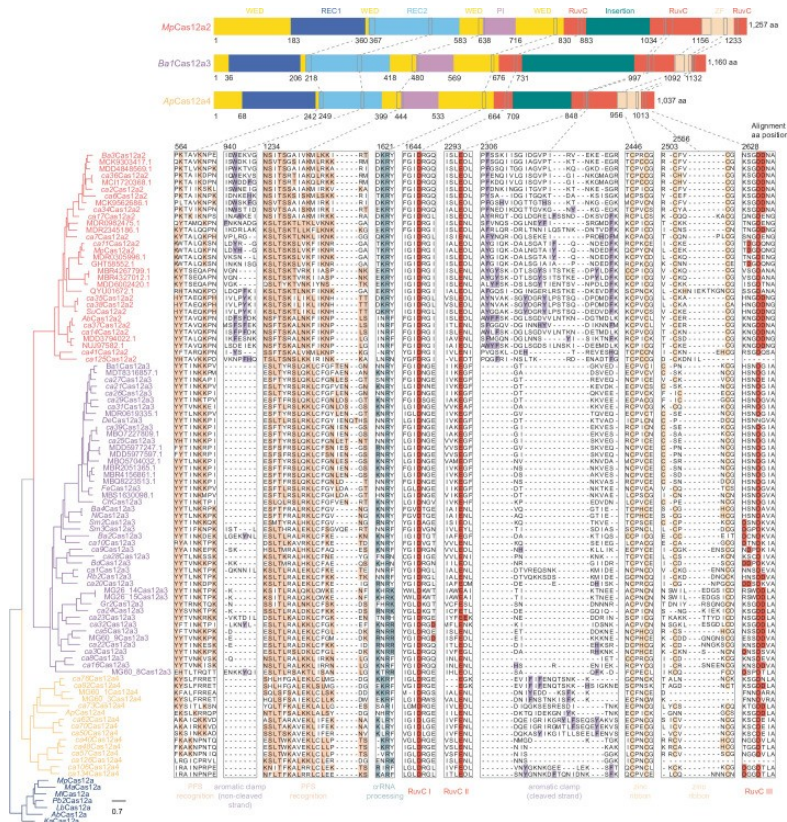
2.1 材料方法与结果： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immuni...

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting summa- ries, source data, extended data, supplementary information, acknowl- edgements, peer...
The making and breaking of t RNAs by ribonucleases.
Direct activation of a bacterial innate immune system by a viral capsid protein.
Phage T4-coded Stp: double-edged effector of coupled DNA and t RNA-restriction systems.

核心结果

are the averages of independent experiments (n = 2).
Note that the PAM for Mp Cas12a is the reverse complement of the sequence commonly reported for Cas12a nucleases (YYV).
d, Schematic (left) and quantification (right) of the assessment of plasmid clearance versus growth arrest in E.
coli based on variations of a plasmid interference assay.
Plasmid clearance and growth arrest were differentiated on the basis of antibiotic selection for the target plasmid or the target plasmid and th...



2.1 讨论与局限： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.

Nature (2026) Experimental | DOI 10.1038/s41586-025-09852-9

讨论与局限

In this work, we have reported the discovery of cr RNA-guided Cas12a3 nucleases that recognize complementary RNA targets and, in response, cleav...

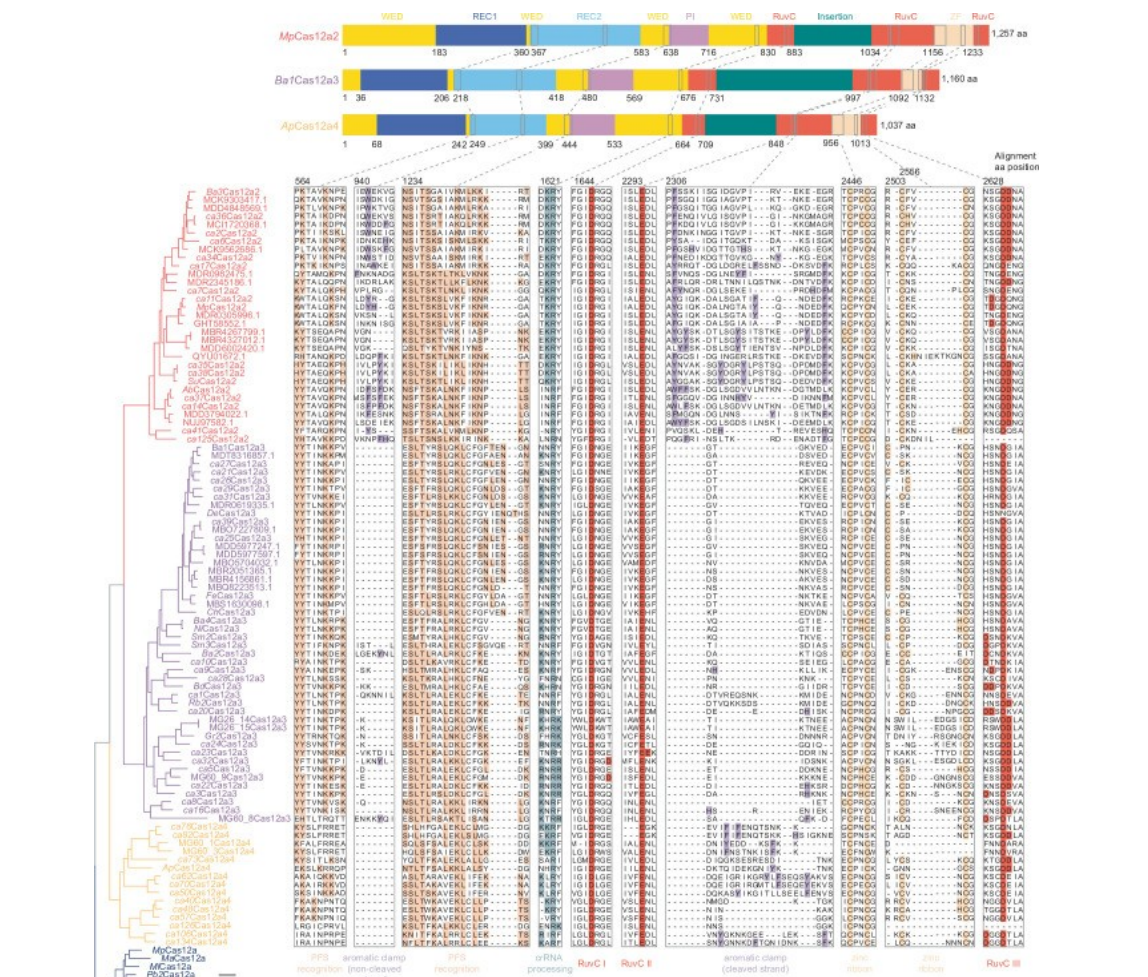
Integrating genetic, biochemical, sequencing and structural studies, we propose a model (Extended Data Fig.

9) in which the nuclease undergoes a large conformational change that can then bind free t RNAs through multiple sequence-specific and shape-spe...

The unique t RLD in the nuclease positions the t RNA tail next to the *RuvC* active site to cleave off the tail, with extensive cleavage of cellu...

可用于本项目的提炼

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.2 文献概览： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

doi:10.1038/s41586-022-04533-3 | experimental | DOI 10.1038/s41586-022-04533-3

研究问题 / 背景

Recombinant adeno-associated virus (r AAV)-based gene therapy 1, 2 typically uses transgenes to produce therapeutic effects, such as delivering...

Although it holds tremendous promise 3, transgene expression at non-physiological levels or in off-target cell and tissue types may cause toxic...

Therefore, harnessing the endogenous gene expression regulation is an attractive approach to gene therapy development.

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.2 材料方法与结果： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

2.2 | Nature | 2022 | experimental | DOI 10.1038/s41586-022-04533-3

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

Gene therapy is a potentially curative medicine for many currently untreatable diseases, and recombinant adeno-associated virus (r AAV) is the m...

However, r AAV-based gene therapy suffers from several limitations, such as constrained DNA cargo size and toxicities caused by non-physiologica...

Here we show that r AAV delivery of a suppressor t RNA (r AAV.sup-t RNA) safely and efficiently rescued a genetic disease in a mouse model carry...

Mechanistically, this was achieved through a synergistic effect of premature stop codon readthrough and inhibition of nonsense-mediated m RNA de...

r AAV.sup-t RNA had a limited effect on global readthrough at normal stop codons and did not perturb endogenous t RNA homeostasis, as determined...

2.2 讨论与局限： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

doi:10.1038/s41586-022-04533-3

讨论与局限

Here, we combined r AAV-based gene delivery and sup-t RNA-mediated PTC readthrough to develop AAV-No STOP, which overcame a pathogenic nonsense...

Importantly, AAV-No STOP demonstrated a good safety profile and stable efficacy more than six months after treatment.

It obviates the need to deliver a full-length c DNA that may exceed the packaging limit of r AAV such as CFTR for cystic fibrosis, or elicit imm...

In contrast to gene replacement therapy, AAV-No STOP operates on endogenous transcripts, and therefore restores gene function under physiologica...

可用于本项目的提炼

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.3 文献概览: Design, construction, and functional characterization of a t RNA neochromosome...

Cell 2023; DOI 10.1016/j.cell.2023.10.015

研究问题 / 背景

Here, we report the design, construction, and characterization of a t RNA neochromosome, a designer chromosome that functions as an additional,...
Intending to address one of the central design principles of the Sc2.0 project, the ~190-kb t RNA neochromosome houses all 275 relocated nuclear...
To maximize stability, the design incorporates orthogonal genetic elements from non-S.

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.3 材料方法与结果： Design, construction, and functional characterization of a t RNA neoc...

Cell (2023) | experimental | DOI 10.1016/j.cell.2023.10.015

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

Here, we report the design, construction, and characterization of a t RNA neochromosome, a designer chromosome that functions as an additional,...

Intending to address one of the central design principles of the Sc2.0 project, the ~190-kb t RNA neochromosome houses all 275 relocated nuclear...

To maximize stability, the design incorporates orthogonal genetic elements from non-S.

Furthermore, the presence of 283 rox recombination sites enables an orthogonal t RNA SCRa Mb LE system.

Following construction in yeast, we obtained evidence of a potent selective force, manifesting as a spontaneous doubling in cell ploidy.

2.3 讨论与局限: Design, construction, and functional characterization of a t RNA neochr...

Cell 2023 | experimental | DOI 10.1016/j.cell.2023.10.015

讨论与局限

Here, we report the design, construction, and characterization of a t RNA neochromosome, a designer chromosome that functions as an additional,...

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可用于本项目的提炼

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.4 文献概览: Quantitative analysis of t RNA abundance and modifications by nanopore RNA...

nanopore RNA... | DOI 10.1038/s41587-023-01743-6

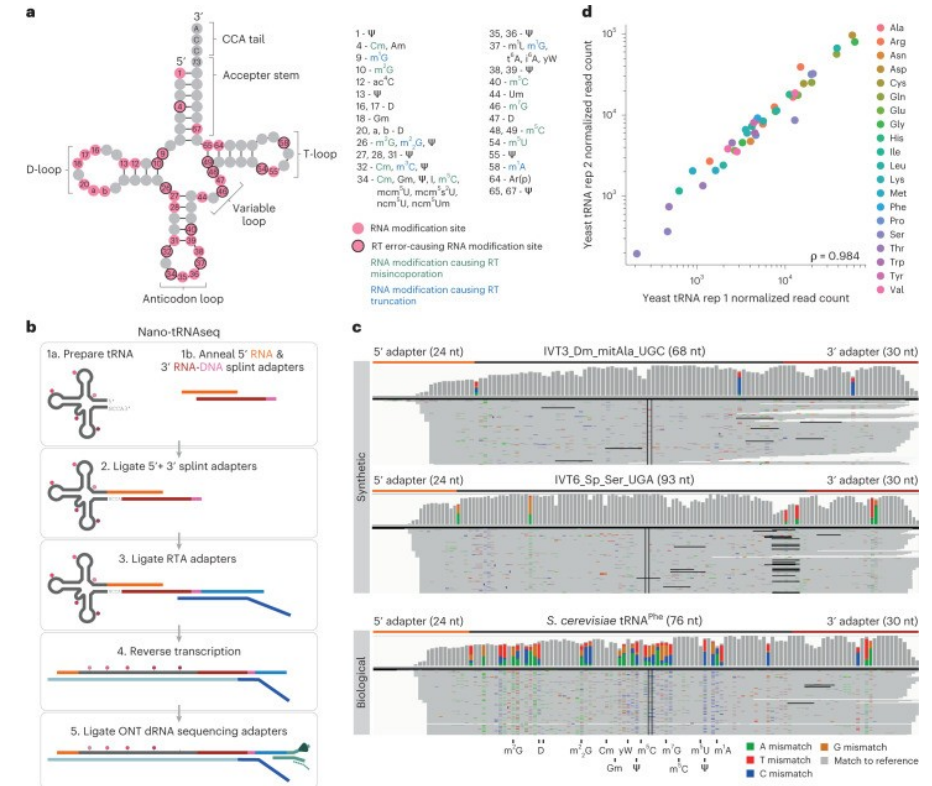
研究问题 / 背景

of both m5U54 and m1A58, and m5U54 positively influencing the introduction of m1A58.

To explore whether our method could capture the effect of Ψ 55 loss on other modifications, we examined the summed basecalling error (base mismat...

4 | Nano-t RNAseq can quantify t RNA abundance and RNA modifications as well as capture modification interdependencies.

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.4 材料方法与结果： Quantitative analysis of t RNA abundance and modifications by nanopor...

2024 | Nature Biotechnology (2024) | experimental | DOI 10.1038/s41587-023-01743-6

材料 / 方法 / 分析路线

Our results show that Nano-t RNAseq, when used with optimized mapping settings and custom Min KNOW configuration, resulted in observed t RNA abu...

We, therefore, wondered whether t RNA abundances predicted using Nano-t RNAseq would correlate well with Illumina-based approaches.

To this end, we compared Nano-t RNAseq S. cerevisiae t RNA abundances to those reported using three different Illumina-based methods: (1) ARM-seq 74 , (2) Hydro-t RNAseq 80 and (3) mim-t...

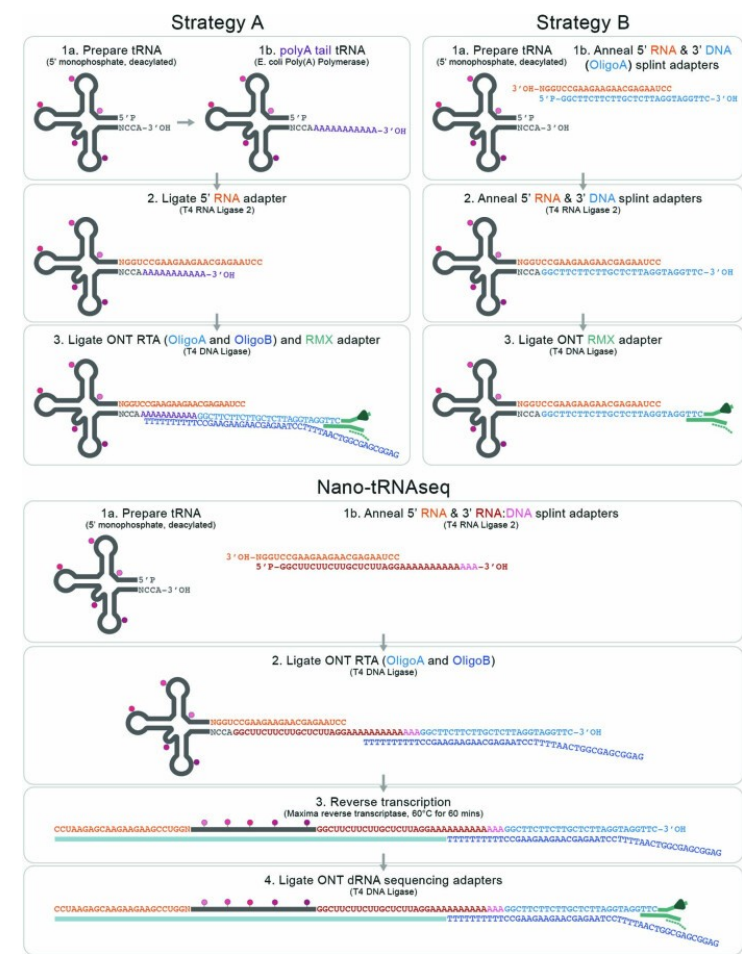
核心结果

Nanopore DRS is a well-established long-read sequencing technology to study RNA molecules, typically polyadenylated m RNAs 93 – 97 . Although several works have shown that this technology can also be used to study short RNA molecules, such as sno RNAs and sn RNAs 96 , 98 , DRS...

In addition, the first ~15 nt at the 5' end of RNA molecules are typically lost in DRS runs 93 , 99 , as this portion cannot be adequately basec...

To overcome these limitations, we reasoned that extension of the 5' and 3' ends of the t RNA would lead to improved sequencing of t RNA molecule...

We first attempted a modified t RNA DRS approach in which a 5' RNA adapter, complementary to the 3' CCA overhang present in mature t RNA molecule...



2.4 讨论与局限： Quantitative analysis of t RNA abundance and modifications by nanopore...

讨论与局限

For many years, t RNAs and their modifications have been primarily viewed as static contributors to gene expression and t RNA structure 127 – 13...

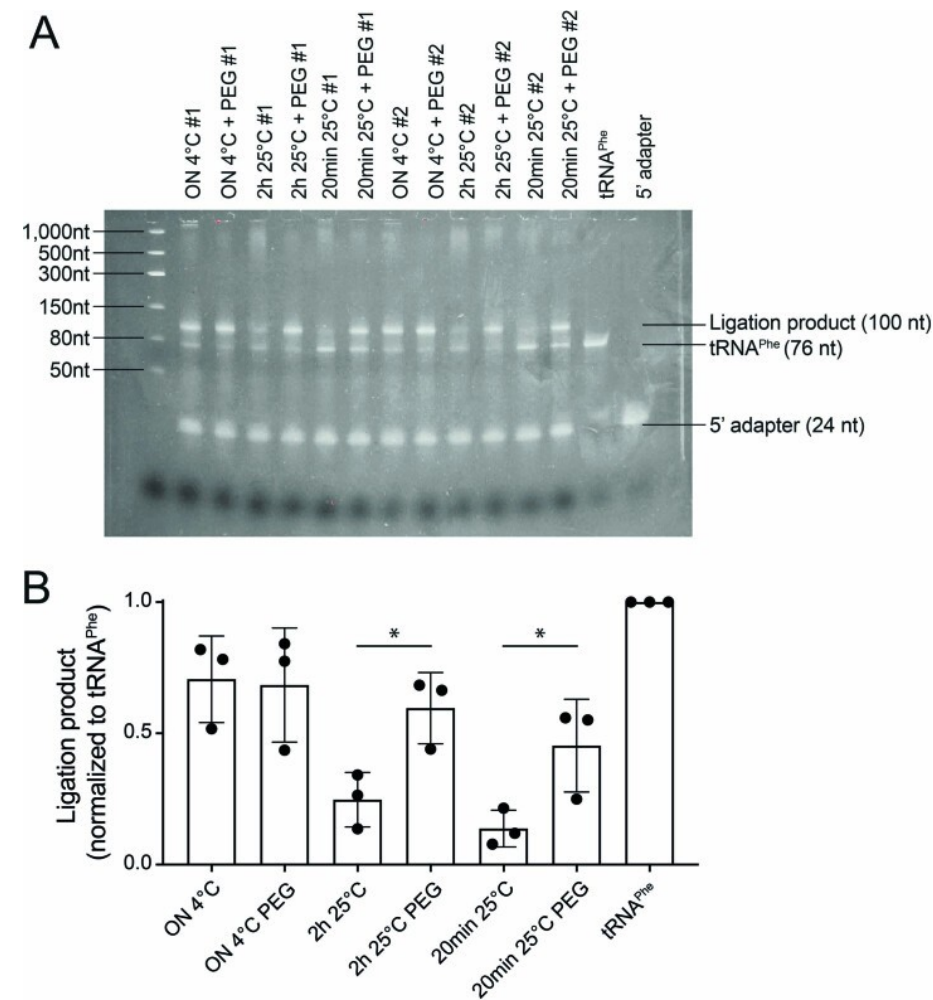
However, multiple studies have shown that t RNA abundances and modification profiles are, in fact, dynamic and can differ in distinct cellular e...

Measuring both t RNA abundances and their modifications with single-transcript resolution has not been feasible due to a lack of available metho...

This has been a substantial limitation in moving forward with studying the biological function and dynamics of t RNA populations and their modif...

可用于本项目的提炼

与本子项目的关系： 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.5 文献概览： Quantifying t RNA abundance by sequencing.

2.5 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

研究问题 / 背景

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.5 材料方法与结果： Quantifying t RNA abundance by sequencing.

2.5 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

材料 / 方法 / 分析路线

核心结果

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

2.5 讨论与局限： Quantifying t RNA abundance by sequencing.

2.5 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

讨论与局限

可用于本项目的提炼

与本子项目的关系： 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.6 文献概览: High-resolution quantitative profiling of t RNA abundance and modification...

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研究问题 / 背景

Measurements of cellular t RNA abundance are hampered by pervasive blocks to c DNA synthesis at modified nucleosides and the extensive similarit...

We overcome these limitations with modification-induced misincorporation t RNA sequencing (mim-t RNAseq), which combines a workflow for full-len...

Our method accurately captures t RNA abundance and modification status in yeast, fly, and human cells and is applicable to any organism with a k...

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2.6 材料方法与结果： High-resolution quantitative profiling of t RNA abundance and modific...

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材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

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2.6 讨论与局限: High-resolution quantitative profiling of t RNA abundance and modification...

2021; Molecular cell 2021; experimental | DOI 10.1016/j.molcel.2021.01.028

讨论与局限

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2.7 文献概览： A ligation-independent sequencing method reveals t RNA-derived RNAs with b...

Molecular cell (2024) | experimental | DOI 10.1016/j.molcel.2024.07.008

研究问题 / 背景

Prokaryotic and eukaryotic cells synthesize a vast range of RNA classes that greatly vary in size and function.

A criterion for RNA classification is their coding capacity, which distinguishes protein-coding RNAs (messenger RNAs; m RNAs) from non-coding RN...

The latter can be subdivided into structural nc RNAs, such as ribosomal RNAs (r RNAs) and transfer RNAs (t RNAs); small nuclear (sn RNAs) and nu...

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.7 材料方法与结果： A ligation-independent sequencing method reveals t RNA-derived RNAs w...

bioRxiv preprint doi: <https://doi.org/10.1101/2024.07.008>; this version posted July 1, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

材料 / 方法 / 分析路线

The synthetic 20 and 50 nts fragments were synthesized by Integrated DNA technologies (IDT) (Table S1).

For 3' biotinylated RNAs, we performed a purification step using Dynabeads My One Streptavidin C1 beads (Invitrogen, Cat. No 65001) followed by Tri Pure (Roche, Cat. No 11667165001) purification to ensure all RNAs used in the reaction contained biotin.

核心结果

Despite the numerous sequencing methods available, the diversity in RNA size and chemical modification makes it difficult to capture all RNAs in...

We developed a method that combines quasi-random priming with template switching to construct sequencing libraries from RNA molecules of any len...

Our ligation-independent detection of all types of RNA (LIDAR) is a simple, effective tool to identify and quantify all classes of coding and no...

With LIDAR, we comprehensively characterized the transcriptomes of mouse embryonic stem cells, neural progenitor cells, mouse tissues, and sperm.

LIDAR detected a much larger variety of t RNA-derived RNAs (t DRs) compared with traditional ligation-dependent sequencing methods and uncovered...

2.7 讨论与局限： A ligation-independent sequencing method reveals t RNA-derived RNAs wit...

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讨论与局限

Here, we presented LIDAR, an RNA-seq technique that utilizes a custom template-switch strategy combined with quasi- random hexamer priming to se...

LIDAR is a powerful, simple, and fast RNA-seq method that allows comprehensive characterization of coding and noncoding transcriptomes from limi...

LIDAR captured a large variety of RNAs in mouse ESCs, NPCs, brain, testis, and sperm, including full-length t RNAs and 3' t DRs with blocked 3' ...

While the functional implications of the presence of these 3' -blocked t DRs in tissues and sperm are currently unknown, we noticed that they are...

可用于本项目的提炼

与本子项目的关系： 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.8 文献概览： A pro-metastatic t RNA fragment drives Nucleolin oligomerization and stability...

Stable cell (2022) | experimental | DOI 10.1016/j.molcel.2022.05.008

研究问题 / 背景

Transfer RNA-derived RNA fragments (t RFs) are a novel class of small non-coding RNAs cleaved from mature or precursor transfer RNAs (t RNAs).

TRFs are classified into distinct groups based on their sites of origin within t RNAs.

5' - and 3' -t RNA halves arise from cleavage within the anticodon loop by Angiogenin, RNase L, or RNase 1 (Donovan et al., 2017 ; Fu et al., 200...

与本子项目的关系：该文可用于提取 **tRNA** 定量、注释、测序信号处理或预测算法的设计原则。

2.8 材料方法与结果： A pro-metastatic t RNA fragment drives Nucleolin oligomerization and...

28 | Molecular cell (2022) | experimental | DOI 10.1016/j.molcel.2022.05.008

材料 / 方法 / 分析路线

FLAG-tagged mouse Nucleolin c DNA was cloned into pc DNA3.

Mouse Mthfd1l or Pafah1b1 c DNAs were cloned into the p LJC6-EF1core-Blast vector.

The 5' UTR of mouse 5' -t RF Cys targets Pafah1b1 and Mthfd1l or non-target GAPDH were cloned into the psi CHECK2 vector (Promega).

Synthetic 5' -t RF-Cys antisense sequence or a scrambled control sequence embedded in the tough decoy backbone (Haraguchi et al.

核心结果

Stress-induced cleavage of transfer RNAs (t RNAs) into t RNA-derived fragments (t RFs) occurs across organisms from yeast to humans; yet, its me...

Small RNA profiling revealed increased abundance of a cysteine t RNA fragment (5'-t RFCys) during breast cancer metastatic progression.

5'-t RFCys was required for efficient breast cancer metastatic lung colonization and cancer cell survival.

We identified Nucleolin as the direct binding partner of 5'-t RFCys.

5'-t RFCys promoted the oligomerization of Nucleolin and its bound metabolic transcripts Mthfd1l and Pafah1b1 into a higher-order transcript sta...

2.8 讨论与局限： A pro-metastatic t RNA fragment drives Nucleolin oligomerization and st...

811 Molecular cell (2022) | experimental | DOI 10.1016/j.molcel.2022.05.008

讨论与局限

In vivo and in vitro experiments revealed that 5' -t RF Cys promoted metastatic lung colonization of breast cancer cells by enhancing cancer cell...

We found that two metabolic enzymes—Pafah1b1 and Mthfd1l—act downstream of 5' -t RF Cys to enhance cancer cell survival and metastasis.

Pafah1b1 and two catalytic enzymes, Pafah1b2 and Pafah1b3, form the type I Platelet-activating factor acetylhydrolase complex (Arai et al.

Pafah1b1 is proposed to control the platelet-activating factor (PAF) levels by regulating the enzymatic activity of Pafah1b2 and Pafah1b3 (Arai...

可用于本项目的提炼

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.9 文献概览: Deciphering t RNA repertoires and translation coordination during mouse ea...

2026 | Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

研究问题 / 背景

Translational control is vital during the maternal-to-zygotic transition (MZT), yet the landscape of embryonic transfer RNA (t RNA) pools has be...

Here, we develop Optimized Reaction for Accurate Capture of Low-input Entities t RNA Sequencing (ORACLE-t RNAseq), enabling robust t RNA profil...

We map t RNA landscapes from oocyte to blastocyst, identifying a distinct transition to embryonic t RNA repertoires and upregulation of t RNA ps...

与本子项目的关系：该文可用于提取 **tRNA 定量、注释、测序信号处理或预测算法的设计原则。**

2.9 材料方法与结果： Deciphering t RNA repertoires and translation coordination during mou...

Source: Nature communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

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We map t RNA landscapes from oocyte to blastocyst, identifying a distinct transition to embryonic t RNA repertoires and upregulation of t RNA ps...

Integrated multi-omics analyses reveal that zygotic t RNA gene activation coincides with zygotic genome activation (ZGA) and correlates with H3K...

By coupling ORACLE-t RNAseq with Ribo-seq, we demonstrate that embryonic t RNA anticodon pools coordinate with high translation-efficiency gene...

2.9 讨论与局限: Deciphering t RNA repertoires and translation coordination during mouse...

2026 | Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-72603-5

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Translational control is vital during the maternal-to-zygotic transition (MZT), yet the landscape of embryonic transfer RNA (t RNA) pools has be...

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与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

2.10 文献概览: In vivo structure profiling reveals human cytosolic and mitochondrial tRNA...

DOI: 10.1038/s41467-025-59435-5 | experimental | DOI 10.1038/s41467-025-59435-5

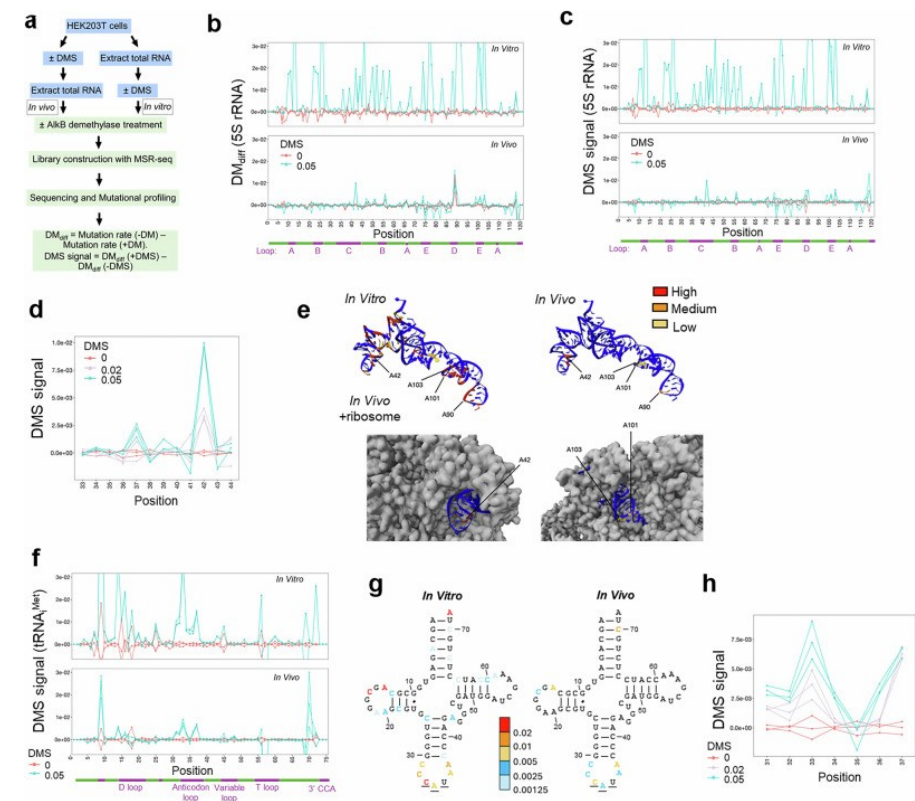
研究问题 / 背景

RNAs are both carriers of genetic information and adapters that mediate biomolecular interactions.

In cells, RNAs fold into secondary and tertiary structures to detect metabolites 1 – 3 , recruit proteins 4 , 5 , and regulate phase separation...

Stable RNA structures also impact RNA–RNA interactions and affect RNA stabilities.

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.10 材料方法与结果: In vivo structure profiling reveals human cytosolic and mitochondrial...

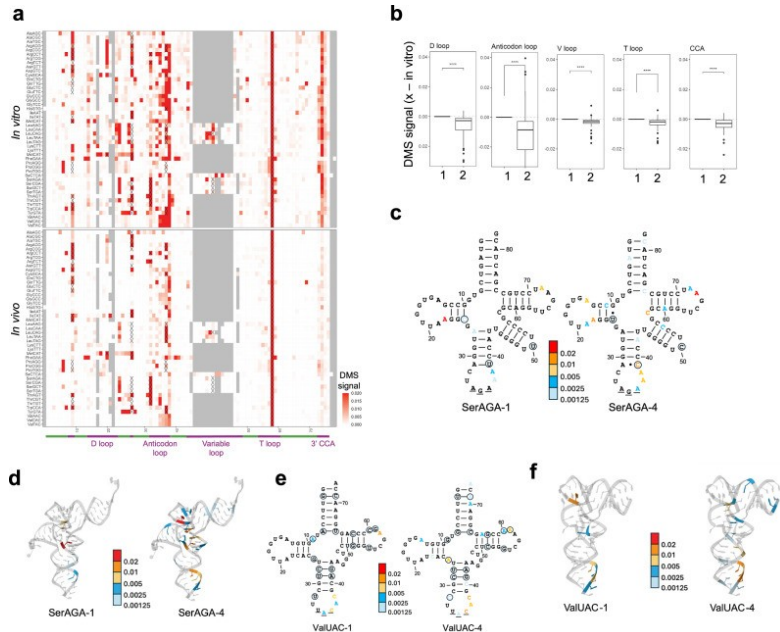
2025 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

材料 / 方法 / 分析路线

, the in vivo structures of t RNAs have been largely taken for granted to resemble their in vitro forms. However, t RNAs have many well-characterized protein interactions and modifications in vivo, which led us to hypothesize that their in vivo str... applied t RNA structure-seq to examine in vivo structures of E. coli t RNAs and revealed dynamics of t RNA folding under heat stress20.

核心结果

reveal variations of t RNA structurome and dynamic interactome that have functional consequences in translational regulation. RNAs are both carriers of genetic information and adapters that mediate biomolecular interactions. In cells, RNAs fold into secondary and tertiary structures to detect metabolites1–3, recruit proteins4,5, and regulate phase separation6. Stable RNA structures also impact RNA–RNA interactions and affect RNA stabilities. In biological processes, including transcription, mRNA splicing, and translation, RNA structures play universal and profound regulatory roles.



2.10 讨论与局限: In vivo structure profiling reveals human cytosolic and mitochondrial

t10. Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

讨论与局限

The rapid development of high-throughput sequencing methods for highly structured RNAs made it possible to examine transcriptome-wide tRNA structure...

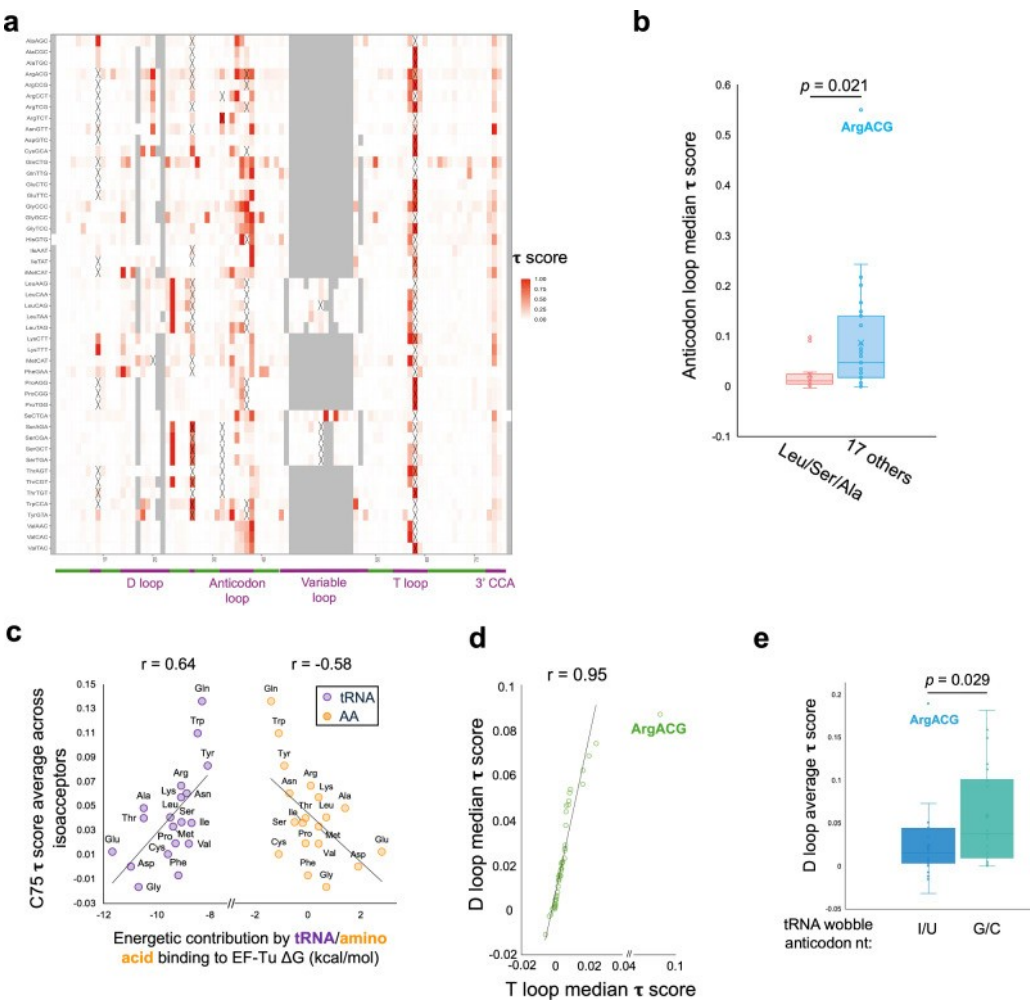
With DM-DMS-Ma Pseq, we present comprehensive in vivo structure and interaction studies for human chromosomal and mitochondrial encoded tRNAs.

Compared to DMS mapping of mRNAs, in vivo DMS signals for tRNA are much lower; therefore, additional considerations need to be implemented to...

This was accomplished here by treating the DMS-reacted RNA samples with a m1A/m3C demethylase before RT and considering only the mutation di...

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2.11 文献概览: Genome-wide profiling of t RNA modifications by Induro-t RNAseq reveals co...

Nature communications (2025) | experimental | DOI 10.1038/s41467-025-56348-1

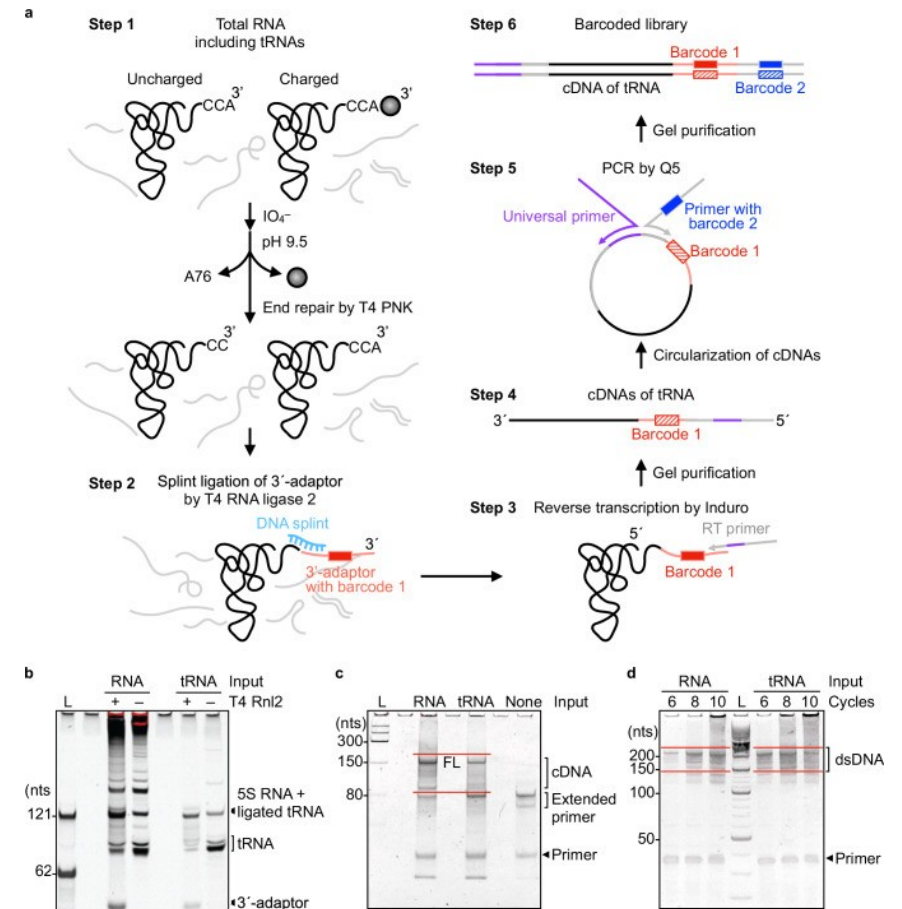
研究问题 / 背景

All native t RNA molecules contain post-transcriptional modifications at 10-20% of the nucleobases across the three domains of life ¹.

Many of these modifications occur in the anticodon stem-loop (ASL) with a role to modulate the quality of anticodon-codon base-pairings, while o...

Whether these two groups of modifications coordinate with each other in the development of a t RNA is unknown.

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.11 材料方法与结果: Genome-wide profiling of t RNA modifications by Induro-t RNAseq reveals...

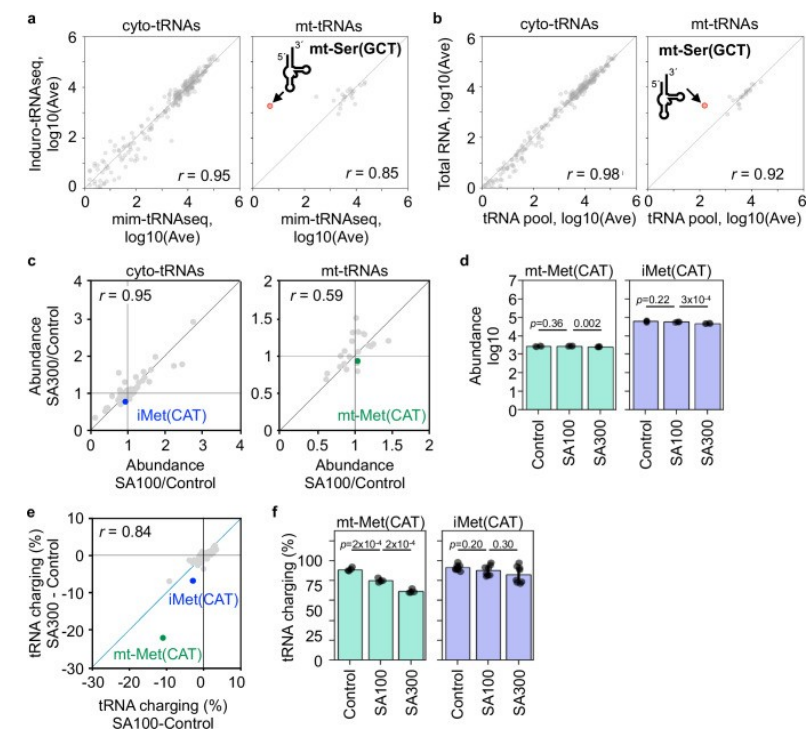
材料 / 方法 / 分析路线

Unless otherwise noted, we followed the manufacturer's instructions of each kit. Additionally, we extracted the TGIRT data from mim-t RNAseq 18 for comparison with the Induro data.

Source: Nature communications 2025; PMID 39865096; PMCID PMC11770116; DOI 10.1038/s41467-025-56348-1

核心结果

A multiplexing Induro-t RNAseq workflow In our development of Induro-t RNAseq, to reduce the number of experimental steps, we used total RNA from... To determine the t RNA charging level of each input, we performed a periodate oxidation of the terminal 2' ,3' -hydroxyls of uncharged t RNAs and... In contrast, charged aminoacyl-t RNA (aa)-t RNAs were resistant to the oxidation and retained the normal CCA-end after deacylation in alkaline (... The CC- and CCA-ending t RNAs, after repair of the 3' -end by T4 PNK, were each joined with a barcoded 3' -adapter via a splint ligation (step 2,... The ligation efficiency was high, typically reaching >75% (quantification in



2.11 讨论与局限: Genome-wide profiling of t RNA modifications by Induro-t RNAseq reveals...

讨论与局限

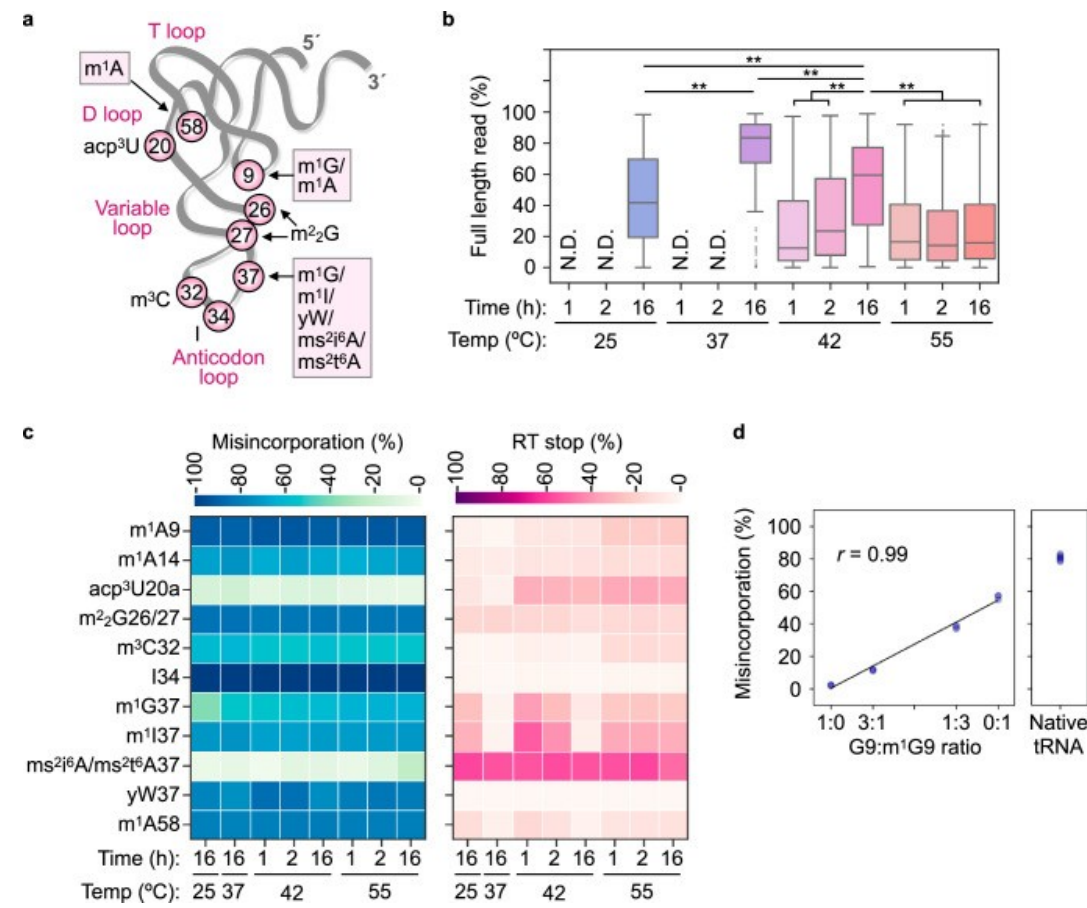
Here we provide a genome-wide t RNAseq dataset that defines the RT signature at each readable modification across the t RNA sequence framework,...

We use the recently developed group-II Induro RT and provide both practical and mechanistic insight into how to maximize its readthrough and how...

Importantly, we produce information on how modifications can undergo a coordinated shift across tissues and cell types, providing a conceptual f...

可用于本项目的提炼

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.12 文献概览： t RNA renovatio: Rebirth through fragmentation.

2.12 | Molecular cell (2023) | review | DOI 10.1016/j.molcel.2023.09.016

研究问题 / 背景

The transfer RNA (t RNA) stood at the center of an entire research era dedicated to the elucidation of the molecular mechanisms underlying cellu...

First discovered in 1958 and identified as the small adaptor molecule that Francis Crick had predicted in his 'adaptor hypothesis' in 1954, t RN...

Contemporaneously, aminoacyl-t RNA synthetases (aa RSs) were identified as the enzymes that catalyze the ATP-dependent matching of each amino ac...

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2.12 综述主要论点: t RNA renovatio: Rebirth through fragmentation.

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主要论点

t RNA function is based on unique structures that enable m RNA decoding using anticodon trinucleotides.

These structures interact with specific aminoacyl-t RNA synthetases and ribosomes using 3D shape and sequence signatures.

Beyond translation, t RNAs serve as versatile signaling molecules interacting with other RNAs and proteins.

Through evolutionary processes, t RNA fragmentation emerges as not merely random degradation but an act of recreation, generating specific short...

These ts RNAs exploit their linear sequences and newly arranged 3D structures for unexpected biological functions, epitomizing the t RNA "renova...

2.12 综合观点与空白： t RNA renovatio: Rebirth through fragmentation.

2.12 | Molecular cell (2023) | review | DOI 10.1016/j.molcel.2023.09.016

综合观点与空白

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2.13 文献概览： t RNA modification dynamics from individual organisms to metaepitranscript...

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研究问题 / 背景

t RNA is the most extensively modified RNA in cells.

On average, a bacterial t RNA contains 8 modifications per molecule and a eukaryotic t RNA contains 13 modifications per molecule.

Recent studies reveal that t RNA modifications are highly dynamic and respond extensively to environmental conditions.

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

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Functions of t RNA modification dynamics include enhanced, on-demand decoding of specific codons in response genes and regulation of t RNA fragm...

This review summarizes recent advances in the studies of t RNA modification dynamics in biological processes, t RNA modification erasers, and hu...

2.13 综合观点与空白： t RNA modification dynamics from individual organisms to metaepitranscr...

综合观点与空白

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2.14 文献概览: Nanopore sequencing of intact aminoacylated t RNAs.

2.14 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

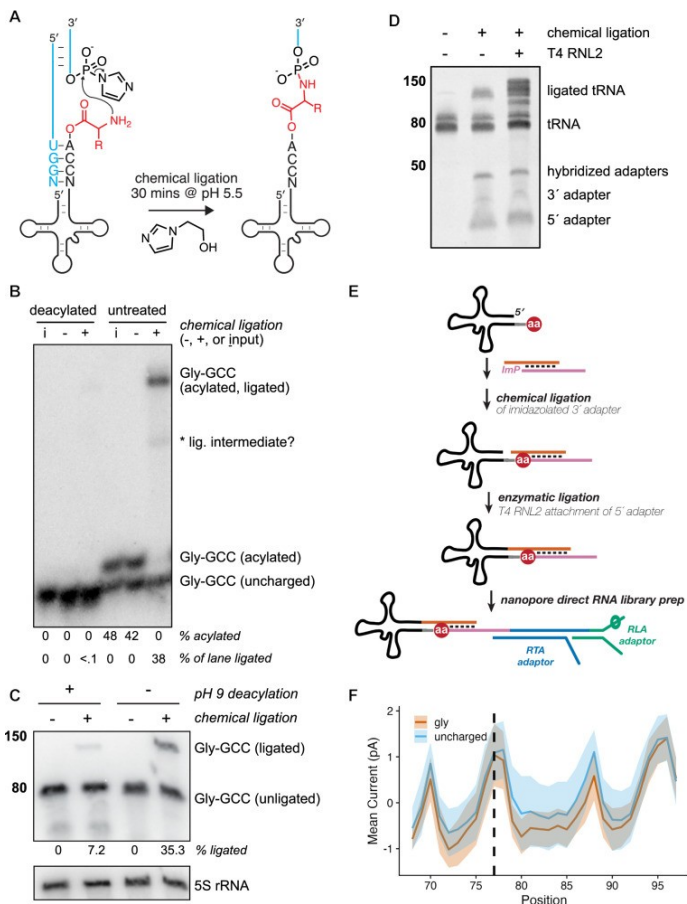
研究问题 / 背景

The intricate landscape of t RNA modification presents persistent analytical challenges, which have impeded efforts to simultaneously resolve se...

To address these challenges, we introduce "aa-t RNA-seq", an integrated method that uses chemical ligation to sandwich the amino acid of a charg...

Our approach reveals the identity of the amino acids attached to all t RNAs in a cellular sample, at the single molecule level.

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.14 综述主要论点: Nanopore sequencing of intact aminoacylated t RNAs.

2.14 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

主要论点

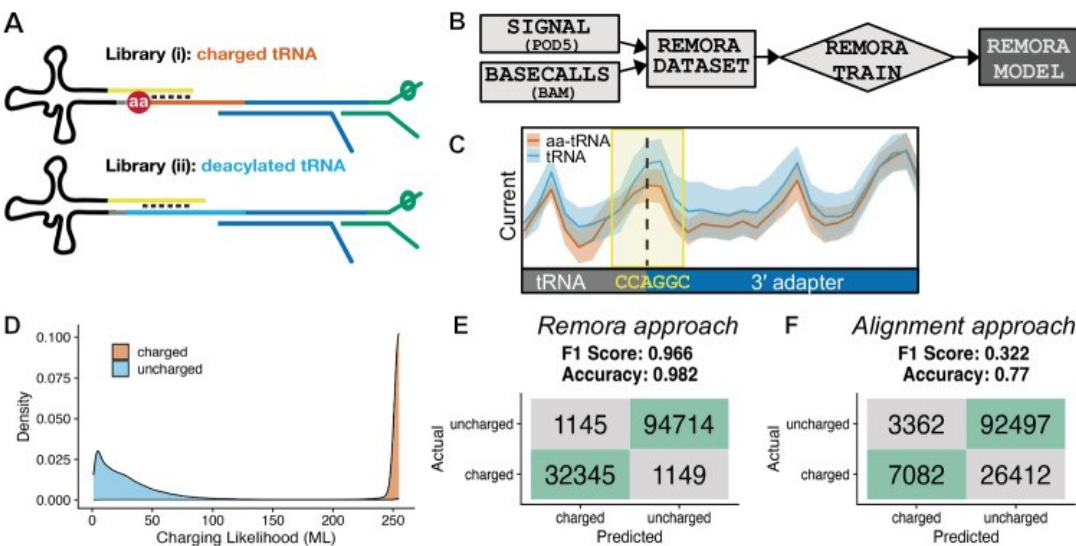
A chemical ligation approach enables selective capture of intact aminoacylated t RNAs While investigating prebiotic roles of aminoacyl-RNAs, we...

We realized that aminoacyl-t RNAs were analogous to these substrates, so we tested and found that a synthetic t RNA-Gly-GCC aminoacylated with...

To accelerate the reaction between the α -amino group of the aminoacyl-t RNA and the activated adapter, we included 1-(2-Hydroxyethyl)imidazole...

1c), and subjected *Saccharomyces cerevisiae* t RNA to a 30 min chemical ligation at pH 5.5.

Under these conditions, we achieved efficient ligation of aminoacylated budding yeast t RNA as measured by acid northern blot, with <0.1% background...



2.14 综合观点与空白： Nanopore sequencing of intact aminoacylated t RNAs.

2.14 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

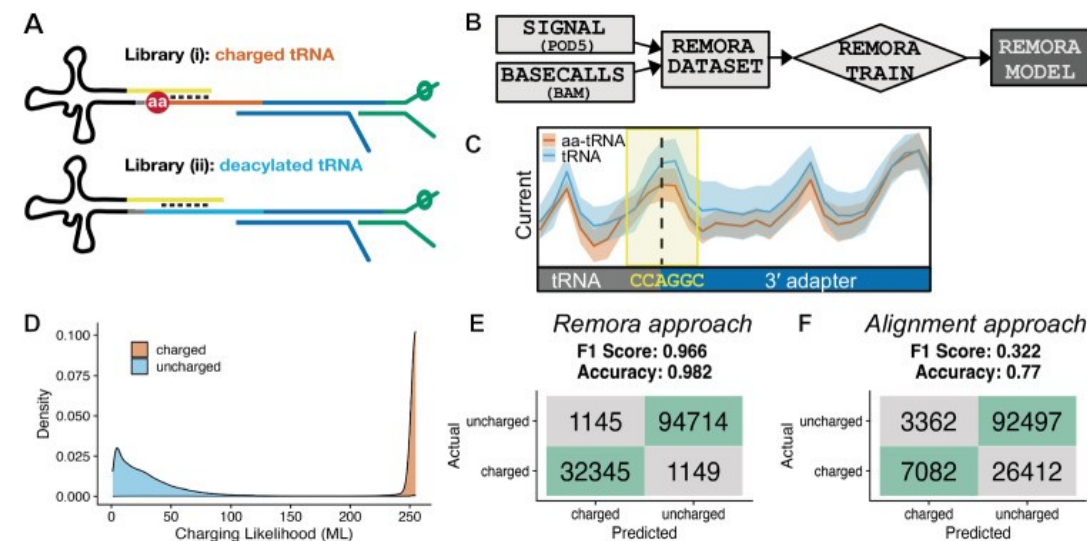
综合观点与空白

Transfer RNAs are long-lived and undergo repeated cycles of charging and deacylation throughout their lifetime.

While t RNA biogenesis is recognized as an intricately coordinated process—spanning tran- scription, processing, modification, and aminoacylation...

We developed a chemical ligation approach to selectively capture aminoacylated t RNAs and used it to study t RNA aminoacyla- tion in a variety o...

Increased accuracy of aa-t RNA identifi- cation may further enhance our understanding of how t RNA modifications act as determinants or anti-deter...



可用于本项目的提炼

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

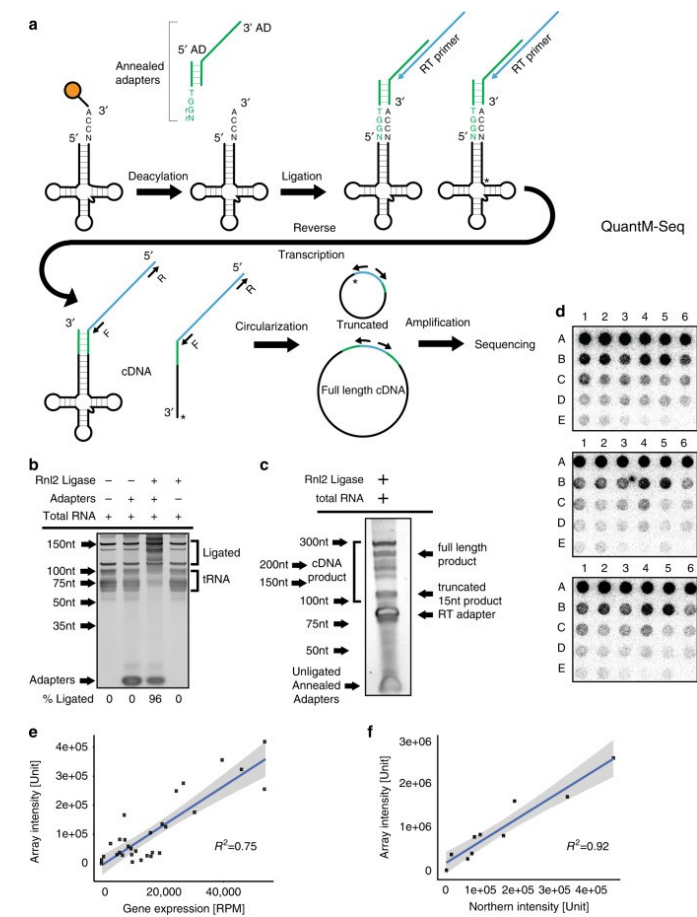
2.15 文献概览: Quantitative t RNA-sequencing uncovers metazoan tissue-specific tRNA regulation

2020 | Nature Communications (2020) | experimental | DOI 10.1038/s41467-020-17879-x

研究问题 / 背景

Translation of the genetic code is clearly critical to all life. While the ribosome is the cellular instrument that executes translation, it relies upon transfer RNA (tRNA) to properly decipher the genetic in... t RNAs parlay codon identity into amino acid identity.

与本子项目的关系：该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。



2.15 材料方法与结果: Quantitative t RNA-sequencing uncovers metazoan tissue-specific t RNA...

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材料 / 方法 / 分析路线

Description of Additional Supplementary Files

核心结果

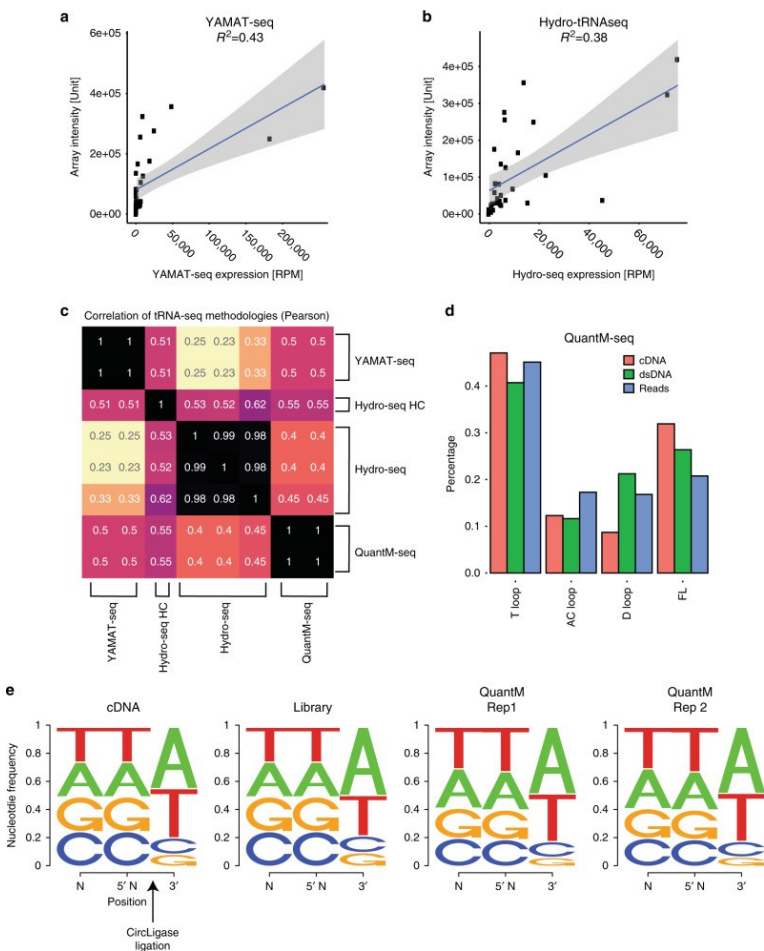
suggest that strong anticodon buffering occurs, reducing the expression of some isodecoders as others increase in expression.

The mechanism and consequence of this buffering is unclear.

In total, we strongly feel that Quant M-t RNA-seq provides the community with a quick and relatively easy means to robustly monitor t RNA levels...

Technologies such as these should greatly accelerate our understanding of how t RNA influences normal and disease states.

Results High-throughput sequencing of mature t RNA by Quant M- t RNA-seq.



2.15 讨论与局限: Quantitative t RNA-sequencing uncovers metazoan tissue-specific t RNA r...

2020 | Nature communications (2020) | experimental | DOI 10.1038/s41467-020-17879-x

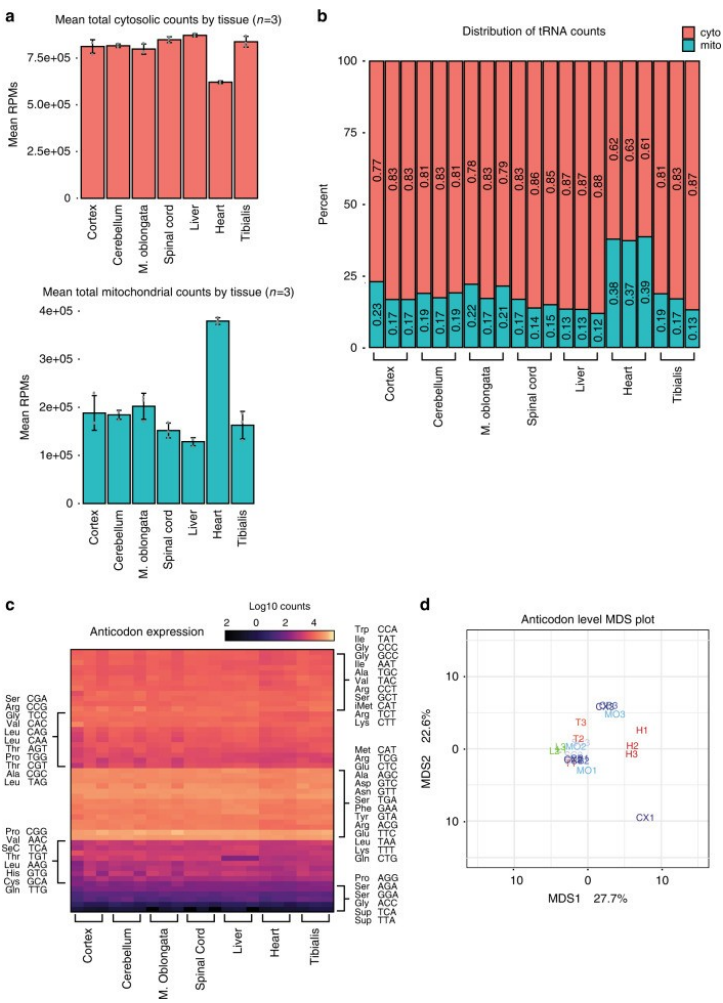
讨论与局限

Historically, t RNAs have been difficult to sequence due to base modifications and extensive structures which impede first-strand synthesis by R...

To circumvent these issues, the few published methods employ clever and diverse library preparation strategies.

However, each protocol has limitations.

YAMAT-seq exhibits variability between technical replicates for some t RNAs and underrepresents many t RNAs.



可用于本项目的提炼

与本子项目的关系: 该文可用于提取 tRNA 定量、注释、测序信号处理或预测算法的设计原则。

Part 3. tRNA 修饰指纹数据库

tRNA mod-fingerprint database

本部分包含 15 篇文献，每篇 3 页。

1. t RNA modifications tune m6A-dependent m RNA decay.
2. Alanyl-t RNA synthetase, AARS1, is a lactate sensor and lactyltransferase that lactylates p53...
3. A cholesterol-responsive hepatic t RNA-derived small RNA regulates cholesterol homeostasis and...
4. Transfer RNA modifications and cellular thermotolerance.
5. NAT10 promotes cancer metastasis by modulating p300/CBP activity through chromatin-associated...
6. t RNA as an assembly chaperone for a macromolecular transcription-processing complex.
7. A ligation-independent sequencing method reveals t RNA-derived RNAs with blocked 3' termini.
8. Structural basis of t RNA recognition by the m3C RNA methyltransferase METTL6 in complex with...
9. Structures of the ribosome bound to EF-Tu-isoleucine t RNA elucidate the mechanism of AUG avoi...
10. t RNA renovatio: Rebirth through fragmentation.
11. Translational regulation by oxidative desulfuration of t RNA modifications.
12. ALKB-1-dependent t RNA methylation is required for efficient paternal mitochondrial eliminatio...
13. The molecular basis of t RNA selectivity by human pseudouridine synthase 3.
14. Perturbation of METTL1-mediated t RNA N7- methylguanosine modification induces senescence and...
15. Coronaviruses reprogram the t RNA epitranscriptome to favor viral protein expression.

3.1 文献概览： t RNA modifications tune m6A-dependent m RNA decay.

3.1 | Cell (2025) | experimental | DOI 10.1016/j.cell.2025.04.013

研究问题 / 背景

The epitranscriptomic mark m 6 A decorates transcripts from thousands of metazoan genes and influences their processing and metabolism 1 , 2 .

This includes pre-m RNA splicing, m RNA export, localization and translation.

By far the most prominent function of m 6 A is, however, its impact on m RNA decay 3 .

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.1 材料方法与结果: t RNA modifications tune m6A-dependent m RNA decay.

3.1 | Cell (2025) | experimental | DOI 10.1016/j.cell.2025.04.013

材料 / 方法 / 分析路线

All unique resources generated in this study are available from the lead contact upon reasonable request with a completed Materials Transfer Agr...

mi CLIP libraries were prepared as described previously 5 .

Raw sequencing reads from all mi CLIP datasets (see the Key Resource Table) were concatenated, aligned with STAR to GRCh38/hg19 and de-duplicat...

Alignment was performed in end-to-end mode to assess the intra-read distribution of C to T transitions.

核心结果

Chemically modified nucleotides in m RNA are critical regulators of gene expression, primarily through interactions with reader proteins that bi...

Here, we present a mechanism by which the epitranscriptomic mark N6-methyladenosine (m6A) is read by t RNAs during translation.

Codons that are modified with m6A are decoded inefficiently by the ribosome, rendering them "non-optimal" and inducing ribosome collisions on ce...

This couples m RNA translation to decay.

5-Methoxycarbonylmethyl-2-thiouridine (mcm5s2U) in the t RNA anticodon loop counteracts this effect.

3.1 讨论与局限： t RNA modifications tune m6A-dependent m RNA decay.

3.1 | Cell (2025) | experimental | DOI 10.1016/j.cell.2025.04.013

讨论与局限

Chemically modified nucleotides tune the activities of different RNA species such as m RNA, t RNA and r RNA.

Here, we show that two such nucleotide modifications in different RNA species interact at the molecular level in cells to exert an in vivo funct...

Using different m 6 A maps at single-nucleotide resolution, we found that m RNA decay correlates with m 6 A levels in the CDS (Figure 1 and S1...

By combining ribosome profiling with the use of a small molecule inhibitor of METTL3, we show that translation of most codons becomes less effic...

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.2 文献概览: Alanyl-t RNA synthetase, AARS1, is a lactate sensor and lactyltransferase...

研究问题 / 背景

Lysine lactylation is a post-translational modification that links cellular metabolism to protein function.

Here, we find that AARS1 functions as a lactate sensor that mediates global lysine lactylation in tumor cells.

AARS1 binds to lactate and catalyzes the formation of lactate-AMP, followed by transfer of lactate to the lysine acceptor residue.

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.2 材料方法与结果： Alanyl-t RNA synthetase, AARS1, is a lactate sensor and lactyltransfe...

Cell 2024; bioRxiv preprint doi: <https://doi.org/10.1016/j.cell.2024.04.002>; this version posted April 1, 2024. The copyright holder for this preprint (which was not certified by peer review) is the author/funder, who has granted bioRxiv a license to display the preprint in perpetuity. It is made available under aCC-BY-NC-ND 4.0 International license.

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

Lysine lactylation is a post-translational modification that links cellular metabolism to protein function.

Here, we find that AARS1 functions as a lactate sensor that mediates global lysine lacylation in tumor cells.

AARS1 binds to lactate and catalyzes the formation of lactate-AMP, followed by transfer of lactate to the lysince acceptor residue.

Proteomics studies reveal a large number of AARS1 targets, including p53 where lysine 120 and lysine 139 in the DNA binding domain are lactylate...

Generation and utilization of p53 variants carrying constitutively lactylated lysine residues revealed that AARS1 lactylation of p53 hinders its...

3.2 讨论与局限: Alanyl-t RNA synthetase, AARS1, is a lactate sensor and lactyltransfera...

Cell 2024; PMID 38653238; PMCID NA; DOI 10.1016/j.cell.2024.04.002

讨论与局限

Lysine lactylation is a post-translational modification that links cellular metabolism to protein function.

Here, we find that AARS1 functions as a lactate sensor that mediates global lysine lacylation in tumor cells.

AARS1 binds to lactate and catalyzes the formation of lactate-AMP, followed by transfer of lactate to the lysince acceptor residue.

Proteomics studies reveal a large number of AARS1 targets, including p53 where lysine 120 and lysine 139 in the DNA binding domain are lactylate...

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.3 文献概览: A cholesterol-responsive hepatic t RNA-derived small RNA regulates cholest...

Source: Nature communications (2025) | experimental | DOI 10.1038/s41467-025-67387-z

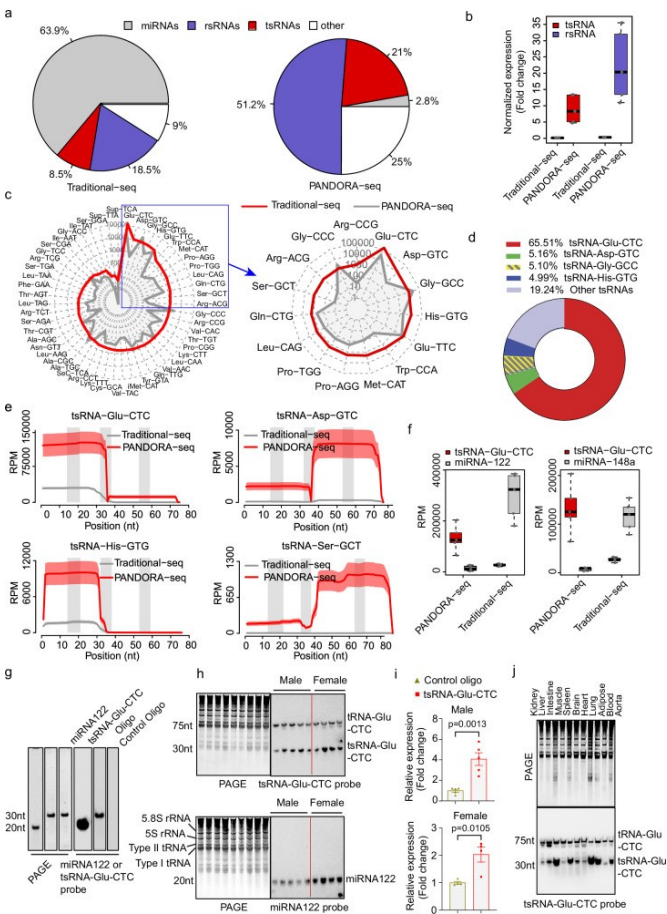
研究问题 / 背景

Cholesterol is an essential component of cell membranes and is also a precursor of numerous steroid hormones and bile acids 1 – 3 .

Cholesterol homeostasis is crucial for normal cellular and biological functions, and aberrant cholesterol homeostasis can lead to the developmen...

Cholesterol concentration is tightly regulated by a feedback control system including transcriptional and post-transcriptional mechanisms 1 , 2 .

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.3 材料方法与结果： A cholesterol-responsive hepatic t RNA-derived small RNA regulates ch...

Source: Nature communications 2025; | experimental | DOI 10.1038/s41467-025-67387-z

材料 / 方法 / 分析路线

Description Of Additional Supplementary File

核心结果

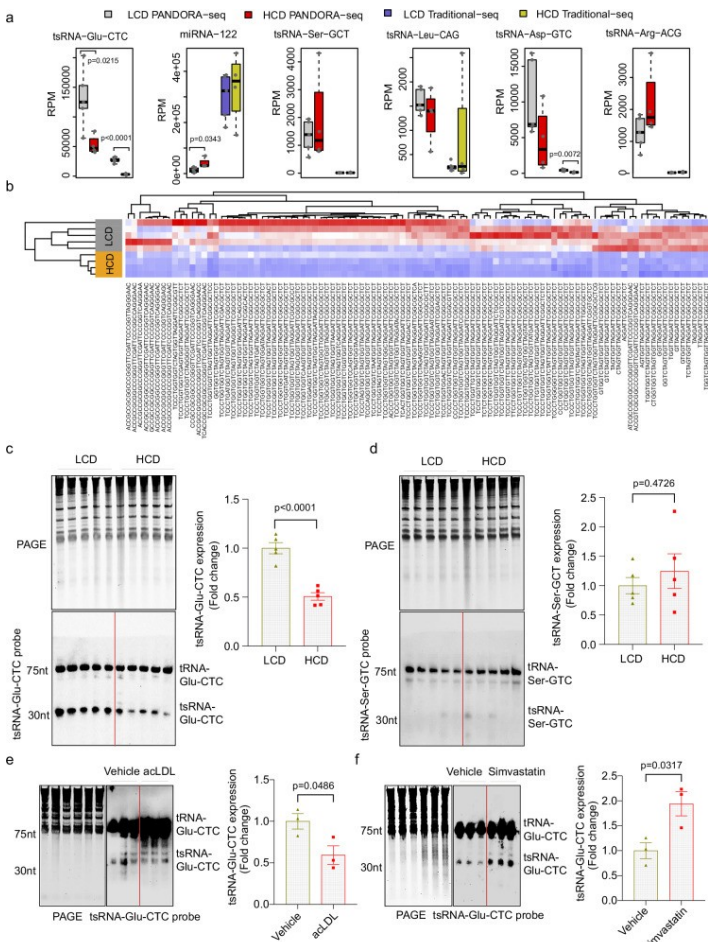
PANDORA-seq but not traditional small RNA-seq identifies ts RNA-Glu-CTC as the most abundant ts RNA in the liver Many snc RNAs including ts RNAs...

To address this obstacle, we have recently developed an advanced PANDORA-seq to eliminate the RNA modification-elicited sequence interferences21,...

Liver is a key organ with numerous key functions including regulating lipid homeostasis.

To reveal the hepatic snc RNA profiles, we isolated total RNAs from mouse liver and performed both PANDORA-seq and traditional small RNA sequenci...

As expected, traditional RNA-seq identified mi RNAs as the most abundant snc RNA species in the liver (Fig.



3.3 讨论与局限: A cholesterol-responsive hepatic t RNA-derived small RNA regulates chol...

讨论与局限

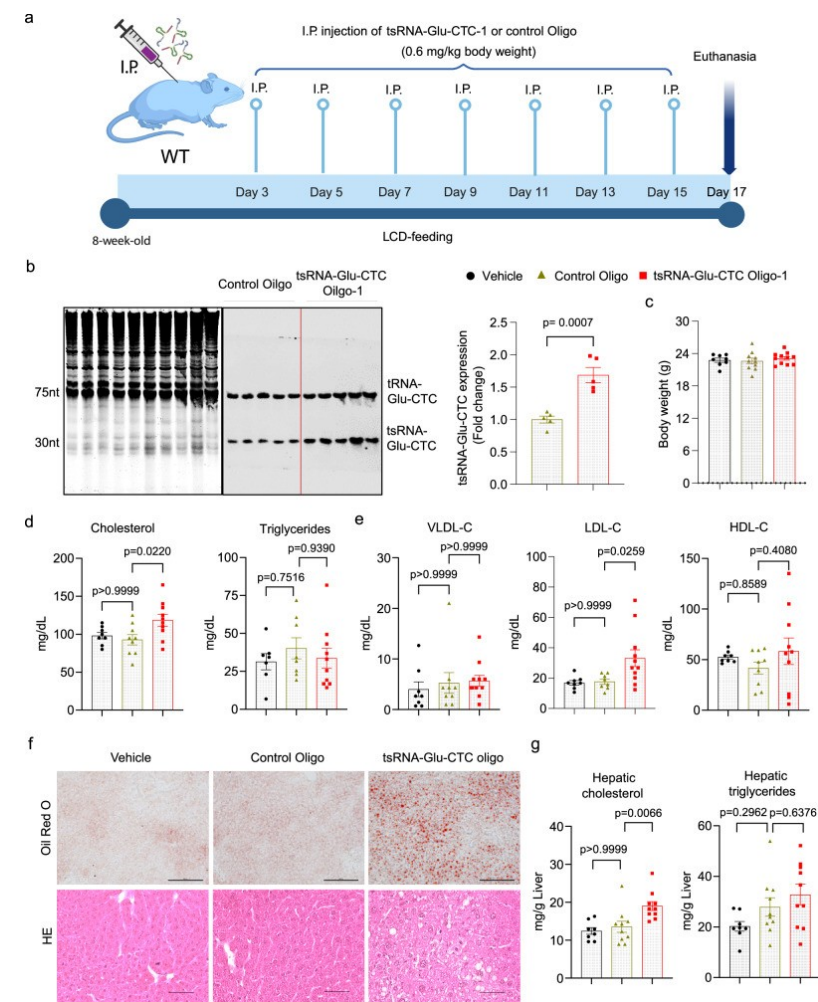
snc RNAs play diverse roles in regulating numerous biological processes in both normal and pathological conditions 63 , 64 .

For CVD research, most studies have focused on investigating the function of mi RNAs, previously considered as the dominant regulatory snc RNAs. These studies have led to the identification of important mi RNAs in the regulation of lipid metabolism and CVD 4 , 7 – 11 , 65 , 66 .

With advanced RNA sequencing methods, recent studies have unveiled surprisingly new snc RNA landscapes in various tissues and cells 21 , 41 , 44...

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.4 文献概览： Transfer RNA modifications and cellular thermotolerance.

3.4 | Molecular cell (2024) | review | DOI 10.1016/j.molcel.2023.11.041

研究问题 / 背景

RNA molecules are modified post-transcriptionally to acquire their diverse functions. Transfer RNA (t RNA) has the widest variety and largest numbers of RNA modifications. t RNA modifications are pivotal for decoding the genetic code and stabilizing the tertiary structure of t RNA molecules.

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.4 综述主要论点: Transfer RNA modifications and cellular thermotolerance.

3.4 | Molecular cell (2024) | review | DOI 10.1016/j.molcel.2023.11.041

主要论点

RNA molecules are modified post-transcriptionally to acquire their diverse functions.

Transfer RNA (t RNA) has the widest variety and largest numbers of RNA modifications.

t RNA modifications are pivotal for decoding the genetic code and stabilizing the tertiary structure of t RNA molecules.

Alternation of t RNA modifications directly modulates the structure and function of t RNAs and regulates gene expression.

Notably, thermophilic organisms exhibit characteristic t RNA modifications that are dynamically regulated in response to varying growth temperat...

3.4 综合观点与空白: Transfer RNA modifications and cellular thermotolerance.

3.4 | Molecular cell (2024) | review | DOI 10.1016/j.molcel.2023.11.041

综合观点与空白

RNA molecules are modified post-transcriptionally to acquire their diverse functions.

Transfer RNA (t RNA) has the widest variety and largest numbers of RNA modifications.

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可用于本项目的提炼

与本子项目的关系: 该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.5 文献概览： NAT10 promotes cancer metastasis by modulating p300/CBP activity through c...

1.5 Molecular Cell 2025 | experimental | DOI 10.1016/j.molcel.2025.11.010

研究问题 / 背景

Metastasis, a complex process by which cancer cells spread to distant organs, remains one of the most formidable challenges in oncology 1 , 2 .

Metastasis is associated with most cancer-related deaths and limits the success of existing therapeutics 3 .

Numerous genetic 1 , 4 , epigenetic 5 , and microenvironmental factors 6 have been implicated in the metastatic cascade, underscoring the need f...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.5 材料方法与结果： NAT10 promotes cancer metastasis by modulating p300/CBP activity thro...

256 Molecular Cell (2025) | Experimental | DOI 10.1016/j.molcel.2025.11.010

材料 / 方法 / 分析路线

The NAT10-d TAG knockin vector was constructed using the protocol described elsewhere 31 .

C-terminal targeting guide RNA was designed using CRISPick webpage.

sg RNA oligonucleotides purchased from IDT were annealed and ligated in the Bbs I restriction site of p X330A-1×2 (Addgene, #58766) plasmid to m...

PITCh sg RNA from p X330S-2-PITCh (Addgene #63670) vector was then inserted into the p X330A-1×2-c NAT10 vector using Golden Gate Assembly (New...

核心结果

Protein and RNA acetylation are crucial for development and cancer progression.

NAT10 is the only known acetyltransferase responsible for N4-acetylcytidine (ac4C) modification of RNA.

However, the mechanism by which NAT10 contributes to cancer progression remains unclear.

Here, we show that NAT10 interacts with a mechanosensitive, metastasis-susceptibility protein complex at the nuclear pore.

Loss of NAT10's acetylation activity significantly reduces lung metastasis in both allograft and genetically engineered mouse models of breast C...

3.5 讨论与局限： NAT10 promotes cancer metastasis by modulating p300/CBP activity through...

Cell Molecular Cell (2025) | experimental | DOI 10.1016/j.molcel.2025.11.010

讨论与局限

In this study, we identify NAT10 as a metastasis-associated factor in breast cancer.

Beyond its canonical nucleolar role, NAT10 is sensitive to extracellular matrix stiffness and interacts with NUP210 at the nuclear pore.

Although the precise mechanistic link between NAT10 and NUP210 remains to be defined, our data indicate that they act within a shared pathway.

NAT10 regulates transcriptional activity at the NUP210 enhancer, and a subset of NAT10 localizes to nuclear pore-associated enhancers enriched f...

可用于本项目的提炼

与本子项目的关系： 该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

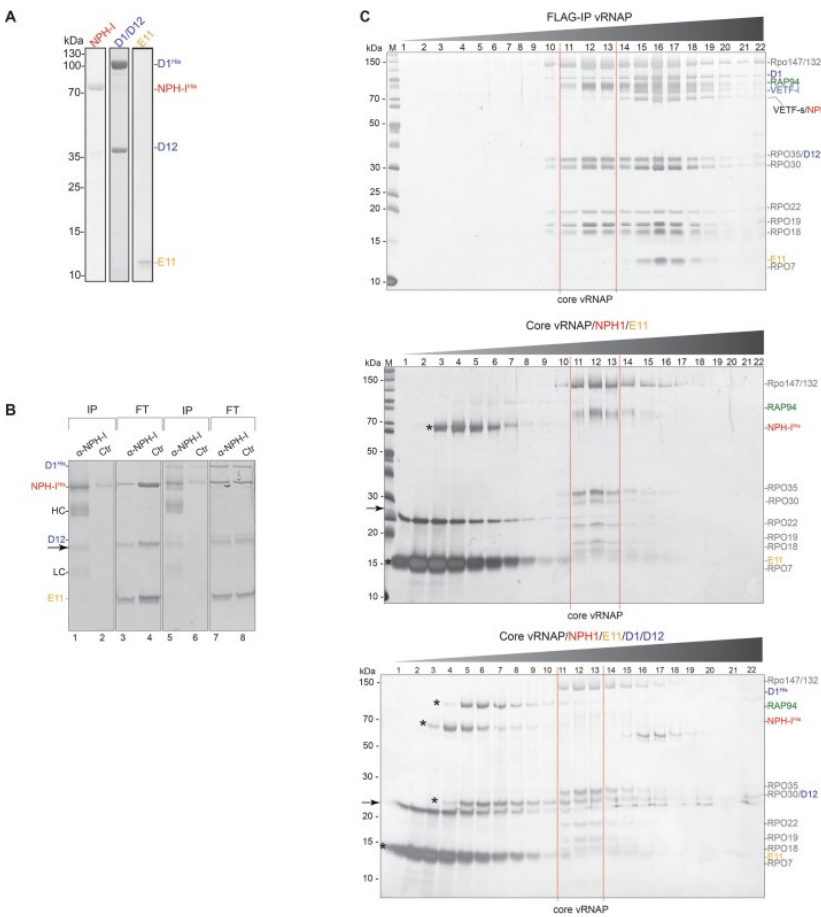
3.6 文献概览: t RNA as an assembly chaperone for a macromolecular transcription-processi...

Source: Nature structural & molecular biology (2025) | review | DOI 10.1038/s41594-025-01653-y

研究问题 / 背景

Transfer RNAs (t RNAs) are widely recognized for their role in translation. Here, we describe a previously unidentified function of t RNA as an assembly chaperone. During poxviral infection, t RNAGln/Arg lacking the anticodon mcm5s2U34 modification is specifically sequestered from the cellular t RNA pool to...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.6 综述主要论点: t RNA as an assembly chaperone for a macromolecular transcription-pro...

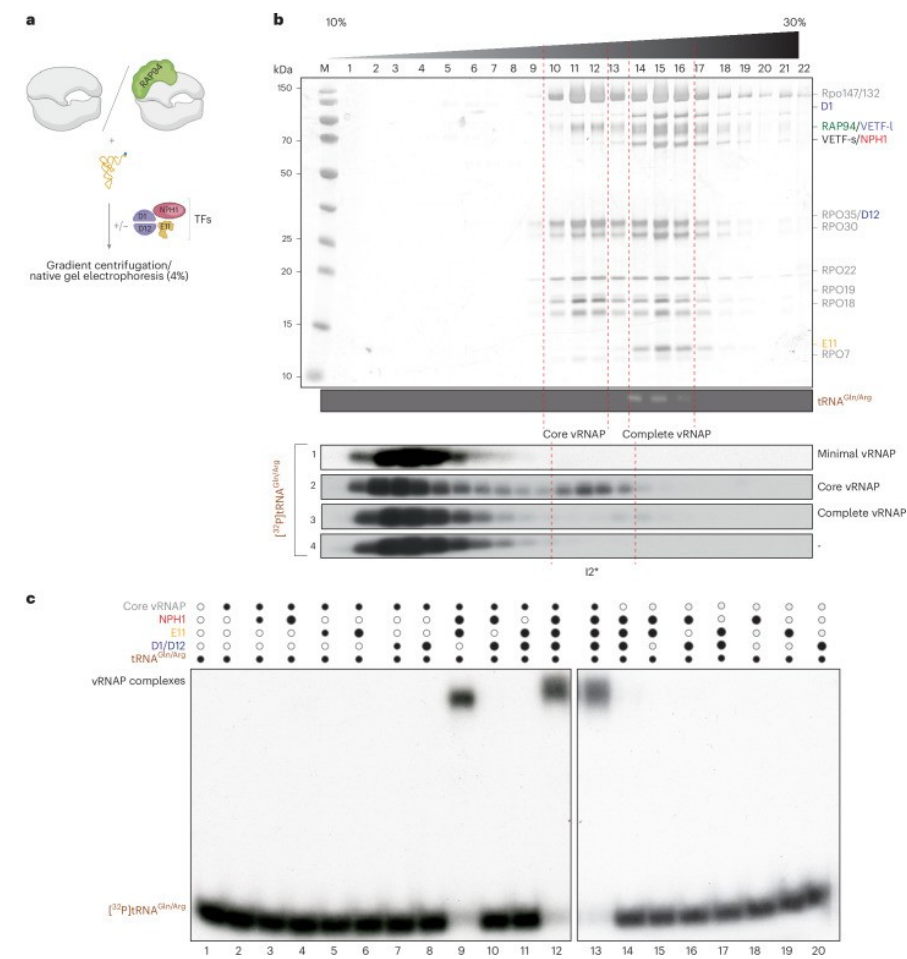
主要论点

Coordinated assembly of Rap94, NPH-I and E11 with v RNAP by t RNAGln/Arg To test whether t RNAGln/Arg acts as an assembly chaperone rather than...

We isolated a minimal version of v RNAP, consisting of the eight Rpo subunits, only as a presumed starting point of assembly.

This unit, along with recombinant transcription factors (Extended Data Fig. 1a) and t RNAGln/Arg isolated from complete v RNAP, was then used to study the formation of complete v RNAP in vitro (Fig.

Cosedimentation and gel mobility shift experiments showed that [32P]t RNAGln/Arg alone failed to bind minimal v RNAP (Fig.



3.6 综合观点与空白： t RNA as an assembly chaperone for a macromolecular transcription-proce...

综合观点与空白

In this study, we dissected the entire assembly pathway of complete v RNAP through a combination of biochemical and structural (cryo-EM) approach...

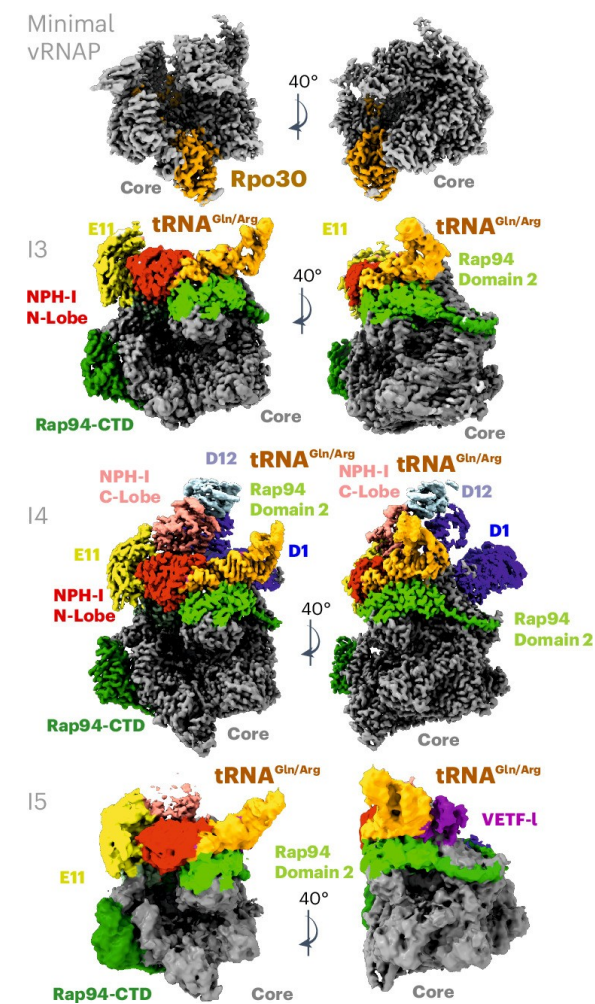
Our findings identify t RNA Gln/Arg as assembly chaperones for a macromolecular transcription and processing complex and, thus, add a novel aspect...

Assembly chaperones are a heterogeneous group of factors that join components with otherwise little or no affinity to each other and typically unassemble...

They have, however, no further role with respect to the function of the assembled unit and often dissociate from the assembled particle.

可用于本项目的提炼

与本子项目的关系： 该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.7 文献概览： A ligation-independent sequencing method reveals t RNA-derived RNAs with b...

Molecular cell (2024) | experimental | DOI 10.1016/j.molcel.2024.07.008

研究问题 / 背景

Despite the numerous sequencing methods available, the diversity in RNA size and chemical modification makes it difficult to capture all RNAs in...

We developed a method that combines quasi-random priming with template switching to construct sequencing libraries from RNA molecules of any len...

Our ligation-independent detection of all types of RNA (LIDAR) is a simple, effective tool to identify and quantify all classes of coding and no...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.7 材料方法与结果： A ligation-independent sequencing method reveals t RNA-derived RNAs w...

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材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

材料 / 方法 / 分析路线

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核心结果

Despite the numerous sequencing methods available, the diversity in RNA size and chemical modification makes it difficult to capture all RNAs in...

We developed a method that combines quasi-random priming with template switching to construct sequencing libraries from RNA molecules of any len...

Our ligation-independent detection of all types of RNA (LIDAR) is a simple, effective tool to identify and quantify all classes of coding and no...

With LIDAR, we comprehensively characterized the transcriptomes of mouse embryonic stem cells, neural progenitor cells, mouse tissues, and sperm.

LIDAR detected a much larger variety of t RNA-derived RNAs (t DRs) compared with traditional ligation-dependent sequencing methods and uncovered...

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3.7 讨论与局限： A ligation-independent sequencing method reveals t RNA-derived RNAs wit...

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讨论与局限

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With LIDAR, we comprehensively characterized the transcriptomes of mouse embryonic stem cells, neural progenitor cells, mouse tissues, and sperm.

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

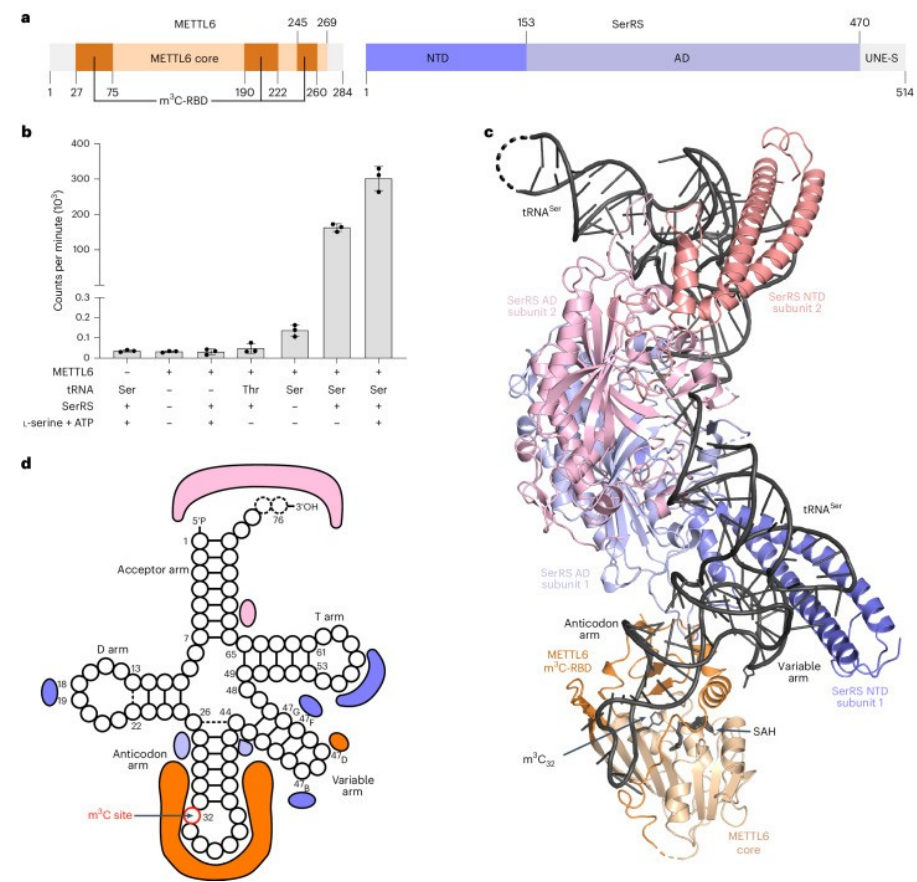
3.8 文献概览: Structural basis of t RNA recognition by the m3C RNA methyltransferase MET...

Source: Nature structural & molecular biology (2024) | experimental | DOI 10.1038/s41594-024-01341-3

研究问题 / 背景

Methylation of cytosine 32 in the anticodon loop of t RNAs to 3-methylcytosine (m3C) is crucial for cellular translation fidelity. Misregulation of the RNA methyltransferases setting this modification can cause aggressive cancers and metabolic disturbances. Here, we report the cryo-electron microscopy structure of the human m3C t RNA methyltransferase METTL6 in complex with seryl-t RNA synthetase (S...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.8 材料方法与结果： Structural basis of t RNA recognition by the m3C RNA methyltransferas...

Source: Nature structural & molecular biology 2024; | experimental | DOI 10.1038/s41594-024-01341-3

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer r...

RNA nucleotide methylation: 2021 update.

Rapid t RNA decay can result from lack of nonessential modifications.

Retrograde transfer RNA nuclear import provides a new level of t RNA quality control in *Saccharomyces cerevisiae*.

核心结果

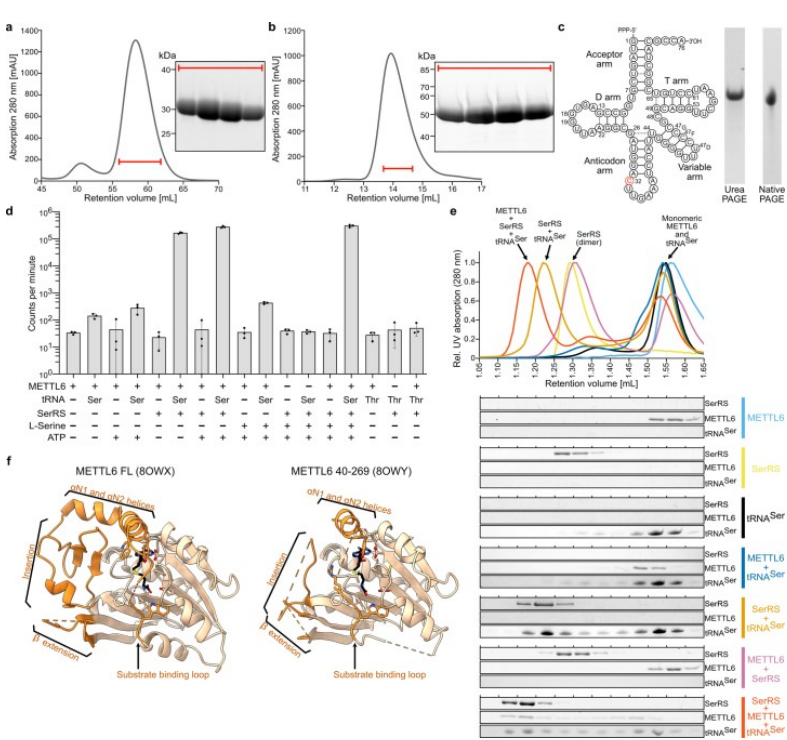
, we identified the methyltransferases METTL11A and METTL11B, which methylate the N-terminal α -amino group of proteins, as relatives with a si...

Overall, our METTL6–Ser RS–t RNASer complex structure identifies the m3C methyltransferase-specific RNA-binding domain, which differs from so-fa...

Acceptor arm D arm T arm Anticodon arm m3C site Variable arm 32 26 22 13 7 65 49 48 53 61 1 76 18 19 44 5'P 3'OH 47B 47D 47F 47G d t RNASer t RN...

1 | The structure of the Ser RS–t RNA–METTL6 complex.

a, Domain organization of human METTL6 and Ser RS.



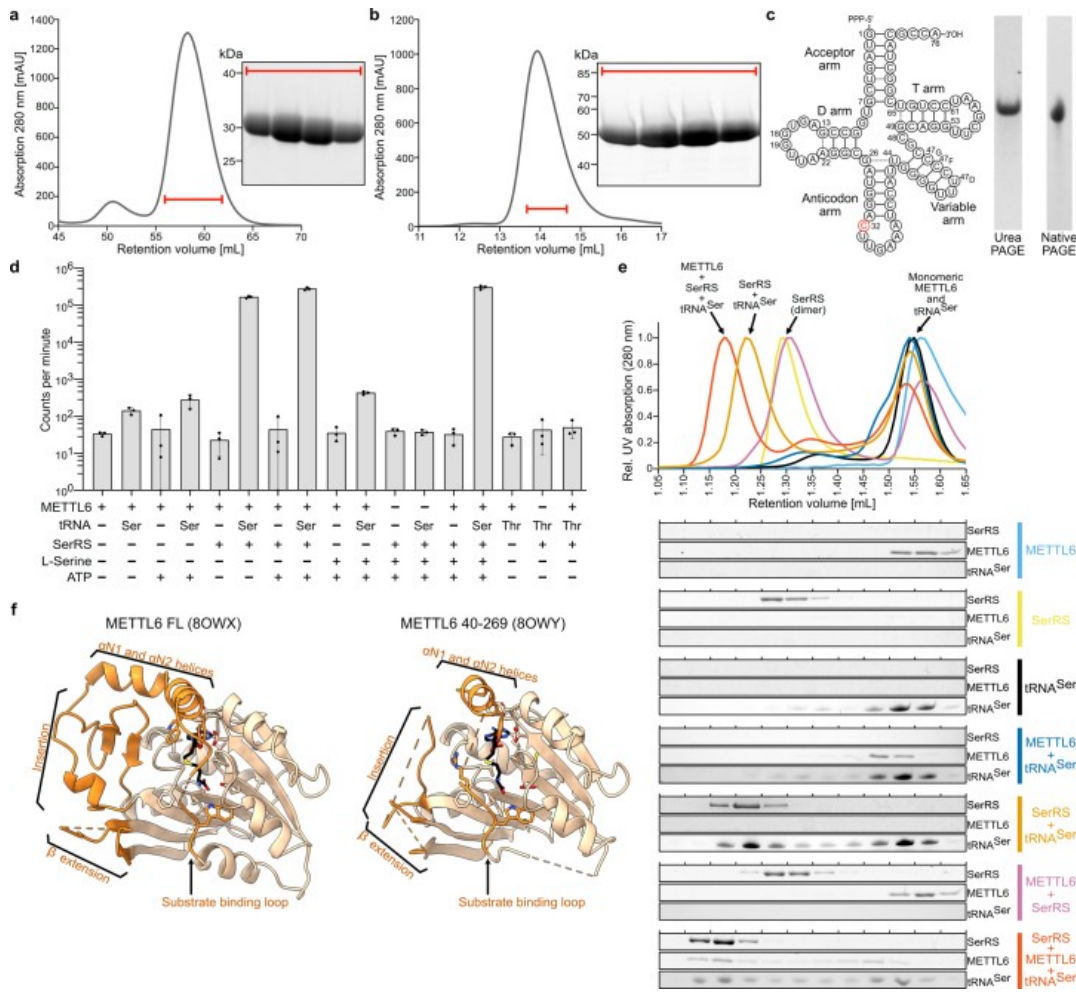
3.8 讨论与局限: Structural basis of t RNA recognition by the m3C RNA methyltransferase...

讨论与局限

In this work, we present the cryo-EM structure of METTL6 in complex with Ser RS bound to the co-products of the methyl transfer, the m3C32-mod... We identify the tripartite t RNA-binding domain of METTL6 and show that Ser RS is necessary as a t RNA selection factor that augments METTL6 act... Aminoacylation-independent functions of aminoacylases have been identified, most commonly as gene-regulatory auto-feedback mechanisms53–56. Ser RS plays a noncanonical role in vascular development that is independent of its aminoacylation activity57–59.

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.9 文献概览： Structures of the ribosome bound to EF-Tu-isoleucine t RNA elucidate the m...

3.9 | Nature structural & molecular biology (2024) | experimental | DOI 10.1038/s41594-024-01236-3

研究问题 / 背景

The frequency of errors upon decoding of messenger RNA by the bacterial ribosome is low, with one misreading event per 1×10^4 codons.

In the universal genetic code, the AUN codon box specifies two amino acids, isoleucine and methionine.

In bacteria and archaea, decoding specificity of the AUA and AUG codons relies on the wobble avoidance strategy that requires modification of C3...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.9 材料方法与结果： Structures of the ribosome bound to EF-Tu-isoleucine t RNA elucidate...

Nature structural & molecular biology (2024) | experimental | DOI 10.1038/s41594-024-01236-3

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

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核心结果

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In the universal genetic code, the AUN codon box specifies two amino acids, isoleucine and methionine.

In bacteria and archaea, decoding specificity of the AUA and AUG codons relies on the wobble avoidance strategy that requires modification of C3...

Bacterial t RNA^{Ala} CAU with 2-lysylcytidine (lysidine) at the wobble position deciphers AUA while avoiding AUG.

Here we report cryo-electron microscopy structures of the Escherichia coli 70S ribosome complexed with elongation factor thermo unstable (EF-Tu)...

3.9 讨论与局限： Structures of the ribosome bound to EF-Tu-isoleucine t RNA elucidate th...

3.9 | Nature structural & molecular biology (2024) | experimental | DOI 10.1038/s41594-024-01236-3

讨论与局限

Structures of the 70S ribosome captured during decoding of the cognate AUA (structure I) and near-cognate AUG (structure II) codons reveal the m...
Ribosome pelleting assays of the complexes show that EF-Tu•GDPCP•Ile-t RNA Ile LAU associates with ribosomes programmed with the AUG codon in th...
Based on this observation, we first collected cryo-EM data of EF-Tu•GDPCP•Ile-t RNA Ile LAU bound to the ribosome in the process of decoding the...
However, despite extensive particle sorting using focused 3D variability analysis and 3D classification, no class average displayed the EF-Tu-te...

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.10 文献概览： t RNA renovatio: Rebirth through fragmentation.

3.10 | Molecular cell (2023) | review | DOI 10.1016/j.molcel.2023.09.016

研究问题 / 背景

t RNA function is based on unique structures that enable m RNA decoding using anticodon trinucleotides.

These structures interact with specific aminoacyl-t RNA synthetases and ribosomes using 3D shape and sequence signatures.

Beyond translation, t RNAs serve as versatile signaling molecules interacting with other RNAs and proteins.

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.10 综述主要论点: t RNA renovatio: Rebirth through fragmentation.

3.10 | Molecular cell (2023) | review | DOI 10.1016/j.molcel.2023.09.016

主要论点

t RNA function is based on unique structures that enable m RNA decoding using anticodon trinucleotides.

These structures interact with specific aminoacyl-t RNA synthetases and ribosomes using 3D shape and sequence signatures.

Beyond translation, t RNAs serve as versatile signaling molecules interacting with other RNAs and proteins.

Through evolutionary processes, t RNA fragmentation emerges as not merely random degradation but an act of recreation, generating specific short...

These ts RNAs exploit their linear sequences and newly arranged 3D structures for unexpected biological functions, epitomizing the t RNA "renova...

3.10 综合观点与空白: t RNA renovatio: Rebirth through fragmentation.

3.10 | Molecular cell (2023) | review | DOI 10.1016/j.molcel.2023.09.016

综合观点与空白

t RNA function is based on unique structures that enable m RNA decoding using anticodon trinucleotides.

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可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.11 文献概览: Translational regulation by oxidative desulfuration of t RNA modifications.

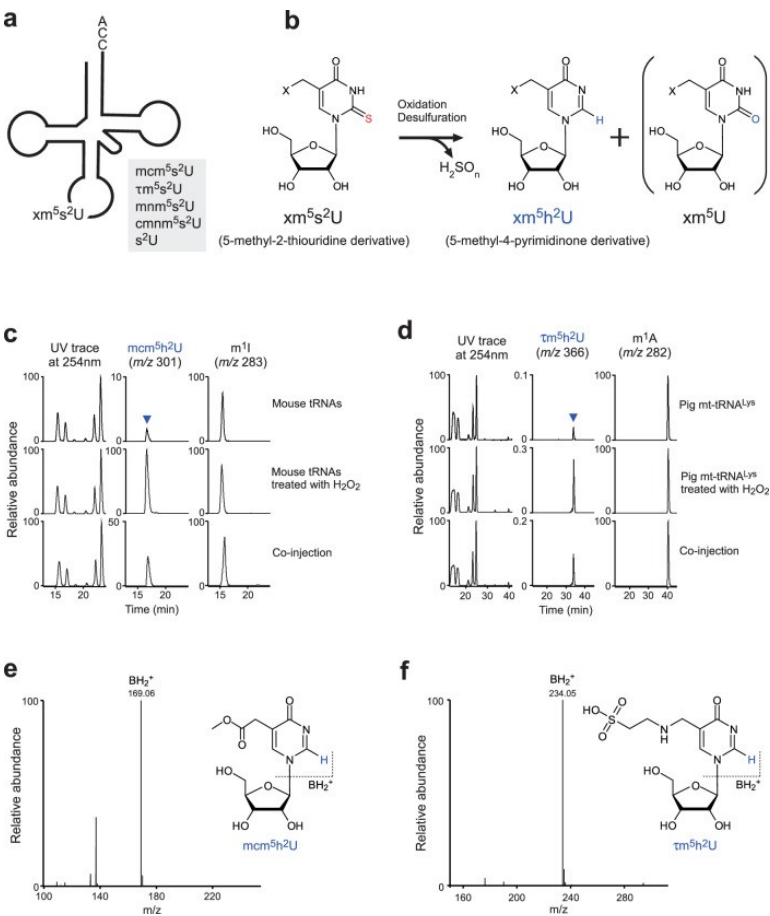
Nature Communications 2026 | experimental | DOI 10.1038/s41467-026-70126-7

研究问题 / 背景

Modifications in the anticodon region of transfer RNA (t RNA) are essential for accurate and efficient protein synthesis.

5-Methyl-2-thiouridine derivatives (xm5s2U) are major modifications at the wobble position of t RNA anticodons decoding purine-ending two-codon...

Although the thiocarbonyl group of xm5s2U enhances decoding efficiency, it is chemically susceptible to oxidative desulfuration, yielding 4-pyri...



与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.11 材料方法与结果: Translational regulation by oxidative desulfuration of t RNA modifica...

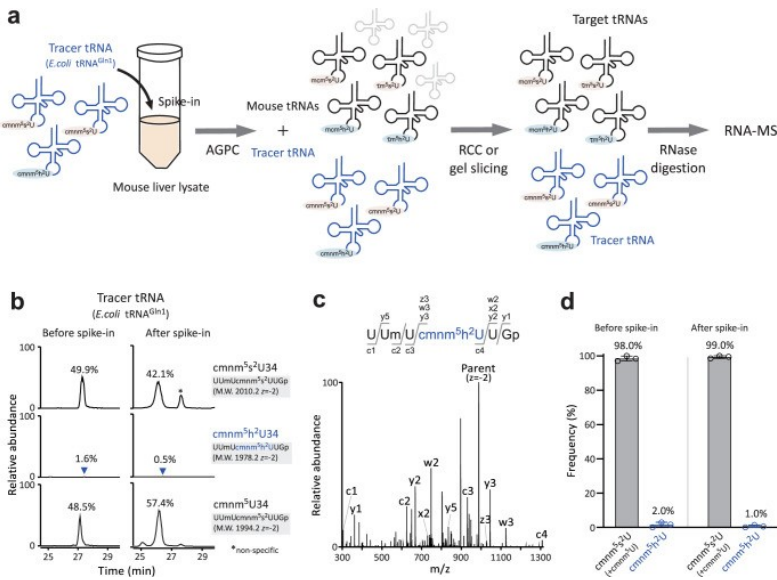
Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-70126-7

材料 / 方法 / 分析路线

coli strain BW25113 was obtained from National Bio Resource Project E.
coli) at the National Institute of Genetics (Mishima, Japan).
The Δmnm A strain was obtained from the E. coli KEIO collection resource (Tokyo, Japan)76.

核心结果

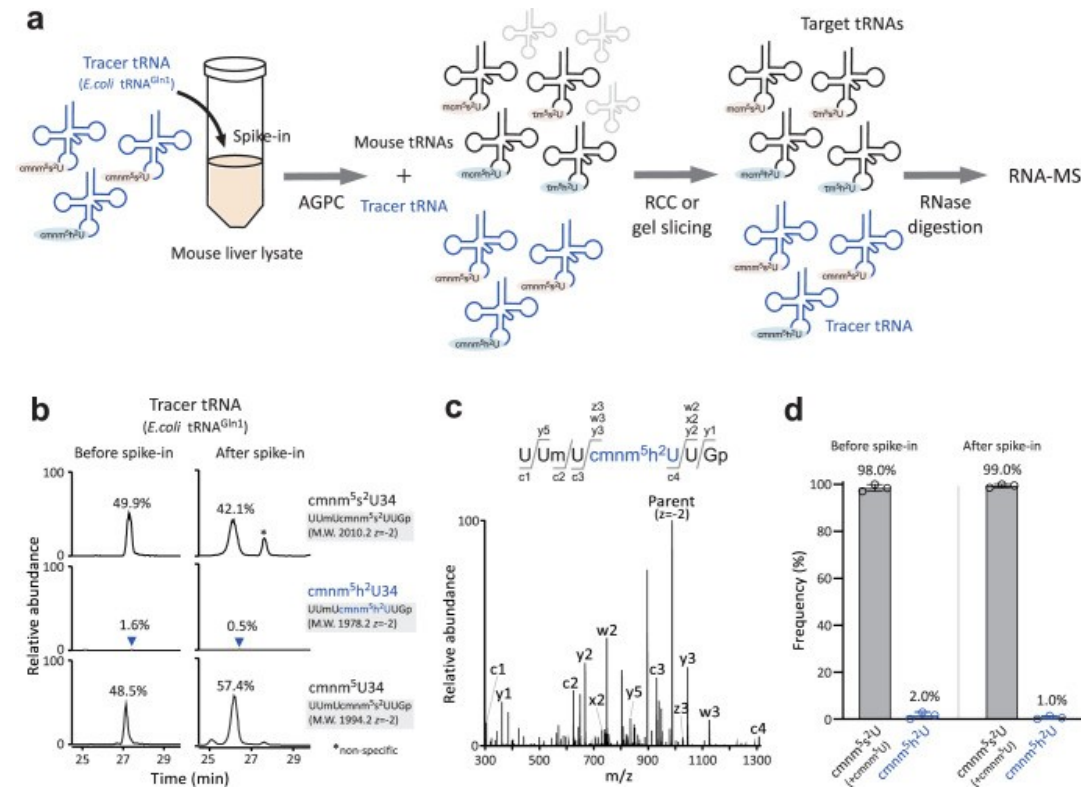
Detection of h2U derivatives in mammalian and bacterial t RNAs We reviewed our previous liquid chromatography-mass spectrometry (LC-MS) data on...
In LC-MS nucleoside analyses of total RNA from mouse livers, we clearly detected unassigned nucleosides with m/z 301 (Fig.
In addition, we also detected nucleosides with m/z 366 in mitochondrial t RNALys isolated from pig liver (Fig.
Collision- induced dissociation (CID) of these nucleosides generated base- related ions (BH2 +) with loss of unmodified ribose (Fig.
1e, f), sug- gesting that the 32 Da loss originates from their bases.



3.11 讨论与局限: Translational regulation by oxidative desulfuration of t RNA modification...

讨论与局限

Meanwhile, mcm5h2U34 pairs with A3 via a unique geometry involving a single H-bond between O4 of mcm5h2U34 and N6 of A3 (Fig. The N3 atom of mcm5h2U no longer acts as a proton donor due to its chemical nature, so that it is not involved in any additional H-bonding (Fig. The observed geometry with a single H-bond could explain why s2U-to-h2U conversion results in loss of the ability to recognize the AAA codon (Fi... 5 | Desulfuration of t RNA modification affects aminoacylation.



可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

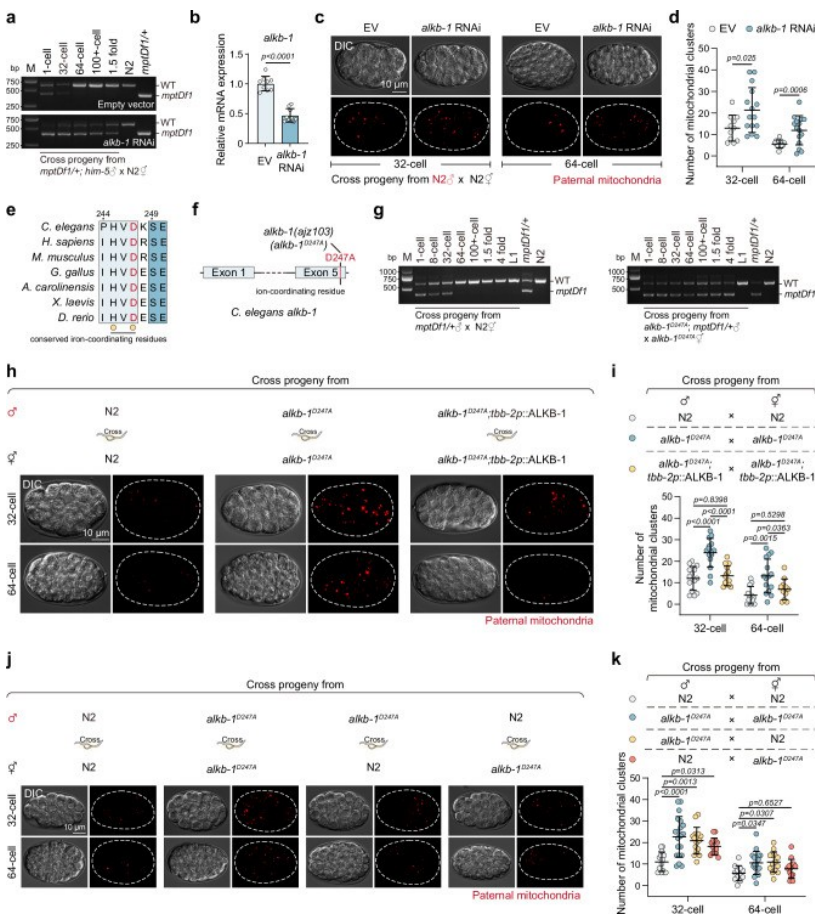
3.12 文献概览: ALKB-1-dependent t RNA methylation is required for efficient paternal mito...

312 | Nature communications (2026) | experimental | DOI 10.1038/s41467-026-68813-6

研究问题 / 背景

The coordination of nuclear and mitochondrial genome expression is crucial for maintaining mitochondrial function and homeostasis, despite their...
Maternal inheritance of the mitochondrial genome (mt DNA) is predominant in eukaryotes, except for certain fungi and mussels.
Paternal mitochondria are selectively eliminated during spermatogenesis and early embryogenesis, a key step for maintaining maternal mt DNA 3 -...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.12 材料方法与结果： ALKB-1-dependent t RNA methylation is required for efficient paternal...

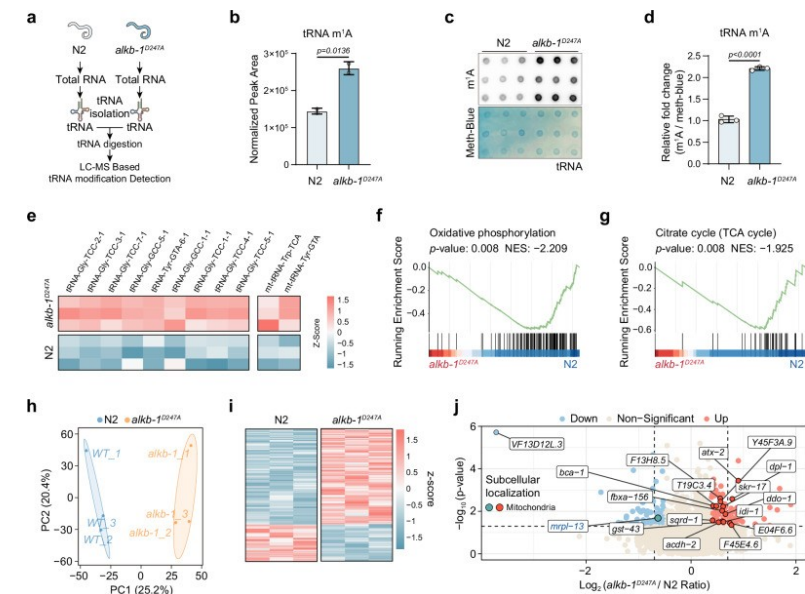
Nature communications (2026) | experimental | DOI 10.1038/s41467-026-68813-6

材料 / 方法 / 分析路线

, which may potentially be insufficient to detect changes in 6 m A levels in the *alkb-1*D247A mutants. To further ascertain the role of 6 m A in PME, we overexpressed the methyltransferase DAMT-1, which is known to increase 6 m A levels in bot... Surprisingly, DAMT-1 overexpression did not affect the PME process (Supplementary Fig. Collectively, these data indicate that the ALKB-1-mediated delay in PME is not related to its DNA 6 m A demethylase activity.

核心结果

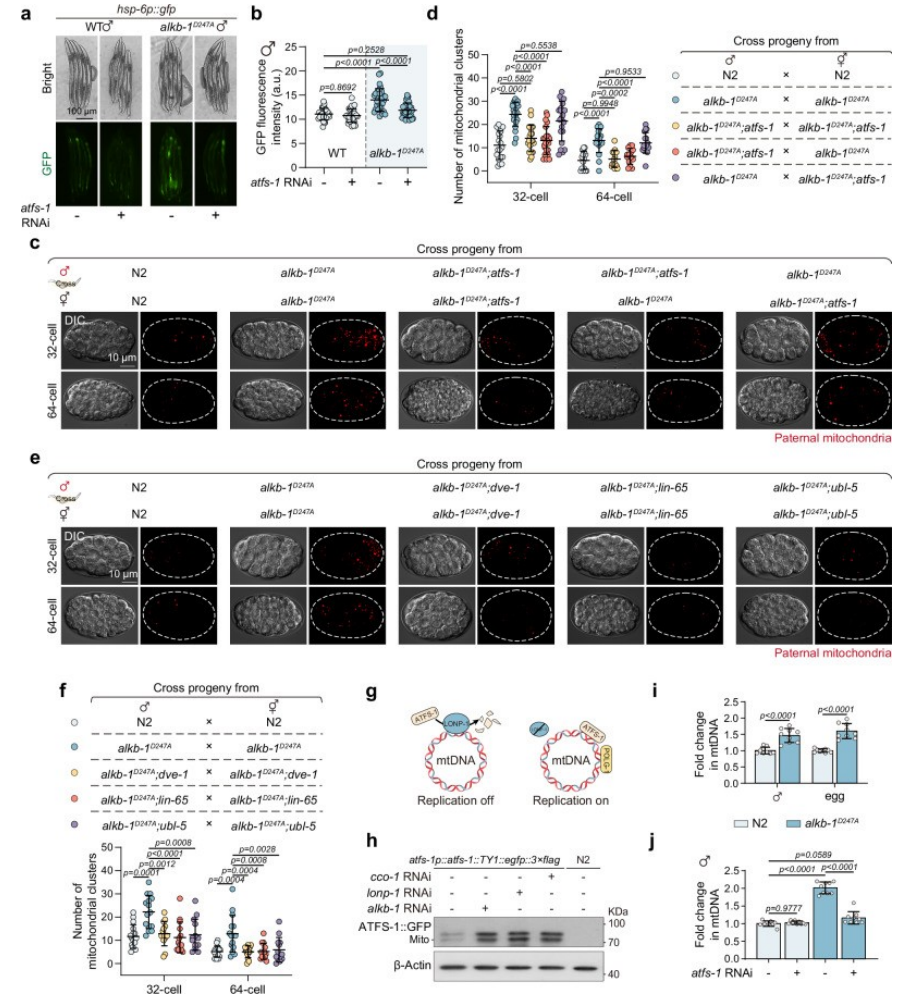
The demethylase ALKB-1 modulates paternal mitochondrial elimination dynamics To monitor PME in embryos, we employed the mpt Df1 mt DNA allele, w... We used a nested PCR assay to track the presence of sperm-derived mt DNA during embryogenesis (Supplementary Fig. Crossing mpt Df1/+ heteroplasmic males with N2 hermaphrodites revealed that mpt Df1 mt DNA is present in early-stage embryos but disappeared bef... 1e), consistent with previous reports of early paternal mt DNA elimination^{21,39}. To investigate the function of methylation in the PME process, we conducted an RNA interference (RNAi) screen against potential homologs of the...



paternal m... (2026) | experimental | DOI 10.1038/s41467-026-68813-6

Maternal mitochondrial inheritance provides evolutionary benefits by limiting the accumulation of mt DNA mutations in offspring 3 . Unraveling the regulatory mechanism underlying the rapid elimination of paternal mitochondria could yield important insights for developing inno... Given the multifaceted roles of methylation patterns in spermatogenesis and embryonic development 24 – 26 , it is of great interest to study how... Our study reveals that ALKB-1 impairment delays PME in C.

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.13 文献概览： The molecular basis of t RNA selectivity by human pseudouridine synthase 3.

Source: Molecular cell 2024; PMID 38996458; PMCID PMC11258540; DOI 10.1016/j.molcel.2024.06.013

研究问题 / 背景

RNA consists of four different nucleotides (A, U, G, and C) that can be modified to over 170 different chemical moieties.

1 Pseudouridine (Ψ), also known as 5-ribosyluracil, is the most abundant RNA modification, affecting approximately 0.5% of all uridines.

2 The formation of Ψ requires the cleavage of the N1–C1' glycosidic bond of uridine, followed by a rotation of the base and its re-attachment to...

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.13 材料方法与结果: The molecular basis of t RNA selectivity by human pseudouridine synth...

Source: Molecular cell (2024) | experimental | DOI 10.1016/j.molcel.2024.06.013

材料 / 方法 / 分析路线

The codon-optimized sequence of a synthetic human PUS1 (Q9Y606) ORF or PUS3 (Q9BZE2) ORF fused with a glutathione- S -transferase (GST)-tag...

PUS1 and PUS3 mutants were generated using the standard mutagenesis PCR with primers that contain the mutation site sequence.

PUS1 was recombinantly expressed in the strain BL21 (DE3) Codon Plus-RIL cells.

Transformed cells were grown in LB broth and protein expressions were induced by isopropyl β -D-1-thiogalactopyranoside IPTG (1 m M) for overnigh...

核心结果

Pseudouridine (Ψ), the isomer of uridine, is ubiquitously found in RNA, including t RNA, r RNA, and m RNA.

Human pseudouridine synthase 3 (PUS3) catalyzes pseudouridylation of position 38/39 in t RNAs.

However, the molecular mechanisms by which it recognizes its RNA targets and achieves site specificity remain elusive.

Here, we determine single-particle cryo-EM structures of PUS3 in its apo form and bound to three t RNAs, showing how the symmetric PUS3 homodime...

Structure-guided and patient-derived mutations validate our structural findings in complementary biochemical assays.

3.13 讨论与局限： The molecular basis of t RNA selectivity by human pseudouridine synthas...

Source: Molecular cell (2024) | experimental | DOI 10.1016/j.molcel.2024.06.013

讨论与局限

PUS enzymes are ubiquitous across the tree of life and catalyze the most abundant RNA modification that regulates splicing rates, 29 m RNA half-...
Here, we present experimental evidence that PUS3 is unique among human PUS enzymes due to its dimerization, which is key to its mechanism of sub...
Our structural, biochemical, and mutational analyses provide a framework for understanding the PUS3-mediated modification mechanism and its link...
Importantly, we show that PUS3 targets t RNAs, whereas PUS1 (like other PUS enzymes) modifies t RNAs as well as m RNAs.

可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

3.14 文献概览： Perturbation of METTL1-mediated t RNA N7- methylguanosine modification ind...

2024 | Nature Communications | 2024 | Experimental | DOI 10.1038/s41467-024-49796-8

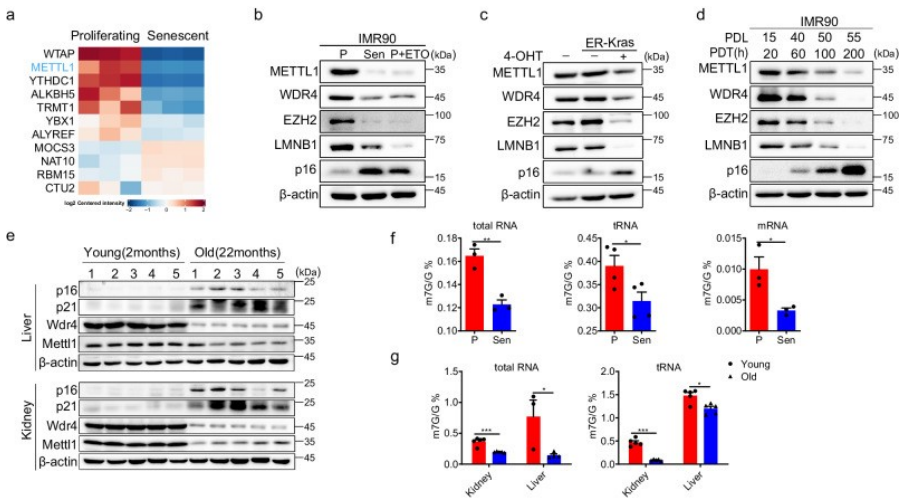
研究问题 / 背景

Cell senescence is characterized by permanent growth arrest while remaining metabolically active 1 .

Accumulation of senescent cells in tissues causes organism aging 2 – 5 .

It has been reported that protein synthesis capacity decreases with senescence and aging 6 – 8 .

与本子项目的关系： 该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



3.14 材料方法与结果: Perturbation of METTL1-mediated t RNA N7- methylguanosine modification...

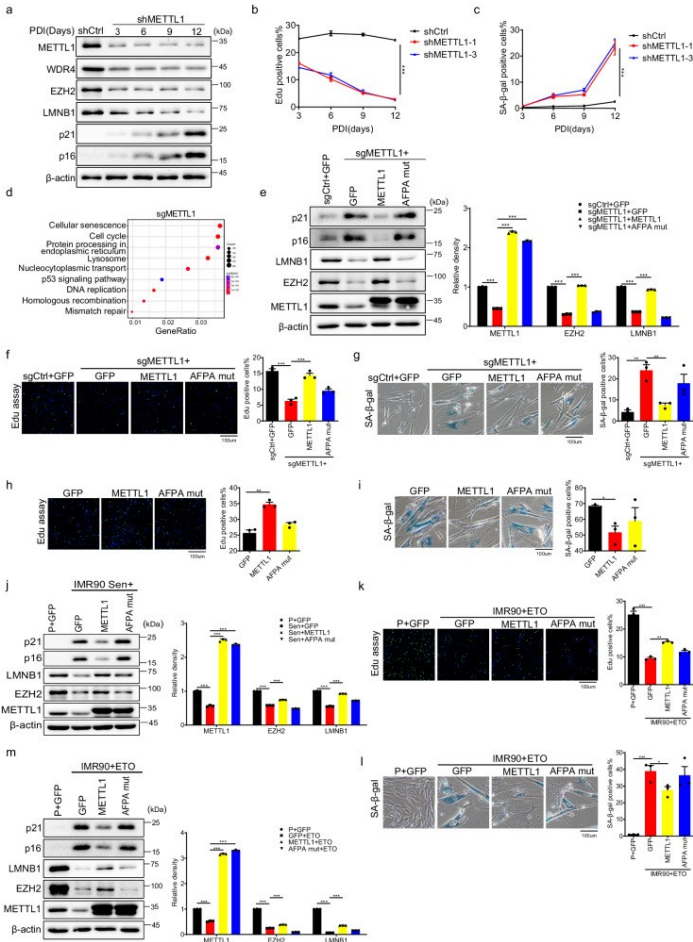
2024 | Nature Communications (2024) | experimental | DOI 10.1038/s41467-024-49796-8

材料 / 方法 / 分析路线

Description of Additional Supplementary Files

核心结果

in a subset of t RNAs being targeted for RTD degradation due to m7G46 hypomethylation. Knockdown of METTL1 drives pro- liferating cells into senescence and significantly shortens the lifespan of mice. Overexpression of METTL1 delays cell senescence and miti- gates tissue damage induced by chemotherapy agents. METTL1 defi- ciency leads to ribosome stalling at specific codons, triggering ribotoxic stress response(RSR) and integrated stress response(ISR). This suppresses global translation initiation and elongation while activating the senescence-associated secretory phenotype (SASP).



3.14 讨论与局限: Perturbation of METTL1-mediated t RNA N7- methylguanosine modification...

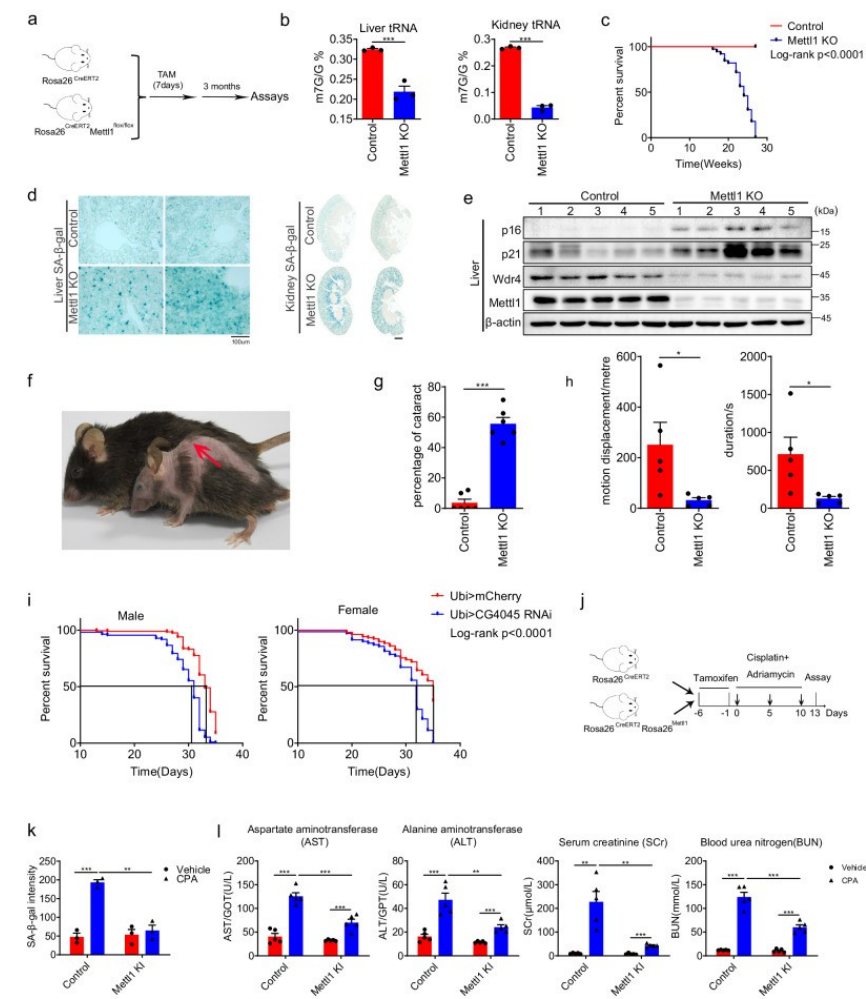
讨论与局限

In the study, we demonstrated the essential role of METTL1-mediated t RNA m7G modification in protein synthesis and its relevance in senescence...

As previously reported, a group of RNAs was recognized as substrates for METTL1 43 , 86 – 89 .

PAR-CLIP revealed that 31.8% of METTL1 binding RNAs belong to t RNAs, which are the most abundant RNA species in the assay 90 .

Our experiments focused on t RNAs to explore the mechanisms of METTL1 mediated senescence.



可用于本项目的提炼

与本子项目的关系：该文可转化为修饰位点、实验条件、检测技术和证据等级字段。

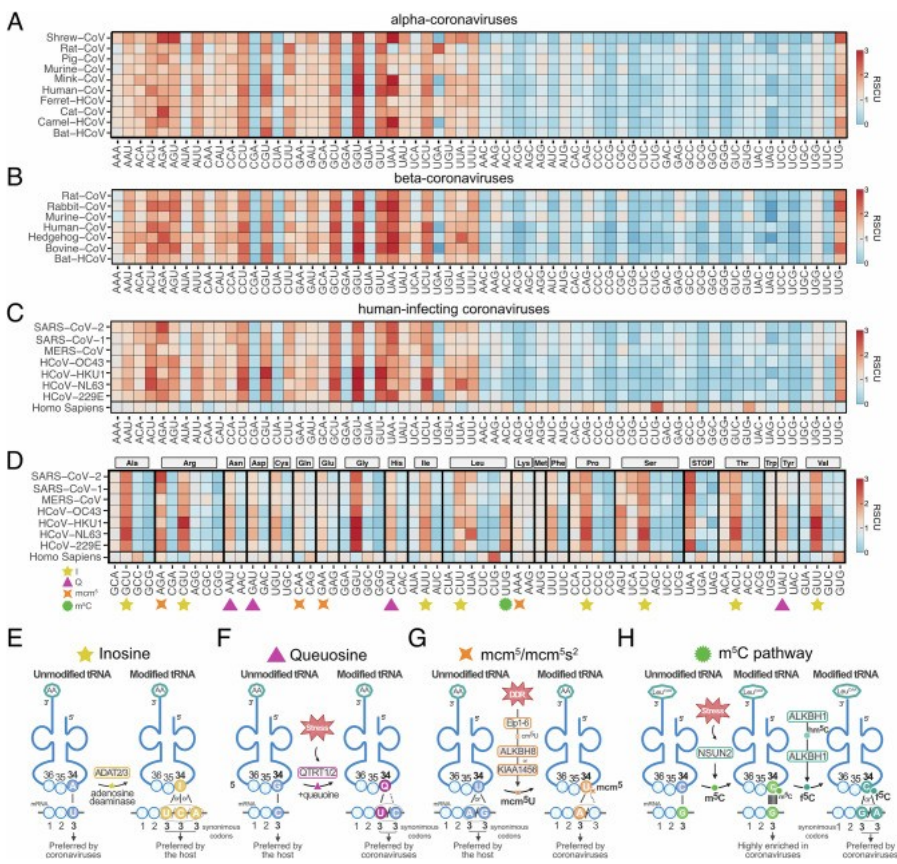
3.15 文献概览: Coronaviruses reprogram the t RNA epitranscriptome to favor viral protein...

Source: Nature communications (2026) | experimental | DOI 10.1038/s41467-026-69700-w

研究问题 / 背景

Coronaviruses pose a major global health threat due to their ability to cross species barriers and establish infections in humans 1 .
These zoonotic transmissions, exemplified by the COVID-19 pandemic, can give rise to novel viral strains with unpredictable consequences 2 .
Gaining insights into how coronaviruses exploit host cell machineries to multiply is crucial for the development of novel antiviral approaches.

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3.15 材料方法与结果: Coronaviruses reprogram the t RNA epitranscriptome to favor viral pro...

Nature communications (2026) | experimental | DOI 10.1038/s41467-026-69700-w

材料 / 方法 / 分析路线

Description of Additional Supplementary Files

核心结果

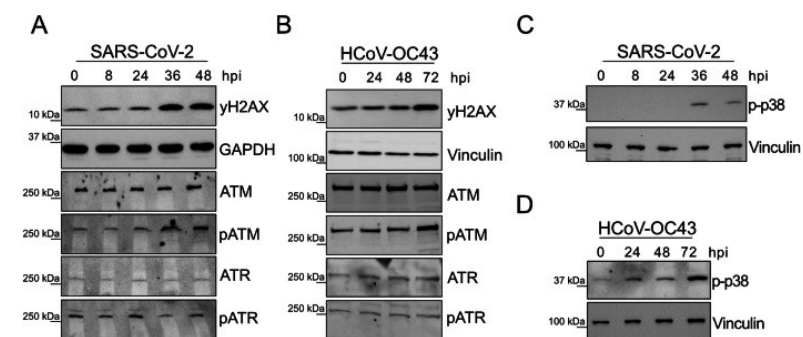
from wobble base pairing between the third position of the codon in the m RNA and the first position of the anticodon in the t RNA (N34), known a...

t RNAs deliver the amino acids to the ribosome by pairing their anticodons (located at positions 34, 35 and 36) with complementary codons in m...

While positions 1 and 2 follow strict Watson-Crick base pairing rules, the third position allows for non-standard base pairing, which can be fur...

For example, uridine at position 34 (U34) can pair with both adenine (A) and guanine (G).

Chemical modifications at U34 modulate this flexibility, either enhancing or restricting non-canonical base pairing.



讨论与局限

Our study reveals that coronavirus infection triggers a profound alteration of the t RNA modification landscape that aligns with the viral codon...

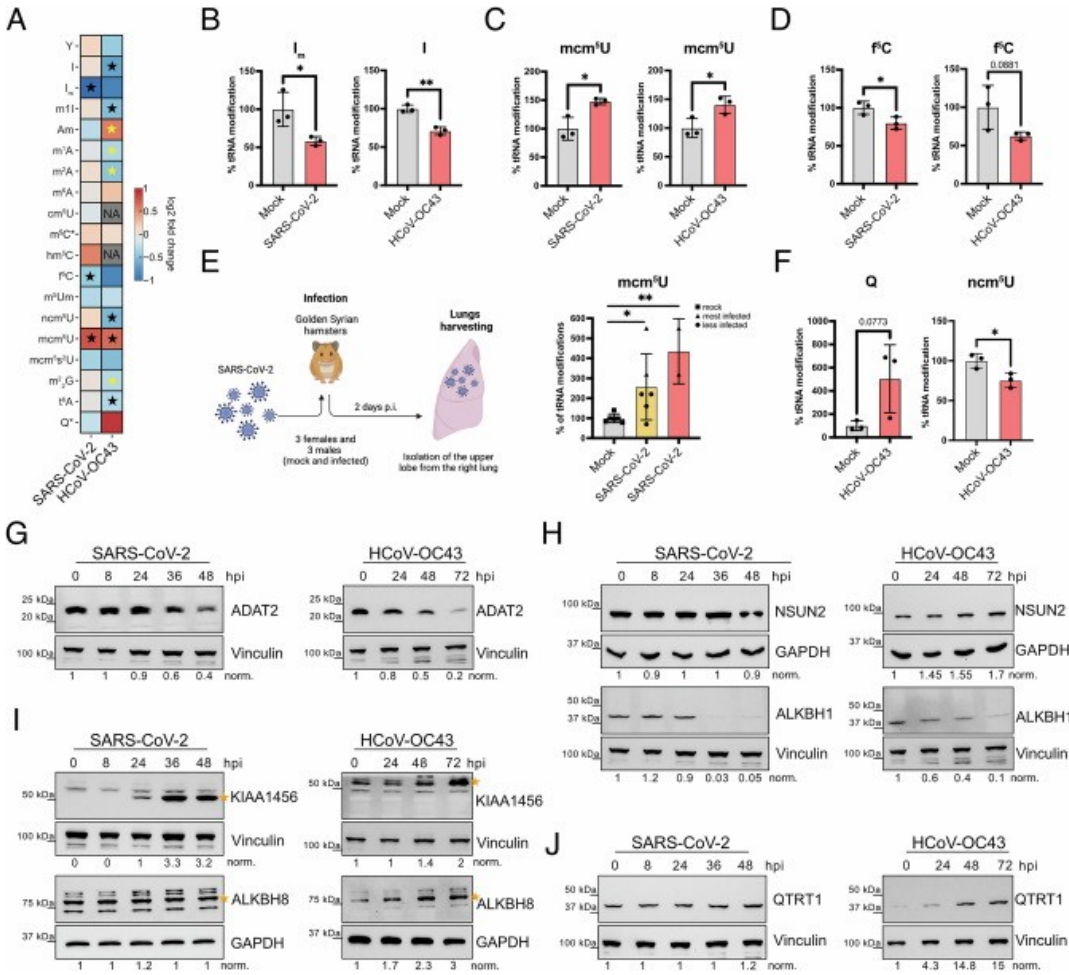
Beyond coronaviruses, many RNA viruses have A/U-rich genomes, a bias suggested to arise from multiple selective pressures.

For example, A/U-rich sequences may enhance RNA structural flexibility, thereby facilitating transitions between different stages of the viral l...

Moreover, the underrepresentation of Cp G dinucleotides likely reflects an evolutionary strategy to evade immune detection 58 .

可用于本项目的提炼

与本子项目的关系： 该文可转化为修饰位点、实验条件、检测技术和证据等级字段。



Part 4. tRNA 相关湿实验

tRNA-related wet-lab experiments

本部分包含 15 篇文献，每篇 3 页。

1. RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.
2. AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.
3. Design, construction, and functional characterization of a t RNA neochromosome in yeast.
4. Quantitative analysis of t RNA abundance and modifications by nanopore RNA sequencing.
5. An engineered UGA suppressor t RNA gene for disease-agnostic AAV delivery.
6. Quantifying t RNA abundance by sequencing.
7. Translational regulation by oxidative desulfuration of t RNA modifications.
8. Rel A-Spo T Homolog toxins pyrophosphorylate the CCA end of t RNA to inhibit protein synthesis.
9. A ligation-independent sequencing method reveals t RNA-derived RNAs with blocked 3' termini.
10. t RNA m1A modification regulate HSC maintenance and self-renewal via m TORC1 signaling.
11. A methyltransferase-independent role for METTL1 in t RNA aminoacylation and oncogenic transfor...
12. Nanopore sequencing of intact aminoacylated t RNAs.
13. t RNA modification dynamics from individual organisms to metaepitranscriptomics of microbiomes.
14. High-resolution quantitative profiling of t RNA abundance and modification status in eukaryote...
15. In vivo structure profiling reveals human cytosolic and mitochondrial t RNA structurome and in...

4.1 文献概览： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.

Nature (2026) | Experimental | DOI 10.1038/s41586-025-09852-9

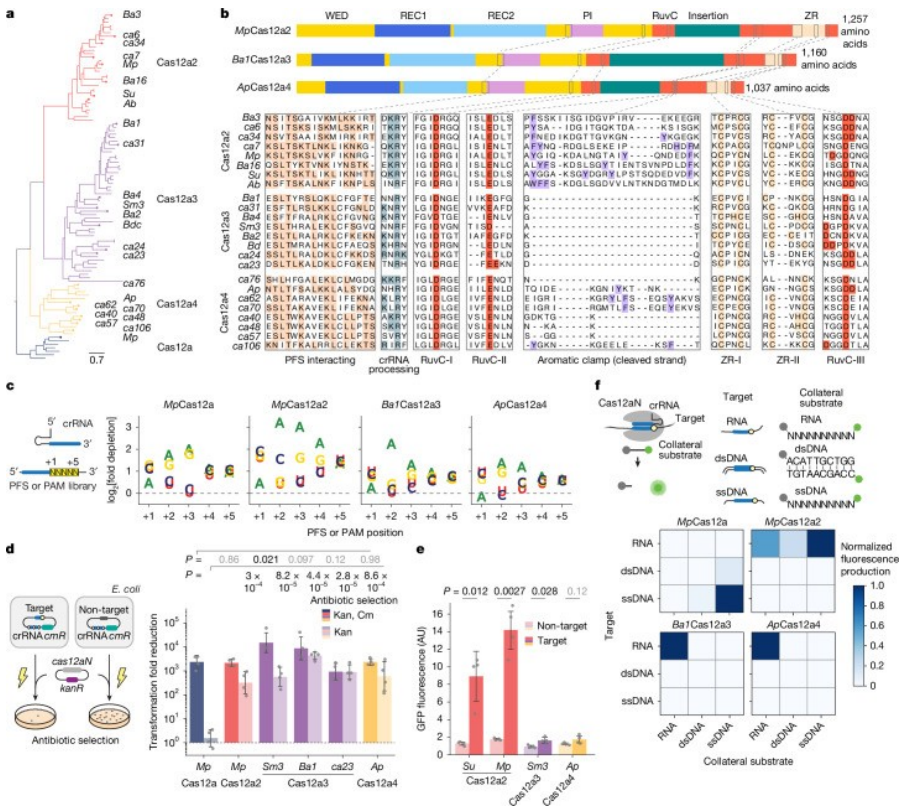
研究问题 / 背景

In all domains of life, t RNAs mediate the transfer of genetic information from m RNAs to proteins.

As their depletion suppresses translation and, consequently, viral replication, t RNAs represent long-standing and increasingly recognized target...

Here we report Cas12a3 effector nucleases from type V CRISPR-Cas adaptive immune systems in bacteria that preferentially cleave t RNAs after rec...

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



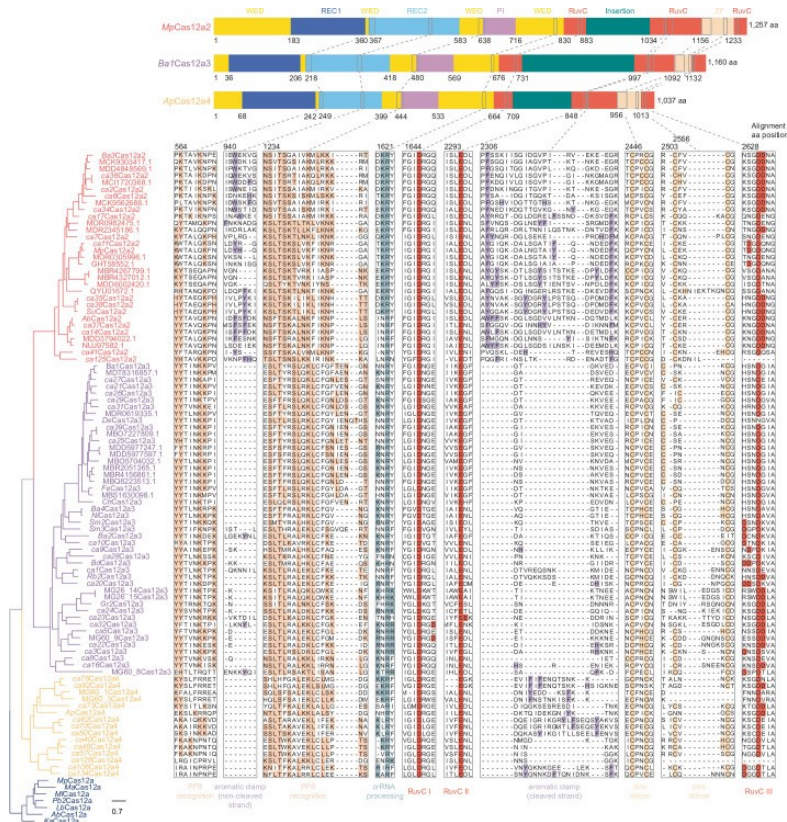
4.1 材料方法与结果： RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immuni...

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting summa- ries, source data, extended data, supplementary information, acknowl- edgements, peer...
The making and breaking of t RNAs by ribonucleases.
Direct activation of a bacterial innate immune system by a viral capsid protein.
Phage T4-coded Stp: double-edged effector of coupled DNA and t RNA-restriction systems.

核心结果

are the averages of independent experiments (n = 2).
Note that the PAM for Mp Cas12a is the reverse complement of the sequence commonly reported for Cas12a nucleases (YYV).
d, Schematic (left) and quantification (right) of the assessment of plasmid clearance versus growth arrest in E.
coli based on variations of a plasmid interference assay.
Plasmid clearance and growth arrest were differentiated on the basis of antibiotic selection for the target plasmid or the target plasmid and th...



4.1 讨论与局限: RNA-triggered Cas12a3 cleaves t RNA tails to execute bacterial immunity.

Nature (2026) | Experimental | DOI 10.1038/s41586-025-09852-9

讨论与局限

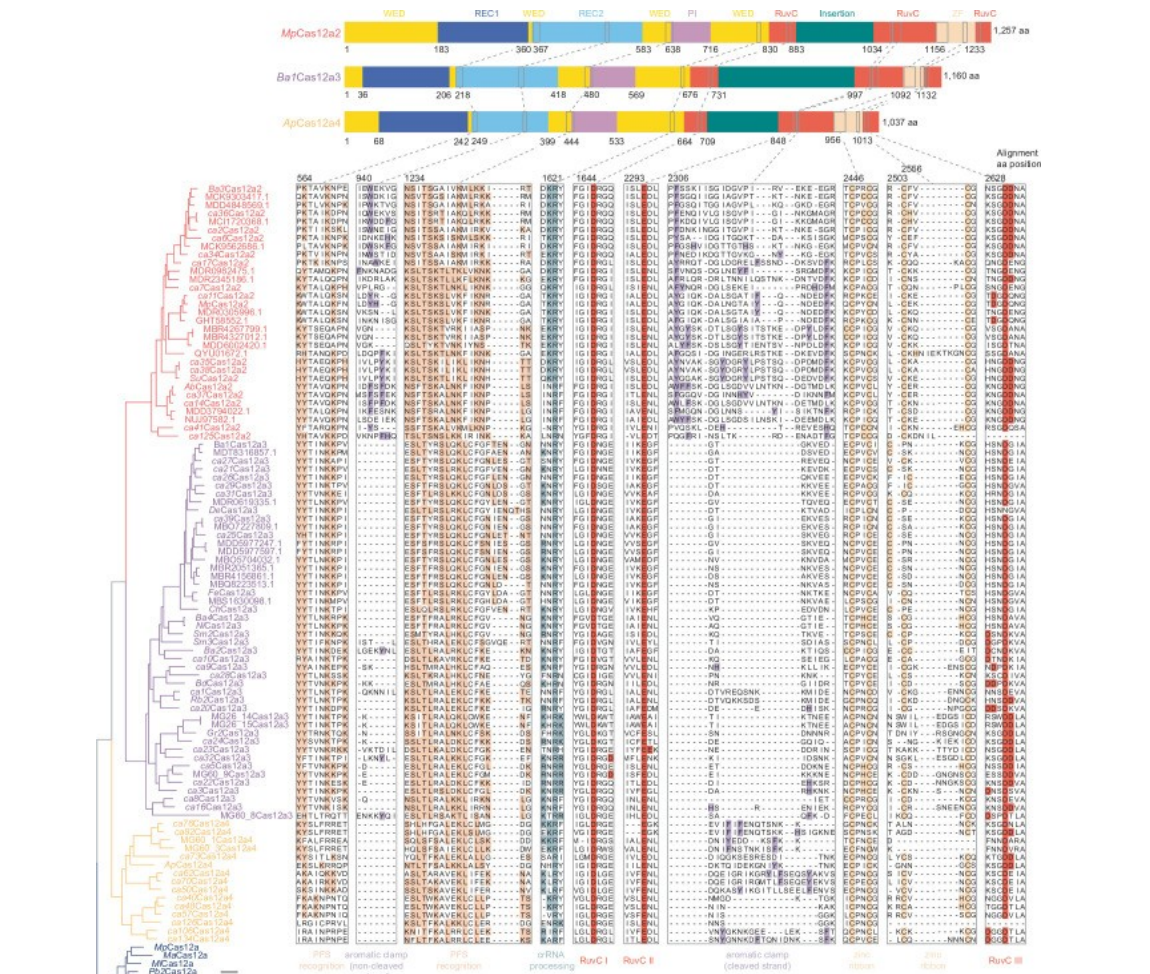
In this work, we have reported the discovery of cr RNA-guided Cas12a3 nucleases that recognize complementary RNA targets and, in response, cleav...

Integrating genetic, biochemical, sequencing and structural studies, we propose a model (Extended Data Fig. 9) in which the nuclease undergoes a large conformational change that can then bind free t RNAs through multiple sequence-specific and shape-spe...

The unique t RLD in the nuclease positions the t RNA tail next to the Ruv C active site to cleave off the tail, with extensive cleavage of cellu...

可用于本项目的提炼

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



4.2 文献概览： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

doi:10.1038/s41586-022-04533-3 | experimental | DOI 10.1038/s41586-022-04533-3

研究问题 / 背景

Gene therapy is a potentially curative medicine for many currently untreatable diseases, and recombinant adeno-associated virus (r AAV) is the m...

However, r AAV-based gene therapy suffers from several limitations, such as constrained DNA cargo size and toxicities caused by non-physiological...

Here we show that r AAV delivery of a suppressor t RNA (r AAV.sup-t RNA) safely and efficiently rescued a genetic disease in a mouse model carrying...

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

4.2 材料方法与结果： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

4.2 Nature 2022 | experimental | DOI 10.1038/s41586-022-04533-3

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

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Mechanistically, this was achieved through a synergistic effect of premature stop codon readthrough and inhibition of nonsense-mediated m RNA de...

r AAV.sup-t RNA had a limited effect on global readthrough at normal stop codons and did not perturb endogenous t RNA homeostasis, as determined...

4.2 讨论与局限： AAV-delivered suppressor t RNA overcomes a nonsense mutation in mice.

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讨论与局限

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4.3 文献概览: Design, construction, and functional characterization of a t RNA neochromosome...

Cell 2023; DOI 10.1016/j.cell.2023.10.015

研究问题 / 背景

Here, we report the design, construction, and characterization of a t RNA neochromosome, a designer chromosome that functions as an additional,...
Intending to address one of the central design principles of the Sc2.0 project, the ~190-kb t RNA neochromosome houses all 275 relocated nuclear...
To maximize stability, the design incorporates orthogonal genetic elements from non-S.

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

材料 / 方法 / 分析路线

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

核心结果

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Furthermore, the presence of 283 rox recombination sites enables an orthogonal t RNA SCRa Mb LE system.

Following construction in yeast, we obtained evidence of a potent selective force, manifesting as a spontaneous doubling in cell ploidy.

4.3 讨论与局限: Design, construction, and functional characterization of a t RNA neochr...

Cell 2023 | experimental | DOI 10.1016/j.cell.2023.10.015

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可用于本项目的提炼

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

4.4 文献概览： Quantitative analysis of t RNA abundance and modifications by nanopore RNA...

DOI 10.1038/s41587-023-01743-6

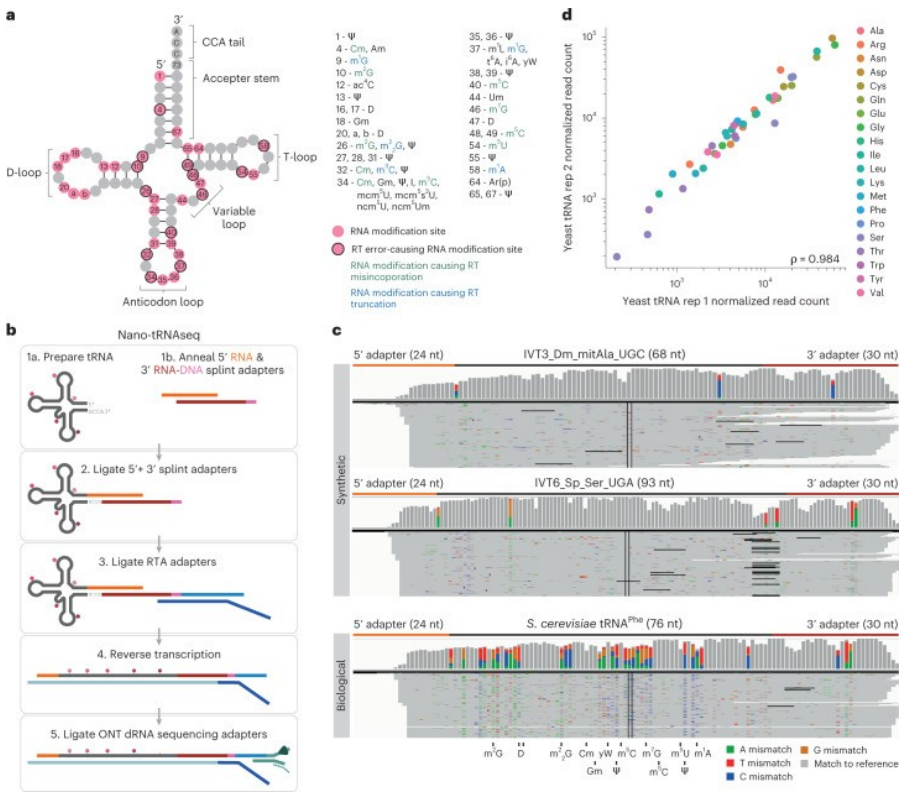
研究问题 / 背景

of both m5U54 and m1A58, and m5U54 positively influencing the intro- duction of m1A58.

To explore whether our method could capture the effect of Ψ55 loss on other modifications, we examined the summed basecalling error (base mismat...

4 | Nano-t RNAseq can quantify t RNA abundance and RNA modifications as well as capture modification interdependencies.

与本子项目的关系： 该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



4.4 材料方法与结果: Quantitative analysis of t RNA abundance and modifications by nanopor...

材料 / 方法 / 分析路线

A set of nine synthetic in vitro transcribed (IVT) t RNAs of various lengths and sequences (Supplementary Table 1) were sequenced using this str...

However, this approach produced poor sequencing yields (56,002 reads; Supplementary Table 2) compared to a standard DRS run (~1–2 million reads).

Moreover, only 7.5% of reads mapped uniquely to t RNAs using minimap2 (ref. 100) with recommended parameters (-ax map -ont -k15) (Supplementary Table 2).

Source: Nature biotechnology 2024; PMID 37024678; PMCID PMC10791586; DOI 10.1038/s41587-023-01743-6

核心结果

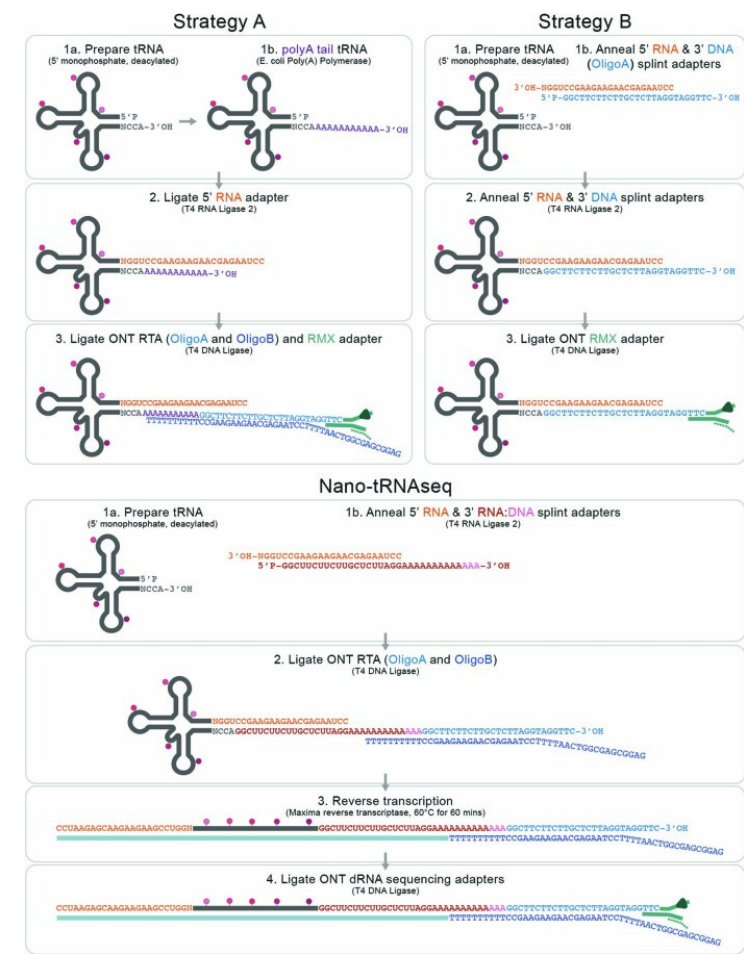
Standard nanopore DRS results in low t RNA sequencing yields Nanopore DRS is a well-established long-read sequencing technology to study RNA m...

Although several works have shown that this technology can also be used to study short RNA molecules, such as sno RNAs and sn RNAs96,98, DRS i...

In addition, the first ~15 nt at the 5' end of RNA molecules are typically lost in DRS runs93,99, as this portion cannot be adequately basecalled...

To overcome these limitations, we reasoned that extension of the 5' and 3' ends of the t RNA would lead to improved sequencing of t RNA molecule...

We first attempted a modified t RNA DRS approach in which a 5' RNA adapter, complementary to the 3' CCA overhang present in mature t RNA molecule...



4.4 讨论与局限: Quantitative analysis of t RNA abundance and modifications by nanopore...

讨论与局限

For many years, t RNAs and their modifications have been primarily viewed as static contributors to gene expression and t RNA struc- ture127–130.

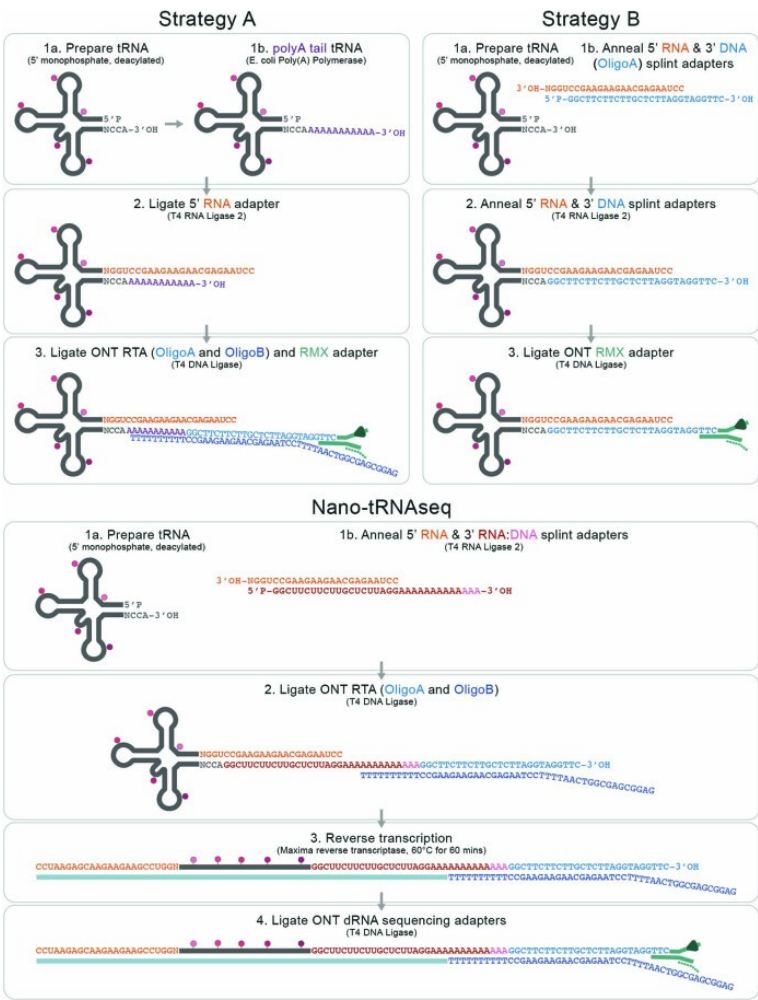
However, multiple studies have shown that t RNA abundances and modification profiles are, in fact, dynamic and can differ in distinct cellular e...

Measuring both t RNA abun- dances and their modifications with single-transcript resolution has not been feasible due to a lack of available met...

This has been a substantial limitation in moving forward with studying the biological function and dynamics of t RNA populations and their modif...

可用于本项目的提炼

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



4.5 文献概览： An engineered UGA suppressor t RNA gene for disease-agnostic AAV delivery.

Source: Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-025-02982-5

研究问题 / 背景

Suppressor transfer RNAs (sup-t RNAs) have the potential to rescue nonsense mutations in a disease-agnostic manner and are an alternative therap...

Among all human pathogenic nonsense variants, approximately 20% arise from C-to-T transitions that convert the CGA arginine codon into a UGA sto...

While recombinant adeno-associated virus (r AAV) has been successfully used to deliver a UAG-targeting sup-t RNA gene in vivo, extending this ap...

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

nature biotechnology

Article

<https://doi.org/10.1038/s41587-025-02982-5>

An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery

Received: 8 October 2025

Accepted: 10 December 2025

Published online: 19 January 2026

Check for updates

Mengyao Xu^{1,6}, Hao Liu^{1,6}, Jiaming Wang^{1,7}, Andre F. C. Vieira^{1,8,9}, Xuntao Zhou¹, Nan Liu¹, Jialing Liang¹, Ailing Du¹, Xiupeng Chen¹, Ruxiao Xing¹, Yang Yang¹, Maria P. Gonzalez-Perez^{1,10}, Vikas Kumar^{1,11} & Dan Wang^{1,12}✉

Suppressor transfer RNAs (sup-tRNAs) have the potential to rescue nonsense mutations in a disease-agnostic manner and are an alternative therapeutic approach for many rare and ultrarare disorders. Among all human pathogenic nonsense variants, approximately 20% arise from C-to-T transitions that convert the CGA arginine codon into a UGA stop codon. While recombinant adeno-associated virus (rAAV) has been successfully used to deliver a UAG-targeting sup-tRNA gene in vivo, extending this approach to UGA-targeting sup-tRNA genes has posed unique challenges related to rAAV vector production. Here, we demonstrate that an engineered UGA-sup-tRNA gene, designed with transcriptional regulatory elements, can be efficiently packaged into rAAV for in vivo delivery. A single administration in mouse models of two distinct lysosomal storage disorders restores enzymatic activity to approximately 10% of normal levels. Comparative analysis reveals differential sup-tRNA expression and aminoacylation patterns across tissue types, which correlate with enhanced therapeutic effects. The applied rAAV-based agents and engineering strategies expand the potential therapeutic scope of sup-tRNA therapies.

Nonsense mutations are point mutations that convert a sense codon into a premature termination codon (PTC) in the mRNA, causing two deleterious events during translation: nonsense-mediated mRNA decay (NMD) and cessation of full-length protein synthesis. Nonsense suppression^{1–4}, also known as stop codon readthrough therapy, allows translation machineries to decode the PTC during protein synthesis as they normally decode sense codons, thereby overcoming its detrimental effects without changing the DNA or mRNA sequences. As nonsense mutations account for approximately 11% of human pathogenic mutations, nonsense suppression has the potential to treat a large population in a gene-agnostic and disease-agnostic manner. This is an attractive feature considering the challenges associated with developing bespoke gene therapies or mutation-specific gene editing therapies for individual rare and ultrarare diseases with a small population^{5,6}.

PTCs, like normal stop codons in mammalian cells, can be broadly categorized into three classes: UAA, UAG and UGA. By mutating the anticodon of a human lysine transfer RNA (tRNA) for decoding UAG, Kan and colleagues generated a suppressor tRNA (sup-tRNA) and demonstrated that it could readthrough a UAG PTC underlying β⁰ thalassemia, partially restoring the full-length β-globin protein in *Xenopus* oocyte⁷. Since then, a body of literature has documented the effectiveness of sup-tRNAs in overcoming various pathogenic nonsense mutations in cell culture models⁸. Recent breakthroughs in developing in vivo sup-tRNA therapy benefited from the nucleic acid delivery technologies that have advanced in the gene therapy field^{9–14}. Albers and colleagues showed that lipid nanoparticles (LNPs) could deliver an in vitro transcribed UGA-targeting sup-tRNA (UGA-sup-tRNA) to the mouse liver and lung and mediated efficient readthrough of the PTC

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Nature Biotechnology

Source: Nature biotechnology 2026; PMID 41555020; PMCID NA; DOI 10.1038/s41587-025-02982-5

4.5 材料方法与结果: An engineered UGA suppressor tRNA gene for disease-agnostic AAV delivery

1 | nature biotechnology (2026) | experimental | DOI 10.1038/s41587-025-02982-5

材料 / 方法 / 分析路线

, additional references, Nature Portfolio reporting
sum- maries, source data, extended data,
supplementary information, acknowledgements, peer
r...

Therapeutics based on stop codon readthrough.

Suppression of nonsense mutations by new emerging
technologies.

Targeting shared molecular etiologies to accelerate
drug development for rare diseases.

核心结果

, h NFS-R8 and m NFS-R8 restored higher GLuc
activity in serum than U6-R8, which
outperformed U6-R7 and U6-R17 (Fig.

Quantifi- cation of the EGFP-FLAG protein in
liver lysates by western blot more consistently
showed the superior performance of h/m NFS-
R8 over...

1d), which was corroborated by
immunohistochemistry analysis with liver tissue
sections (Fig.

Taken together, these results identified R8 in NFS
as the leading can- didate to readthrough UGA
PTC and we chose to focus on 2×h NFS-R8 in
the...

Silencing UGA-sup-t RNA expression to improve
r AAV production We previously showed that
standard triple transfection in HEK293 cells
resulted i...

nature biotechnology



Article

<https://doi.org/10.1038/s41587-025-02982-5>

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Nature Biotechnology

4.5 讨论与局限： An engineered UGA suppressor t RNA gene for disease-agnostic AAV delivery...

Source: Nature biotechnology (2026) | experimental | DOI 10.1038/s41587-025-02982-5

讨论与局限

Here, we report the development of AAV-No STOP(UGA) that installs arginine at a pathogenic UGA PTC during translation and, therefore, can potent...

It builds on multiple optimization steps to improve PTC readthrough efficiency, r AAV pro- duction yield and vector genome homogeneity.

These efforts led us to demonstrate efficacy in two disease mouse models, underscoring the gene-agnostic and disease-agnostic feature of sup-t R...

Using a mouse model with a nonsense mutation in a native, nuclear gene is a rigorous experimental approach to evaluate in vivo sup-t RNA therapy.

可用于本项目的提炼

与本子项目的关系： 该文可用于设计样本处理、 检测平台、 关键对照和结果验证流程。

Source: Nature biotechnology 2026; PMID 41555020; PMCID NA; DOI 10.1038/s41587-025-02982-5

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Article

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Nature Biotechnology

4.6 文献概览： Quantifying t RNA abundance by sequencing.

4.6 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

研究问题 / 背景

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

4.6 材料方法与结果： Quantifying t RNA abundance by sequencing.

4.6 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

材料 / 方法 / 分析路线

核心结果

全文方法段未能自动稳定抽取；本页基于摘要、题名、图像和可下载全文信息概括实验 / 分析路线。

4.6 讨论与局限： Quantifying t RNA abundance by sequencing.

4.6 | Nature genetics (2023) | experimental | DOI 10.1038/s41588-023-01404-z

讨论与局限

可用于本项目的提炼

与本子项目的关系： 该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

4.7 文献概览: Translational regulation by oxidative desulfuration of t RNA modifications.

4.7 Nature Communications (2025) | experimental | DOI 10.1038/s41467-026-70126-7

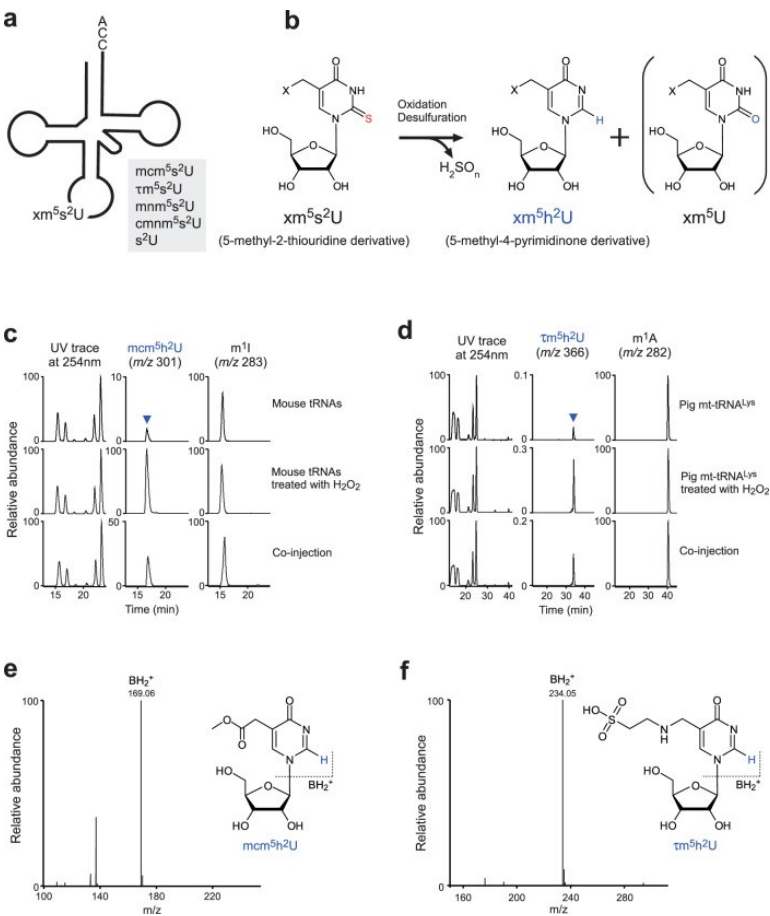
研究问题 / 背景

Higher-order biological processes emerge from sophisticated regulation of gene expression at multiple stages at both transcriptional and post-tr...

In protein synthesis, transfer RNA (t RNA) anticodons recognize and decode messenger RNA (m RNA) codons on the ribosome.

t RNAs undergo the most extensive and diverse range of post-transcriptional modifications, particularly within the anticodon region.

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



4.7 材料方法与结果: Translational regulation by oxidative desulfuration of t RNA modifica...

4.7 | Nature Communications (2026) | experimental | DOI 10.1038/s41467-026-70126-7

材料 / 方法 / 分析路线

Description of Additional Supplementary Files

核心结果

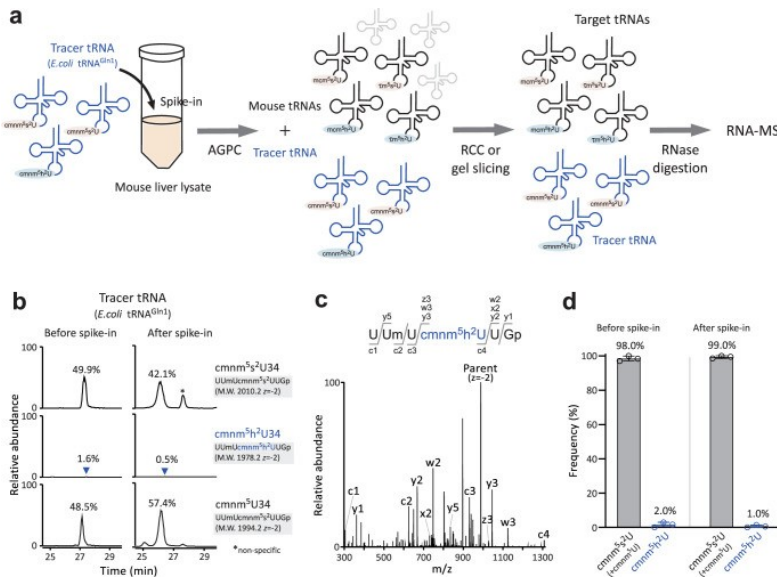
Detection of h2U derivatives in mammalian and bacterial t RNAs We reviewed our previous liquid chromatography-mass spectrometry (LC-MS) data on...

In LC-MS nucleoside analyses of total RNA from mouse livers, we clearly detected unassigned nucleosides with m/ z 301 (Fig.

In addition, we also detected nucleosides with m/z 366 in mitochondrial t RNALys isolated from pig liver (Fig.

Collision- induced dissociation (CID) of these nucleosides generated base- related ions (BH2 +) with loss of unmodified ribose (Fig.

1e, f), sug- gesting that the 32 Da loss originates from their bases.



4.7 讨论与局限: Translational regulation by oxidative desulfuration of t RNA modificati...

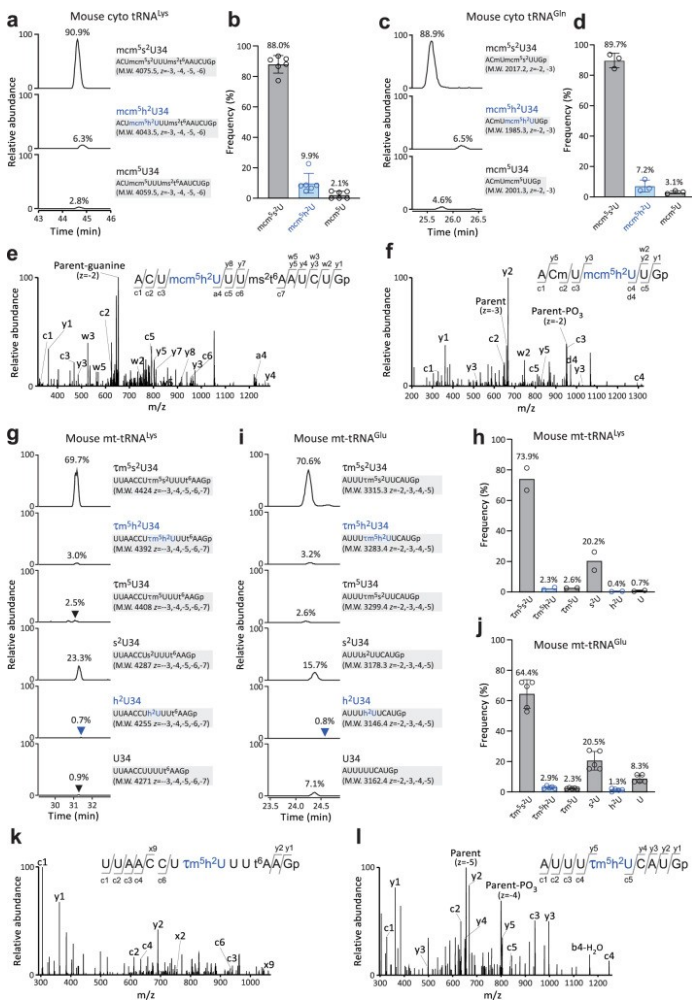
DOI: 10.1038/s41467-026-70126-7 | experimental | DOI 10.1038/s41467-026-70126-7

讨论与局限

Through a spike-in experiment, we demonstrated that 2-thiouridine derivatives in t RNA are desulfurized in cells and converted to h 2 U derivati...

3b, d), and τm 5 h 2 U was detected in only a few percent of mitochondrial t RNAs with τm 5 s 2 U (Fig.

However, the abundance of h 2 U derivatives varies in different cell types. Approximately 15% of t RNA Lys in He La cells contained mcm 5 h 2 U, whereas little was detected in A549 and T24 cells (Supplementary Fig.



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4.8 文献概览： Rel A-Spo T Homolog toxins pyrophosphorylate the CCA end of t RNA to inhib...

Cell Molecular Cell (2021) | experimental | DOI 10.1016/j.molcel.2021.06.005

研究问题 / 背景

Rel A-Spo T Homolog (RSH) enzymes control bacterial physiology through synthesis and degradation of the nucleotide alarmone (p)pp Gpp.

We recently discovered multiple families of small alarmone synthetase (SAS) RSH acting as toxins of toxin-antitoxin (TA) modules, with the Fa Re...

Here we probe the mechanism of growth arrest used by four experimentally unexplored subfamilies of tox SAS: Fa Rel2, Ph Rel, Ph Rel2, and Cap Re...

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4.8 材料方法与结果: Rel A-Spo T Homolog toxins pyrophosphorylate the CCA end of tRNA to...

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材料 / 方法 / 分析路线

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核心结果

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Surprisingly, all these toxins specifically inhibit protein synthesis.

To do so, they transfer a pyrophosphate moiety from ATP to the tRNA 3' CCA.

4.8 讨论与局限： Rel A-Spo T Homolog toxins pyrophosphorylate the CCA end of t RNA to in...

Cell Molecular cell (2021) | experimental | DOI 10.1016/j.molcel.2021.06.005

讨论与局限

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4.9 文献概览： A ligation-independent sequencing method reveals t RNA-derived RNAs with b...

Molecular cell (2024) | experimental | DOI 10.1016/j.molcel.2024.07.008

研究问题 / 背景

Despite the numerous sequencing methods available, the diversity in RNA size and chemical modification makes it difficult to capture all RNAs in...

We developed a method that combines quasi-random priming with template switching to construct sequencing libraries from RNA molecules of any len...

Our ligation-independent detection of all types of RNA (LIDAR) is a simple, effective tool to identify and quantify all classes of coding and no...

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With LIDAR, we comprehensively characterized the transcriptomes of mouse embryonic stem cells, neural progenitor cells, mouse tissues, and sperm.

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DOI 10.1016/j.molcel.2024.07.008 | experimental | DOI 10.1016/j.molcel.2024.07.008

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4.10 文献概览: t RNA m1A modification regulate HSC maintenance and self-renewal via mTOR...

via mTOR... | Nature communications (2024) | experimental | DOI 10.1038/s41467-024-50110-9

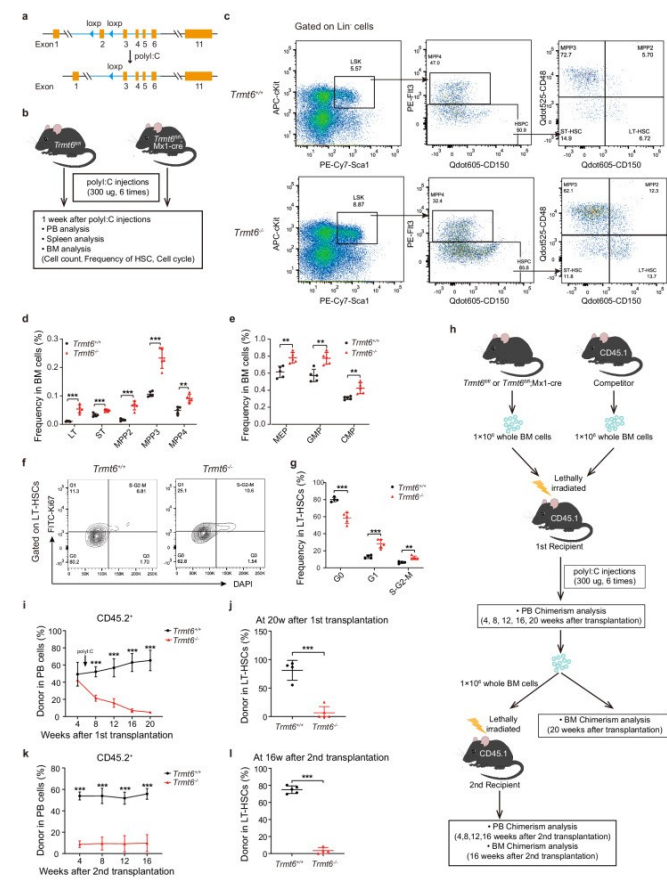
研究问题 / 背景

Haematopoietic stem cells (HSCs) are a rare population of bone marrow-resident adult stem cells that have an extensive capacity for self-renewal...

In order to maintain lifelong stem cell function and cope with various internal and external stresses such as replicative, metabolic, oxidative...

While considerable insights have been generated through studies targeting specific genes or pathways in HSCs 4 , 6 , 9 , 14 – 21 , the mechanism...

与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。



4.10 讨论与局限: t RNA m1A modification regulate HSC maintenance and self-renewal via m...

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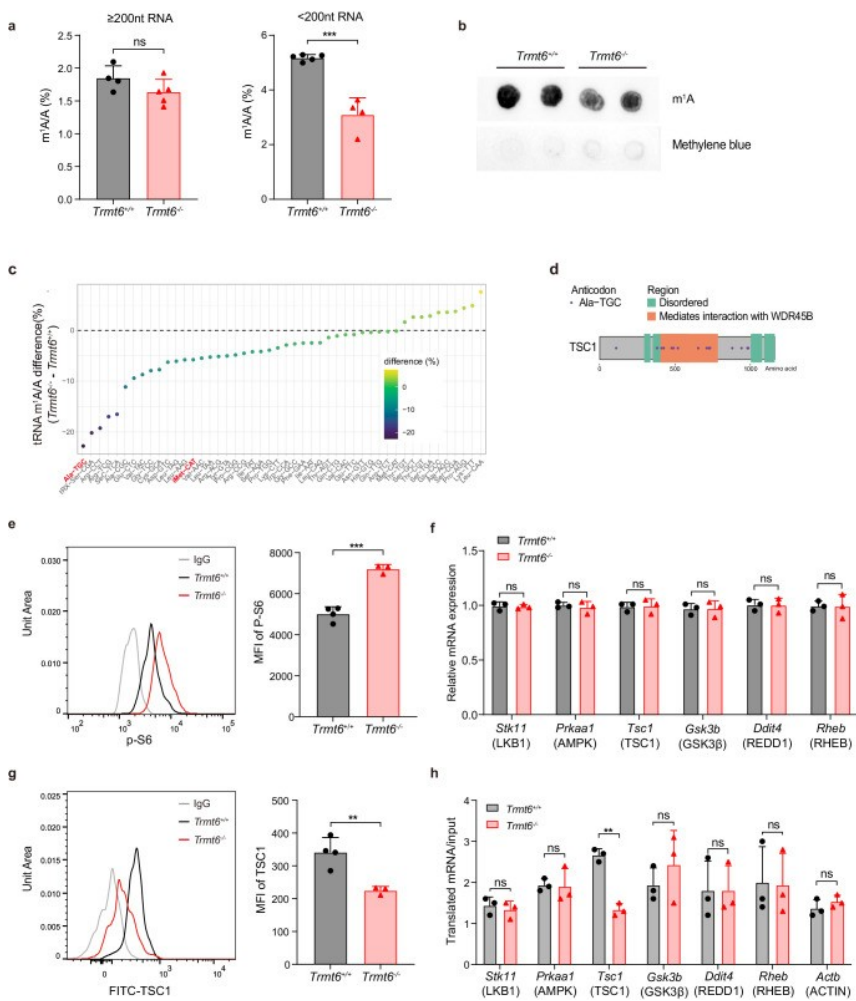
讨论与局限

Maintaining the homeostasis and function of adult stem cells is essential to replenish somatic cells lost under physiological or pathological co...

How transcription and translation of HSCs are controlled is a central issue in haematopoietic homeostasis.

Both the Wnt signaling pathway and the m TOR signaling pathway play important roles in the homeostasis and regeneration of HSC 3 , 8 , 17 , 24 , ...

In our previous study, we found that the histone deacetylase Sirt6 8 and RNA m 6 A reader Ythdf2 20 , maintains the function of HSCs by regulati...



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4.11 文献概览: A methyltransferase-independent role for METTL1 in t RNA aminoacylation an...

Source: Molecular cell 2025; PMID 39892392; PMCID PMC11925124; DOI 10.1016/j.molcel.2025.01.003

研究问题 / 背景

Cancer genomes are shaped by mutational processes that change the sequence or structure of DNA, coupled to selection for mutations that increase...

1 The identification of recurrent mutations affecting individual genes led to the discovery of many of the best-known oncogenes and tumor suppre...

2 – 5 However, genetic lesions affecting multiple genes, such as aneuploidy or focal copy-number alterations of chromosomal segments, are highly...

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4.11 材料方法与结果: A methyltransferase-independent role for METTL1 in t RNA aminoacylati...

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材料 / 方法 / 分析路线

Original plasmids used in this manuscript have been submitted to addgene.org (see key resources table for unique identifiers).

We assessed DNA copy number in human 12q-amplified liposarcomas by leveraging single nucleotide polymorphism (SNP) arrays performed on a cohort...

Six of these 50 cases lacked evidence of 12q amplification, and so were excluded from our analysis (cases excluded: PT1DD, PT13DD, PT20DD, PT21D...

Minimal amplicons were defined as a region of genomic amplification, defined as log2 DNA copy number ratio ≥ 1 , in probes spanning at least 100...

核心结果

Amplification of chromosomal material derived from 12q13-15 is common in human cancer and believed to result in overexpression of multiple colla...

To define the oncogenes involved, we overexpressed genes recurrently amplified in human liposarcoma using a zebrafish model of the disease.

We found several genes whose overexpression collaborated with AKT in sarcomagenesis, including the t RNA methyltransferase METTL1.

This was surprising, because AKT phosphorylates METTL1 to inactivate its enzymatic activity.

Indeed, phosphomimetic S27D or catalytically dead alleles phenocopied the oncogenic activity of wild-type METTL1.

4.11 讨论与局限: A methyltransferase-independent role for METTL1 in t RNA aminoacylation...

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讨论与局限

The oncogenes driving selection for some of the most common copy-number aberrations in human cancer remain incompletely defined.

Here, we used a zebrafish model to define oncogenes within 12q13–15 amplicons, a common aberration in multiple tumor types, including liposarcom...

6 , 12 , 32 We confirmed the long-suspected roles of CDK4 , MDM2 , and HMGA2 as bona fide liposarcoma oncogenes and also found genes whose contr...

We were most surprised to find that overexpression of METTL1 collaborates with AKT activation in sarcomagenesis because AKT phosphorylates and i...

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与本子项目的关系：该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

4.12 文献概览： Nanopore sequencing of intact aminoacylated t RNAs.

4.12 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

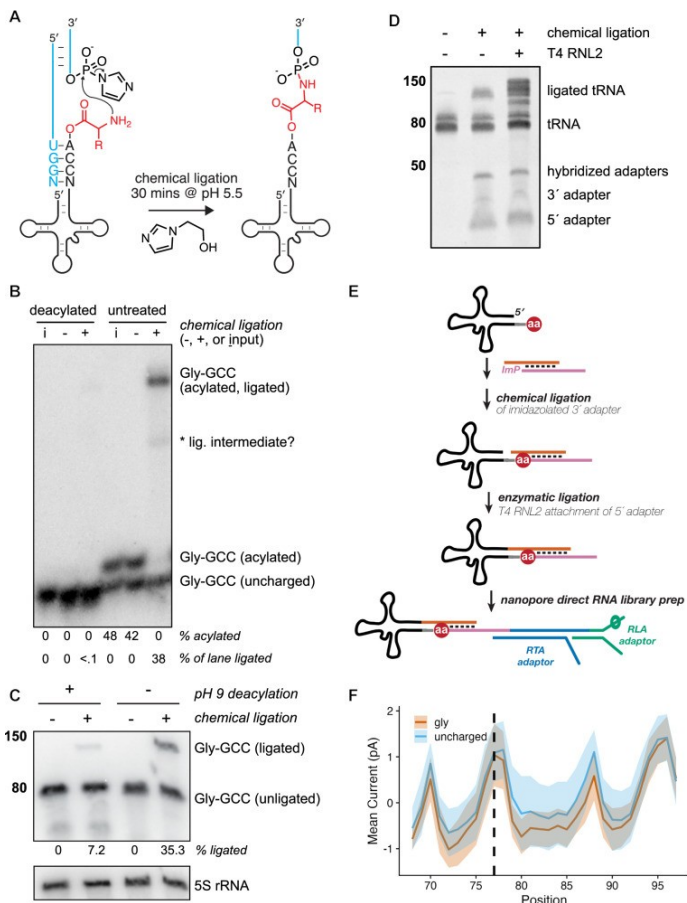
研究问题 / 背景

Transfer RNAs (t RNAs) are the fundamental adaptors of translation, linking m RNA decoding and polypeptide synthesis 1 .

Biogenesis of t RNAs is complex: following transcription, precursor t RNA molecules undergo multiple trimming and modification events, yielding...

t RNA modifications can profoundly impact t RNA function through their impacts on stability 3 , aminoacylation efficiency 4 , 5 , and expanded d...

与本子项目的关系： 该文可用于设计样本处理、检测平台、关键对照和结果验证流程。

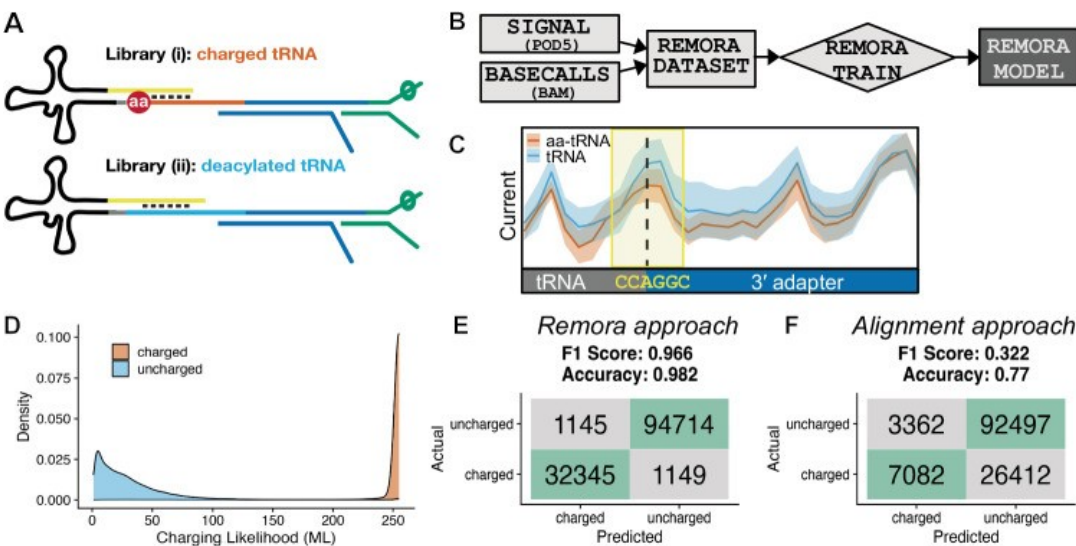


4.12 综述主要论点: Nanopore sequencing of intact aminoacylated t RNAs.

4.12 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

主要论点

A chemical ligation approach enables selective capture of intact aminoacylated t RNAs While investigating prebiotic roles of aminoacyl-RNAs, we... We realized that aminoacyl-t RNAs were ana- logous to these substrates, so we tested and found that a synthetic t RNA-Gly-GCC aminoacylated with... To accelerate the reaction between the α -amino group of the aminoacyl-t RNA and the activated adapter, we included 1-(2- Hydroxyethyl)imidazole... 1c), and subjected *Saccharomyces cerevisiae* t RNA to a 30 min chemical ligation at p H 5.5. Under these conditions, we achieved efficient ligation of aminoacylated budding yeast t RNA as measured by acid northern blot, with <0.1% backgro...



4.12 综合观点与空白： Nanopore sequencing of intact aminoacylated t RNAs.

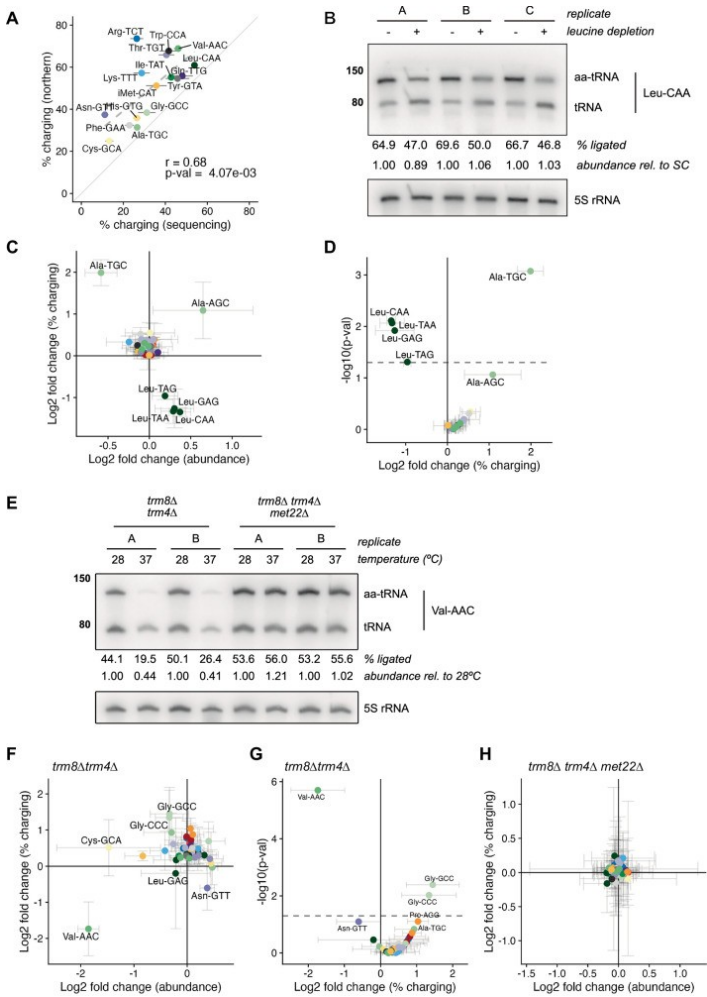
4.12 | Nature communications (2025) | review | DOI 10.1038/s41467-025-62545-9

综合观点与空白

Transfer RNAs are long-lived and undergo repeated cycles of charging and deacylation throughout their lifetime. While t RNA biogenesis is recognized as an intricately coordinated process—spanning transcription, processing, modification, and aminoacylation—... We developed a chemical ligation approach to selectively capture aminoacylated t RNAs and used it to study t RNA aminoacylation in a variety of... Increased accuracy of aa-t RNA identification may further enhance our understanding of how t RNA modifications act as determinants or anti-deter...

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4.13 文献概览： t RNA modification dynamics from individual organisms to metaepitranscript...

Source: Molecular cell 2022; PMID 35032425; PMCID PMC8897278; DOI 10.1016/j.molcel.2021.12.007

研究问题 / 背景

t RNA is the most extensively modified RNA in cells.

On average, a bacterial t RNA contains 8 modifications per molecule and a eukaryotic t RNA contains 13 modifications per molecule.

Recent studies reveal that t RNA modifications are highly dynamic and respond extensively to environmental conditions.

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4.13 综述主要论点: t RNA modification dynamics from individual organisms to metaepitrans...

主要论点

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Functions of t RNA modification dynamics include enhanced, on-demand decoding of specific codons in response genes and regulation of t RNA fragm...

This review summarizes recent advances in the studies of t RNA modification dynamics in biological processes, t RNA modification erasers, and hu...

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4.14 文献概览: High-resolution quantitative profiling of t RNA abundance and modification...

4.14 Molecular cell 2021; DOI 10.1016/j.molcel.2021.01.028

研究问题 / 背景

Transfer RNAs (t RNAs) are short, abundant molecules required for translating genetic information into protein sequences.

The composition of cellular t RNA pools is critical for efficient m RNA decoding and proteome integrity.

t RNA expression is dynamically regulated in different tissues and during development (Dittmar et al., 2006 ; Ishimura et al., 2014 ; Kutter et...

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4.14 材料方法与结果: High-resolution quantitative profiling of t RNA abundance and modific...

2021 | Molecular cell (2021) | experimental | DOI 10.1016/j.molcel.2021.01.028

材料 / 方法 / 分析路线

RNA from *Drosophila* BG3-c2, HEK293T, and human iPS cells was isolated with Trizol (Sigma Aldrich) according to the manufacturer's instructions.

For total RNA isolation from yeast, frozen cells were resuspended in 100 mM sodium acetate pH = 4.5, 10 mM EDTA pH = 8.0, 1% SDS (1 mL per...

An equal volume of hot acid phenol (pH = 4.3) was added, and the cell suspension was vortexed vigorously followed by incubation at 65°C for 5 m...

After addition of 1/10 volume 1-Bromo-3-chloropropane (BCP, Sigma Aldrich), samples were centrifuged at 10,000 x g for 5 min and the aqueous pha...

核心结果

Measurements of cellular tRNA abundance are hampered by pervasive blocks to cDNA synthesis at modified nucleosides and the extensive similarit...

We overcome these limitations with modification-induced misincorporation tRNA sequencing (mim-tRNAseq), which combines a workflow for full-len...

Our method accurately captures tRNA abundance and modification status in yeast, fly, and human cells and is applicable to any organism with a k...

We applied mim-tRNAseq to discover a dramatic heterogeneity of tRNA isodecoder pools among diverse human cell lines and a surprising interdepe...

4.14 讨论与局限: High-resolution quantitative profiling of t RNA abundance and modification...

2021; Molecular cell 2021 | experimental | DOI 10.1016/j.molcel.2021.01.028

讨论与局限

The abundance, charging, and modification status of individual t RNA species can differ in distinct cellular environments.

Measuring these properties on a global scale, however, has not been feasible because of technical limitations.

No library construction method so far has allowed the efficient reverse transcription of these highly modified RNAs, while the lack of computati...

We describe conditions that permit near-complete t RNA modification readthrough by TGIRT, dramatically improving c DNA yield and the fraction of...

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4.15 文献概览: In vivo structure profiling reveals human cytosolic and mitochondrial tRNAs...

2025 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

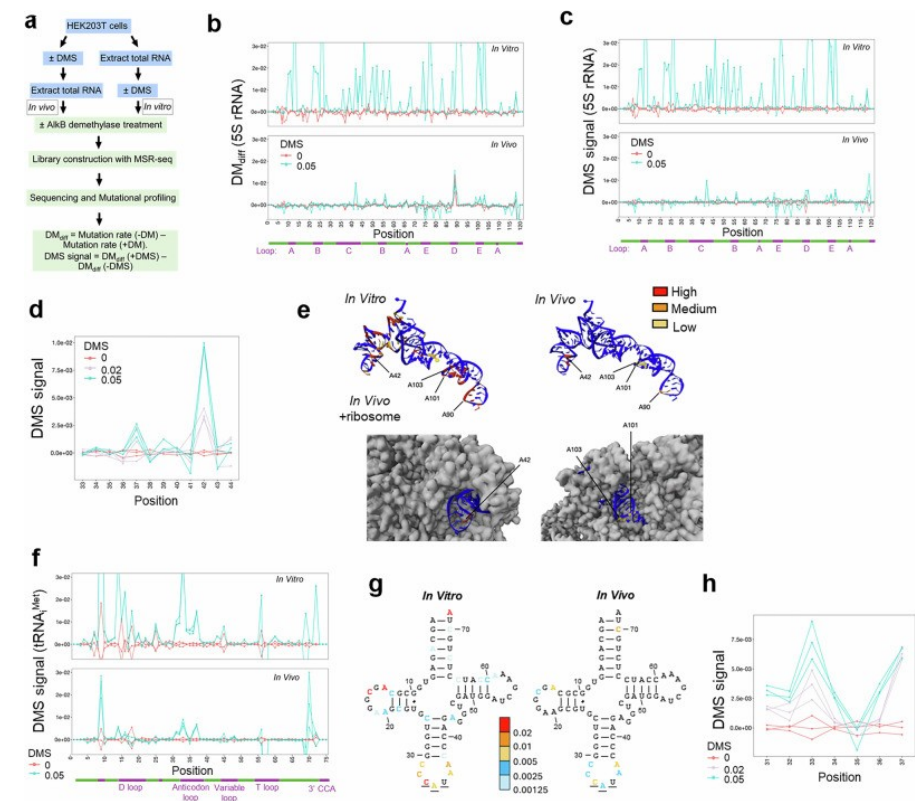
研究问题 / 背景

RNAs are both carriers of genetic information and adapters that mediate biomolecular interactions.

In cells, RNAs fold into secondary and tertiary structures to detect metabolites 1 – 3 , recruit proteins 4 , 5 , and regulate phase separation...

Stable RNA structures also impact RNA–RNA interactions and affect RNA stabilities.

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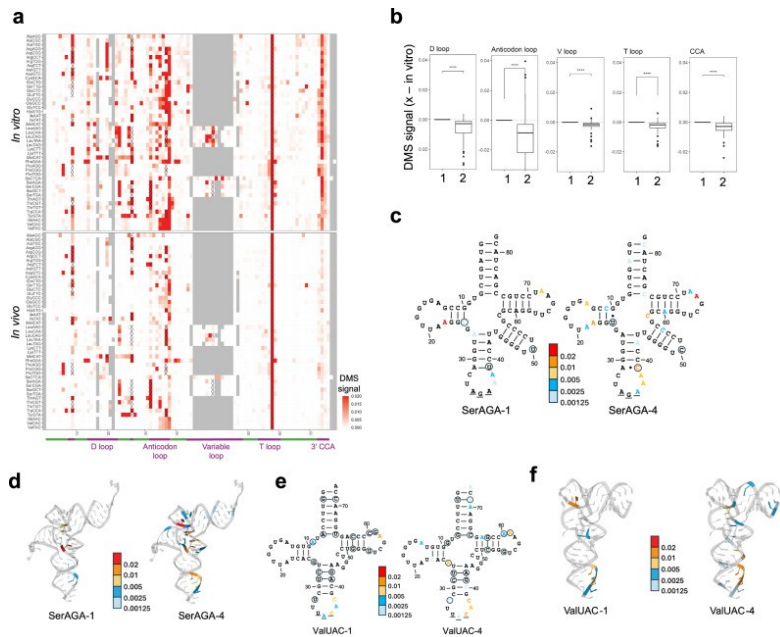
2025 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

材料 / 方法 / 分析路线

, the in vivo structures of t RNAs have been largely taken for granted to resemble their in vitro forms. However, t RNAs have many well-characterized protein inter- actions and modifications in vivo, which led us to hypothesize that their in vivo str... applied t RNA structure-seq to examine in vivo structures of E. coli t RNAs and revealed dynamics of t RNA folding under heat stress20.

核心结果

reveal varia- tions of t RNA structurome and dynamic interactome that have functional consequences in translational regulation. RNAs are both carriers of genetic information and adapters that mediate biomolecular interactions. In cells, RNAs fold into secondary and tertiary structures to detect metabolites1–3, recruit proteins4,5, and regulate phase separation6. Stable RNA structures also impact RNA–RNA interactions and affect RNA stabilities. In biological pro- cesses, including transcription, m RNA splicing, and translation, RNA structures play universal and profound regulatory roles.



4.15 讨论与局限: In vivo structure profiling reveals human cytosolic and mitochondrial tRNA

4.15 | Nature communications (2025) | experimental | DOI 10.1038/s41467-025-59435-5

讨论与局限

The rapid development of high-throughput sequencing methods for highly structured RNAs made it possible to examine transcriptome-wide tRNA structure...

With DM-DMS-Ma Pseq, we present comprehensive in vivo structure and interaction studies for human chromosomal and mitochondrial encoded tRNAs.

Compared to DMS mapping of mRNAs, in vivo DMS signals for tRNA are much lower; therefore, additional considerations need to be implemented to...

This was accomplished here by treating the DMS-reacted RNA samples with a m1A/m3C demethylase before RT and considering only the mutation di...

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